

Financial-Conditions Targeting and the Erosion of Treasury Privilege*

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August 2, 2026

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Abstract

Recent evidence shows that Treasury convenience yields have declined while demand for dollars remain strong. This paper develops a model of risk-bearing and service to show how monetary policy targeting financial conditions can explain this decoupling. When targeting reduces the conditional variance of equity returns, it lowers tail risk and increases the pledgeable collateral value of equity. In a closed economy, equity-backed safe-claim supply expands and the Treasury convenience wedge falls. We then allow foreign investors to hold U.S. equity and Treasuries. In an open economy, the Treasury convenience yield falls when domestic collateral capacity growth exceeds foreign absorption, and this capacity expansion can generate a negative valuation effect that deteriorates the U.S. net foreign asset position.

JEL Codes: E44, E52, F30, G12.

*We are grateful to XXX.. and participants at various seminars for helpful comments. Henry Gold and Wajdi Maaloul provided excellent research assistance. All errors are my own.

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1 Introduction

This paper studies whether monetary policy that targets financial conditions can decrease the Treasury convenience yield by expanding the collateral value of equity. Recent evidence documents a decoupling: the median five-year Treasury convenience premium fell from 20 basis points pre-2008 to -25 basis points in 2021–2025, while the dollar convenience premium remained near 20 basis points (Du et al., 2025). Foreign investors have accumulated massive U.S. asset positions, currently holding \$23.2 trillion in U.S. equity and \$9.35 trillion in Treasuries.

To evaluate how stabilization policy connects these facts, we bridge two distinct literatures. Models of financial-conditions targeting show that a preannounced target reduces asset-price risk (Caballero et al., 2024; Caballero and Simsek, 2026), while holding the supply of safe, liquid claims fixed. Conversely, models of Treasury convenience yields characterize the demand for safe assets, but treat the service capacity of private claims as exogenous (Krishnamurthy and Vissing-Jorgensen, 2012; Jiang, 2024). We connect these frameworks by developing a model of risk-bearing where equity’s pledgeable collateral value responds endogenously to monetary policy. We measure equity pledgeability as the maximum loan-to-value ratio that keeps the probability of a collateral shortfall below a fixed threshold, allowing equity to indirectly produce collateral services by backing information-insensitive debt.

We show that a stronger financial-conditions target decreases the Treasury convenience yield when equity collateral capacity is active and responsive to tail risk. First, a stronger target lowers equity-return variance when it shifts return exposure away from macroeconomic shocks and the equilibrium remains stable. Second, when this lower variance improves the lower tail of the equity payoff, it raises equity pledgeability. The quantity of equity-backed safe claims expands, their scarcity value falls, and both the Treasury convenience yield and the Treasury–equity convenience spread decline. Third, we show this transmission relies strictly on collateral substitution: without equity services, lower equity risk leaves the Treasury convenience yield unchanged.

The economic mechanism relies on how return dispersion shapes collateral quality. Creditors care about recovery values in adverse states. By reducing the conditional variance of equity returns, financial-conditions targeting compresses the lower tail of the payoff distribution. This higher downside recovery value allows the same unit of equity to support a larger volume of information-insensitive debt. As the supply of equity-backed safe claims expands, Treasuries face greater substitution. The marginal value of safety falls, and investors therefore require a smaller return concession to hold Treasuries.

In an open economy, a policy change that lowers risk and raises pledgeability creates opposing portfolio forces: lower risk and greater pledgeability increase equity’s appeal to Foreign investors, while the compressed expected equity premium decreases it. The net effect on Foreign dollar equity demand is ambiguous and depends on which force dominates. Foreign

share ownership rises if and only if this dollar equity demand grows faster than the equity price. Because collateral services accrue to the beneficial owner, the Treasury convenience yield falls if and only if the expansion in domestic collateral capacity exceeds Foreign service absorption and any induced increase in Home consumption demand.

Finally, we evaluate the external accounts and Home asset-holder utility. The fixed-price acquisition component of a Foreign purchase of U.S. claims leaves U.S. net foreign assets unchanged. However, if the collateral expansion raises the equity price, the appreciation of inherited Foreign-owned shares creates a negative valuation effect that lowers total U.S. net foreign assets unless offset by the current account (Atkeson et al., 2025). Home asset-holder utility depends strictly on consumption and retained convenience services; a lower Treasury convenience yield improves Home welfare when it reflects a greater abundance of retained safe claims rather than an offsetting cross-border transfer.

Related Literature

Policy puts, financial conditions, and identification. A growing empirical literature documents feedback from equity prices to monetary policy using reaction functions, high-frequency identification, and textual measures (Rigobon and Sack, 2003; Bjørnland and Leitemo, 2009; Kurov et al., 2022; Shapiro and Wilson, 2022; Cieslak and Vissing-Jorgensen, 2021). This evidence need not imply an independent equity-price objective. It is consistent with the Federal Reserve responding to the implications of stock prices for its standard macroeconomic goals (Fuhrer and Tootell, 2008), including the effects of stock-market wealth on consumption and real activity (Chodorow-Reich et al., 2021; Di Maggio et al., 2020).

In the other direction, monetary-policy surprises move equity valuations, perceived risk, and risk appetite (Bernanke and Kuttner, 2005; Bekaert et al., 2013; Bauer et al., 2023; Cieslak et al., 2019). Option prices provide direct evidence of a policy put, although the estimated effect is concentrated before 2008 (Dahiya et al., 2026). Interpreting announcement-window movements requires separating policy shocks from central-bank information and common-news responses (Jarociński and Karadi, 2020; Nakamura and Steinsson, 2018; Bauer and Swanson, 2023; Miranda-Agrippino and Ricco, 2021). Target and path factors and the transmission of high-frequency innovations to credit costs provide complementary links from policy news to financial conditions (Gürkaynak et al., 2005; Gertler and Karadi, 2015). Conditional asset-purchase promises can also affect prices by changing expected support specifically in bad states (Haddad et al., 2025). We build on the recruitment mechanism and operational protocol in Caballero et al. (2024); Caballero and Simsek (2026) and on models with limited specialist risk-bearing capacity (Shleifer and Vishny, 1997; Gabaix and Koijen, 2021). Our contribution is to add endogenous usable collateral value: by improving equity's lower tail, policy raises pledgeability and equity collateral-service capacity, which substitutes for Treasury services and lowers the Treasury service wedge.

Treasury and dollar convenience. Treasury convenience reflects government-debt supply and the demand for safety and liquidity (Krishnamurthy and Vissing-Jorgensen, 2012; Du et al., 2018; Nagel, 2016). Information-insensitive collateralized claims provide a private substitute (Gorton and Pennacchi, 1990; Dang et al., 2015). The recent evidence separates the persistent convenience of the dollar from the erosion of Treasury-specific convenience (Du et al., 2025) and attributes much of the latter to fiscal supply (Jiang et al., 2025). We complement those explanations with a policy-sensitive supply of equity collateral services. Equity itself is risky; its pledgeable portion supplies Treasury-equivalent services.

International portfolios and external valuation. Delegated demand, intermediary capacity, and currency denomination jointly shape international portfolios, asset prices, and exchange rates (Maggiori, 2017; Maggiori et al., 2020; Camanho et al., 2022; Kojien and Yogo, 2026). We add policy-sensitive equity risk and pledgeability, making portfolio rotation, dollar demand, and beneficial ownership distinct. The external-accounts literature in turn separates flows from valuation and emphasizes the growing role of U.S. equity liabilities (Gourinchas and Rey, 2007; Bertaut et al., 2024; Atkeson et al., 2025; Tabova and Warnock, 2025). Our contribution is to connect the ownership margin to future valuation exposure while tracking collateral-service incidence without inferring Home asset-holder utility from NFA alone.

Section 2 presents the motivating empirical facts. Section 3 presents the closed-economy model. Section 4 introduces the operational policy protocol and derives mandate improvement, recruitment, and service-wedge erosion. Section 5 studies open financial conditions, foreign portfolios, the joint equity–FX equilibrium, external accounts, and asset-holder and participation incidence.

2 Motivating Evidence

This section documents two empirical facts that motivate the model. First, financial accommodation is followed by a compression of the option-implied distribution of future equity returns. Second, a compressed option-implied equity return distribution predicts a lower Treasury convenience yield, but not a lower dollar convenience yield.

2.1 Data and Measurement

Policy events and accommodation measures. The sample consists of scheduled FOMC announcements. As controls for conventional monetary policy shocks, we use the high-frequency monetary and central-bank-information surprises of Jarociński and Karadi (2020). This decomposition exploits the joint movement of interest rates and stock prices around an announcement: a monetary tightening raises interest rates and lowers stock prices, whereas favorable central-bank information raises both. We reverse the monetary component so a positive value denotes

accommodation, and we standardize both components over the sample.

We construct a full-day financial-accommodation proxy using the sign-restricted stock–bond decomposition of [Cieslak and Pang \(2021\)](#). This proxy isolates the equity-price contribution of the common- and hedging-premium components derived from innovations to the S&P 500 and the two-, five-, and ten-year Treasury yields. A positive value indicates that risk-premium compression increased equity prices on the FOMC day. While the full-day window matches the close-to-close timing of the option outcomes, it also captures non-policy news arriving on the day. To separate the proxy from conventional policy surprises, regressions using this proxy control for the high-frequency [Jarociński and Karadi \(2020\)](#) surprises.

Option-implied equity risk. We construct a daily, constant-30-day risk-neutral return distribution from OptionMetrics SPX quotes from January 1996 through August 2025. For each date and expiration, we recover the forward price from put–call parity, invert implied volatilities from option midquotes, and smooth the volatility smile using out-of-the-money puts and calls. We interpolate total variance between the expirations bracketing 30 days. The derivatives of the resulting interpolated price curve yield the continuous risk-neutral density and cumulative distribution functions.

To hold the measurement domain constant across daily variations in strike availability, we enforce a fixed moneyness window of $K/F \in [0.80, 1.05]$. We drop crossed and nonpositive quotes, impose no-arbitrage and density-positivity conditions, and require both endpoints of any measured change to pass these quality filters. We anchor event horizons to exact SPX trading days; the endpoints $d - 1$ and $d + h$ are located on the unconditional trading calendar before applying data filters, ensuring the event window remains fixed.¹

We define five measures to characterize the 30-day risk-neutral return distribution. At-the-money (ATM) volatility is the implied volatility at strike price $K = F$. Downside volatility is the put-option contribution to total risk-neutral volatility. The 5 and 2.5 percent return quantiles, $q_{.05}$ and $q_{.025}$, measure left-tail risk; an increase in these quantiles (a less negative return) indicates lower downside risk. The metric $p_{10} = \Pr^Q(R_{30} \leq -0.10)$ is the risk-neutral probability of a 30-day loss exceeding 10 percent.

Treasury and dollar convenience. Treasury convenience, denoted TCY , is the three-month government-bond covered-interest-parity wedge of [Du et al. \(2018, 2025\)](#). A higher TCY indicates a larger convenience yield for U.S. Treasuries relative to currency-matched foreign government bonds, measured in basis points. We construct a corresponding three-month dollar

¹Robustness specifications apply stricter support rules that exclude 30-day maturity and outcome-specific strike extrapolations.

convenience yield, $DCY_{c,t}$, using public money-market rates:

$$DCY_{c,t} = 100 \left(i_{c,t} - i_{\$,t} - \rho_{c,t}^{IBOR} \right), \quad (1)$$

where $i_{c,t}$ and $i_{\$,t}$ are the foreign-currency and dollar three-month cash rates, respectively, and $\rho_{c,t}^{IBOR}$ is the forward premium; the factor 100 converts the measure to basis points.

The daily panel covers eight currencies: AUD, CAD, DKK, EUR, GBP, JPY, NOK, and SEK. We restrict the sample to dates with at least five available currencies. For formal comparisons, we average TCY and DCY over identical currency–date observations. We sample the last joint observation in each Friday-ending week to construct a panel of 1,148 matched weeks from January 2000 through December 2021. We build these weekly variables prior to merging with option data to ensure option-quality filters do not alter the observation horizons.

2.2 Two Motivating Facts

Fact 1: Financial accommodation compresses the option-implied equity distribution. For outcome Y , event date d , and horizon $h \in \{0, 1, 5, 10, 20\}$, we estimate the local projection:

$$Y_{d+h} - Y_{d-1} = \alpha_h + \beta_h RP_d + \gamma_h A_d + \delta_h I_d + \varepsilon_{d,h}, \quad (2)$$

where RP_d is the standardized financial-accommodation proxy, A_d is the standardized [Jarociński and Karadi \(2020\)](#) monetary-accommodation surprise, and I_d is the corresponding standardized information surprise. We calculate standard errors using HC1 at $h = 0$ and Newey–West with lag h at longer horizons.

The event-response horizon h is measured in trading days, whereas the option outcomes describe a constant 30-calendar-day horizon. Table 1 reports the responses at the one- and ten-trading-day horizons. The coefficients represent the percentage-point change in the option outcome associated with a one-standard-deviation increase in the financial-accommodation proxy.

At the one-day horizon, the estimates have the predicted signs, but only the 2.5 percent return quantile is individually significant, increasing by 0.18 percentage points. By the ten-day horizon, the effects are precisely estimated: a one-standard-deviation increase in financial accommodation decreases ATM and downside volatility by 0.22 and 0.24 percentage points, respectively. It increases the 5 and 2.5 percent return quantiles by 0.17 and 0.29 percentage points, reflecting reduced downside risk, and decreases the risk-neutral probability of a 10 percent monthly loss by 0.16 percentage points.

At the 20-day horizon, the point estimates attenuate and partially reverse. Because estimates at this horizon are highly sensitive to the endpoint support filters, we cannot conclude that the effects persist uniformly to 20 days. Estimates using the VIX index confirm the decline in

volatility. A one-standard-deviation increase in the full-day financial-accommodation proxy decreases the close-to-close VIX by 0.20 points (standard error 0.09). We use the full-day proxy as the benchmark to align with the close-to-close timing of the VIX. Estimating the response of the close-to-close VIX to the narrow announcement-window proxy yields a 0.63-point increase ($p = .038$); however, this specification is not the benchmark because the intraday treatment window is mechanically mismatched with the daily outcome window.

Table 1: Financial Accommodation and the Option-Implied Equity Distribution

Outcome	$h = 1$	$h = 10$
ATM implied volatility	−0.172 (.112)	−0.223 (.016)
Downside volatility	−0.131 (.188)	−0.239 (.002)
5% return quantile	0.126 (.118)	0.172 (.010)
2.5% return quantile	0.185 (.044)	0.293 (< .001)
$\Pr^Q(R_{30} \leq -10\%)$	−0.094 (.160)	−0.164 (.003)
Observations	192	191

Notes. The table reports β_h from equation (2). Quantiles, probabilities, and volatility measures are multiplied by 100. Parentheses report Newey–West p-values with a lag equal to the horizon h . Both the initial ($d - 1$) and terminal ($d + h$) observation dates must pass the baseline quality filter.

Although scale compression is the primary result, financial accommodation also predicts changes in some shape statistics. Across 30 scale-free tests of risk-neutral skewness, kurtosis, quantiles measured in volatility units, and fixed-sigma loss probabilities, two financial-accommodation coefficients survive a 5-percent Benjamini–Hochberg correction. Same-day skewness decreases—opposite to the safer-tail prediction—while the 2.5-sigma loss probability falls at $h = 10$. Controlling directly for the contemporaneous change in ATM volatility attenuates most tail coefficients. Because ATM volatility is a post-treatment outcome, we report this conditional specification as a descriptive exercise. Overall, financial accommodation compresses total option-implied risk but provides mixed evidence of changes in the standardized shape of the distribution.

Because we do not observe pledgeability directly in the data, we define the tail-implied recovery factor, $\tilde{m}_t^H$, and the capacity index, $\tilde{X}_t^H$, to map the empirical estimates to the model:

$$\tilde{m}_t^H = 1 + q_{.01,t} \sqrt{H/30}, \quad \tilde{X}_t^H = \tilde{m}_t^H P_t, \quad H \in \{2, 10\}, \quad (3)$$

where $q_{.01,t}$ is the 1 percent return quantile and P_t is the underlying price level. At the ten-day horizon ($h = 10$), a one-standard-deviation increase in financial accommodation raises $\log \tilde{m}^H$ by 0.072 to 0.169 percent ($p = .014$ and $.015$), raises $\log P$ by 0.243 percent ($p = .046$), and raises $\log \tilde{X}^H$ by 0.315 to 0.412 percent ($p = .015$ and $.005$). These ranges correspond to the chosen margin horizon H . Because $\tilde{m}^H$ transforms option prices rather than measuring an observed

haircut, these indices provide a model-consistent translation rather than a direct empirical measure.

Fact 2: Decreases in option-implied equity risk predict lower Treasury convenience, but not lower dollar convenience. To relate the option-implied equity distribution to convenience yields, we estimate the forecasting regression:

$$CY_{t+h}^j - CY_{t-1}^j = a_h^j + \eta_h^j \Delta Z_t + e_{t,h}^j, \quad h \in \{0, 1, 4, 8, 12\} \text{ weeks}, \quad (4)$$

where $\Delta Z_t = Z_t - Z_{t-1}$ is the change between option observations matched to consecutive Friday-ending weeks, and $j \in \{TCY, DCY, TCY - DCY\}$. We calculate Newey–West standard errors with lag $\max\{h, 1\}$. The three outcomes use identical currency–date observations and the same regression sample. We match each week to the most recent valid option observation on or before Friday, subject to a maximum age of seven calendar days.

Table 2 reports the four-week ($h = 4$) coefficients. We reorient and standardize this weekly change so that a one-standard-deviation increase denotes safer equity. A one-standard-deviation reduction in equity risk predicts a 2.1 to 2.7 basis point decline in Treasury convenience. In contrast, the corresponding *DCY* coefficients are not statistically significant. Because the regressors and sample are identical across equations, the *TCY* – *DCY* coefficient equals the exact difference between the *TCY* and *DCY* estimates.

Table 2: Option-Implied Equity Safety and Matched Convenience Yields

Option-implied safety measure	<i>TCY</i>	<i>DCY</i>	<i>TCY</i> – <i>DCY</i>
5% return quantile	–2.08** (0.81)	0.14 (0.72)	–2.22* (1.27)
Probability of a 10% loss	–2.73*** (0.99)	0.72 (0.74)	–3.45** (1.42)
Downside volatility	–2.54*** (0.92)	0.60 (0.77)	–3.14** (1.41)
ATM implied volatility	–2.74*** (0.92)	0.79 (0.81)	–3.53** (1.44)
Observations	1,096	1,096	1,096

Notes. The table reports η_4^j from equation (4). The predictor ΔZ_t is reoriented so that a positive change corresponds to a reduction in equity risk, and is standardized over the common regression sample. Coefficients represent basis-point changes from week $t - 1$ to week $t + 4$. All outcomes are means over identical currency–date observations. Parentheses report Newey–West standard errors with four lags. We omit the 1 percent quantile because the sample size with valid deep-strike support is substantially smaller.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The relationship between equity risk and Treasury convenience persists at the eight- and twelve-week horizons. Across all five option measures, a one-standard-deviation reduction in

equity risk predicts a *TCY* decline between 1.17 and 2.74 basis points ($p \leq .043$), whereas the corresponding *DCY* estimates remain statistically insignificant. The estimated *TCY* – *DCY* gap retains the predicted sign but does not survive a Holm multiple-hypothesis correction across horizons. The *TCY* association holds in the post-2008 subsample. Applying strict support filters preserves the results for crash probability, downside volatility, and ATM volatility at all horizons, as well as the 5 percent quantile at the eight- and twelve-week horizons.

Results using the VIX index confirm this pattern. ATM implied volatility and the VIX exhibit correlations of .992 in levels and .959 in daily changes. Consistent with Table 2, a one-standard-deviation decline in ATM volatility predicts *TCY* declines of 2.74, 2.13, and 1.55 basis points at four, eight, and twelve weeks ($p = .003, .002, .004$), while the matched *DCY* estimates are not statistically significant. Regressing the three-month G10-median *TCY* directly on the public VIX yields a consistent estimate: a one-standard-deviation increase in the VIX predicts a 1.96 basis point increase in *TCY* after twelve weeks ($p = .075$). Because this specification does not use the matched *DCY* panel, it serves as a directional benchmark rather than a formal comparison.

2.3 Interpretation

Together, the two motivating facts establish an empirical sequence: financial accommodation is followed by lower option-implied equity risk, and lower equity risk predicts a decline in Treasury convenience without a corresponding decline in dollar convenience.

The theoretical framework in Section 3 generates this joint empirical response. In the model, lower equity risk increases downside recovery values, which increases the pledgeability of equity collateral. This expanded pledgeability allows the financial sector to supply more information-insensitive claims. Because investors treat these private claims as substitutes for U.S. Treasuries, the increase in private supply reduces the scarcity value of Treasury collateral, which decreases the Treasury convenience yield. The model formally characterizes this collateral-substitution channel and the resulting foreign portfolio reallocation.

3 A Closed Economy with Convenience Services

We first study a closed economy with two equilibrium assets. Households directly hold and price Treasuries, own equity, and delegate equity trading to portfolio managers. A Treasury dollar has service intensity one, while a dollar of equity ownership delivers m_t Treasury-equivalent service units. Section 5 then allows foreign investors to own U.S. assets and receive their dollar services.

3.1 Environment

Time is discrete, $t = 0, 1, \dots$. The economy contains hand-to-mouth workers, asset-holding households, a representative firm, portfolio managers, a fiscal authority, and a central bank. Households own the managers and all domestic assets. As in [Caballero et al. \(2024\)](#), workers separate labor income from household consumption, households drive aggregate demand, and managers choose households' equity exposure. Uppercase letters denote level quantities and lowercase letters their log deviations unless otherwise stated.

Supply side. Output is produced from capital and labor, $Y_t = Z_t K^\nu L_t^{1-\nu}$, with capital fixed $K = 1$. Nominal prices are fully sticky, so output is demand determined. Assume that potential output $Y_t^* = \bar{Y} \exp(y_t^*)$ follows

$$y_t^* = y_{t-1}^* + \varepsilon_{z,t}, \quad \varepsilon_{z,t} \sim N(0, \sigma_z^2), \quad (5)$$

where $\varepsilon_{z,t}$ is i.i.d. Capital and labor receive their factor shares. Workers supply all labor and spend hand-to-mouth.

Assets and ownership. There are two assets. The *market portfolio* (equity) is a claim to capital's share of output, $d_{P,t} = \nu Y_t$. Its supply is one and its ex-dividend price is P_t . Households own it but delegate its allocation to the managers of Section 3.3. The *Treasury bond* is the economy's one-period default-free claim. Let B_t denote its outstanding quantity. The fiscal authority maintains a fixed positive quantity $B_t = \bar{B} > 0$. The bond trades at price $q_{B,t}$. Households hold and price Treasuries directly; portfolio managers use the Treasury return as the safe benchmark for equity exposure.

The Treasury defines the economy's observed short safe rate,

$$r_t \equiv y_{B,t} \equiv -\log q_{B,t}, \quad \hat{r}_t \equiv r_t - \bar{r}, \quad (6)$$

where $\bar{r}$ is the deterministic steady-state return. The central bank chooses r_t .

Household accounting and delegation. Households hold Treasuries directly, but delegate equity holdings to portfolio managers introduced in Section 3.3. Managers choose equity exposure, taking into account the cash flow and nonpecuniary benefits that accrue to households. For the household, uses of funds equal cash on hand:

$$C_t + P_t + q_{B,t} \bar{B} = \nu Y_t + P_t + \bar{B}. \quad (7)$$

Equity price appears on both sides because households roll over their ownership.

3.2 Convenience and the Price of Safety

Preferences. Households have preferences

$$\mathbb{E}_0 \sum_{t \geq 0} \beta^t \left[\log C_t + \xi v \left(\frac{S_t}{C_t} \right) \right], \quad (8)$$

where $\beta \in (0, 1)$ is the discount factor, C_t is consumption, and S_t is a service aggregate that captures safety, liquidity, and access to secured financing. The parameter $\xi \geq 0$ governs demand for these services. We assume that $v' > 0$, $v'' < 0$ and period utility is jointly concave in (C_t, S_t) .

Service supply. A dollar of Treasury has unit service intensity. A dollar of beneficial equity ownership has service intensity $m_t \in [0, 1]$, which Section 3.4 derives as the pledgeable share of equity, inseparable from ownership. Let $X_t \equiv m_t P_t$ denote this usable equity collateral capacity. Aggregate household services are therefore

$$S_t = \underbrace{q_{B,t} \bar{B}}_{\text{Treasury services}} + \underbrace{X_t}_{\text{equity collateral services}}, \quad \omega_p^S \equiv \frac{X_t}{S_t} \in [0, 1], \quad (9)$$

with $\omega_p^S < 1$ whenever Treasury supply is positive. We write $s_t \equiv \log(S_t/\bar{S})$.

Service demand and pricing. The household's first-order condition for Treasuries is

$$q_{B,t} = \frac{Q_{B,t}}{1 - \lambda_t^S}, \quad \lambda_t^S \equiv \frac{U_S(C_t, S_t)}{U_C(C_t, S_t)} = \frac{\xi v'(z_t)}{1 - \xi z_t v'(z_t)}, \quad (10)$$

where $z_t \equiv S_t/C_t$ is service supply relative to consumption and $Q_{B,t} \equiv \mathbb{E}_t[\Lambda_{t+1}]$ is the pecuniary value of the Treasury's unit payoff.² Equity pricing is determined by managers.

Equation (10) defines the marginal service wedge λ_t^S . Because it is the value of one unit of services, it acts as the household's inverse demand curve. Holding consumption fixed, strict joint concavity implies that λ_t^S falls as service supply rises. The wedge increases with ξ and vanishes when $\xi = 0$.

Treasuries. Because Treasury intensity is one, its marginal service wedge is $\lambda_{T,t}^S \equiv \lambda_t^S < 1$. Let r_t^{pec} denote the return required from the Treasury's cash flow after stripping out these services. The pecuniary value of its payoff is $e^{-r_t^{\text{pec}}}$, so $q_{B,t} = e^{-r_t^{\text{pec}}} / (1 - \lambda_{T,t}^S)$, or equivalently $r_t^{\text{pec}} = r_t - \log(1 - \lambda_{T,t}^S)$. The Treasury convenience yield, understood as its return concession

²The exact pecuniary stochastic discount factor is $\Lambda_{t+1} = \beta(C_t/C_{t+1})[1 - \xi z_{t+1} v'(z_{t+1})] / [1 - \xi z_t v'(z_t)]$. When v is logarithmic, it reduces to $\beta C_t/C_{t+1}$.

relative to the pecuniary value of the same default-free payoff, is

$$r_t^{\text{pec}} - r_t = -\log(1 - \lambda_{T,t}^S). \quad (11)$$

The deterministic steady state satisfies $\bar{r}^{\text{pec}} = \rho \equiv -\log \beta$ and $\bar{r} = \rho + \log(1 - \bar{\lambda}_T^S)$. The corresponding pecuniary rate gap is

$$\hat{r}_t^{\text{pec}} \equiv r_t^{\text{pec}} - \rho = \hat{r}_t - \log\left(\frac{1 - \lambda_{T,t}^S}{1 - \bar{\lambda}_T^S}\right).$$

The central bank chooses the quoted rate r_t ; the service market determines the convenience-stripped rate r_t^{pec} .

Equity. Standard models treat equity as a pecuniary claim with no service value, but financial institutions routinely pledge equities as collateral. We endogenize the usable share m_t in Section 3.4. Because one dollar of beneficial equity ownership raises usable collateral value by m_t , equity has the marginal service wedge

$$\lambda_{P,t} = m_t \lambda_t^S. \quad (12)$$

Relative to a pecuniary clone, equity's exact service-induced log valuation wedge is $-\log(1 - \lambda_{P,t})$. To first order, the valuation gap between Treasuries and equity is the difference in their marginal service wedges, $\lambda_{T,t}^S(1 - m_t)$. A higher m_t raises usable collateral value per dollar of equity, increases the equity service wedge, and narrows this valuation gap.

3.3 Arbitrageurs and Endogenous Volatility

Households delegate equity allocation to a unit mass of portfolio managers who do not consume. A fraction η are noise traders whose equity weight differs from the benchmark by μ_t^p / η ; a mass $1 - \eta - \alpha_p$ are inelastic funds that hold the benchmark; and a mass α_p are arbitrageurs (elastic funds) that choose their weights optimally. Equity-market clearing requires arbitrageurs to absorb aggregate noise:

$$\alpha_p \omega_t^A + \eta \left(1 + \frac{\mu_t^p}{\eta}\right) + (1 - \eta - \alpha_p) = 1 \implies \omega_t^A = 1 - \frac{\mu_t^p}{\alpha_p}. \quad (13)$$

Financial noise follows

$$\mu_t^p = \varphi_\mu \mu_{t-1}^p + \varepsilon_{\mu,t}, \quad \varphi_\mu \in [0, 1), \quad \varepsilon_{\mu,t} \sim N(0, \sigma_\mu^2),$$

independently of supply and demand shocks.

Arbitrageurs choose portfolios after observing r_t and current financial noise. Their delegated mandate maximizes expected log portfolio value while internalizing the household's marginal value of asset services. Let $r_{p,t+1}^P$ denote the traded log equity return, and define the excess return and its conditional variance as

$$r_{p,t+1}^x \equiv r_{p,t+1}^P - r_t, \quad x \equiv \text{Var}_t(r_{p,t+1}^x).$$

Assume returns are conditionally homoskedastic following [Caballero et al. \(2024\)](#). Under a local lognormal approximation (derived in Appendix A.2), the arbitrageur's objective is

$$\max_{\omega_t^A} \left\{ \omega_t^A \left[\mathbb{E}_t r_{p,t+1}^x + \lambda_{P,t} - \lambda_{T,t}^S \right] + \frac{1}{2} \omega_t^A (1 - \omega_t^A) x \right\}.$$

The service differential is expressed in return units. The first-order condition is

$$\mathbb{E}_t r_{p,t+1}^x + \lambda_{P,t} - \lambda_{T,t}^S = \left(\omega_t^A - \frac{1}{2} \right) x.$$

Combining this condition with equity-market clearing yields the required traded return:

$$\mathbb{E}_t r_{p,t+1}^x = \underbrace{\frac{x}{2}}_{\text{risk compensation}} - \underbrace{\mathcal{I}_P \mu_t^p}_{\text{noise absorption}} + \lambda_{T,t}^S - \lambda_{P,t}, \quad \mathcal{I}_P \equiv \frac{x}{\alpha_p}. \quad (14)$$

Greater pledgeability lowers the required equity return by increasing the collateral services it provides. Conversely, a higher Treasury service wedge raises the required equity return because equity is priced relative to Treasuries.

Log-linearizing the total-payoff identity and substituting the manager's first-order condition yields the equity-pricing equation in deviations from the balanced-growth path:

$$p_t = \omega_P \mathbb{E}_t p_{t+1} + \omega_D \mathbb{E}_t y_{t+1} - \hat{r}_t + \mathcal{I}_P \mu_t^p + (\lambda_{P,t} - \bar{\lambda}_P) - (\lambda_{T,t}^S - \bar{\lambda}_T^S), \quad (15)$$

where the payoff weights are

$$\omega_D \equiv \frac{1 - \beta - \bar{\lambda}_P}{1 - \bar{\lambda}_P}, \quad \omega_P \equiv \frac{\beta}{1 - \bar{\lambda}_P}.$$

A rise in the equity service wedge raises its price, while a rise in the Treasury service wedge raises the return required on equity and lowers its price. Appendix A.2 details this derivation.

Two forces make return risk endogenous. First, higher x raises $\mathcal{I}_P$, so arbitrageurs require a larger price concession to absorb a given noise flow. Second, the larger price response raises

return risk. The policy rule for r_t helps determine this scalar fixed point. We take the path of r_t as given here and solve for policy and x jointly in Section 4.

3.4 Pledgeability Microfoundation

We microfound equity pledgeability m as the maximum loan-to-value ratio that sustains information-insensitive borrowing. Consider an undrawn one-period collateralized credit line backed by one unit of equity. If drawn to its limit mP_t , its promised repayment is $e^{r_t}mP_t$, where the safe rate r_t determines the cost of debt. Because dividends accumulate, the collateral payoff is $P_{t+1} + d_{P,t+1}$.

Let $\kappa_C \in (0, 1)$ denote liquidation and operating costs per unit of collateral. Following Gorton and Pennacchi (1990) and Dang et al. (2015), the debt remains information-insensitive only if its loss probability does not exceed a tolerance $\bar{p} < 1/2$. This tolerance marks the threshold below which creditors do not find it profitable to acquire private information about the collateral's payoff.

Let $Q_{\bar{p}}^x$ denote the lower $\bar{p}$ -quantile of the log excess equity return $r_{p,t+1}^x$ at the balanced-growth point. The maximum advance rate consistent with information-insensitive borrowing is

$$m = \max \left\{ 0, \exp \left\{ Q_{\bar{p}}^x \right\} - \kappa_C \right\}. \quad (16)$$

Equity cannot be used as collateral when $Q_{\bar{p}}^x \leq \log \kappa_C$. We maintain the nonsaturation condition (A.22), $Q_{\bar{p}}^x < \log(1 + \kappa_C)$, which guarantees $m < 1$: a dollar of equity never provides more collateral services than a dollar of Treasuries.

Under lognormal returns, the lower $\bar{p}$ -quantile of the log excess return is $Q_{\bar{p}}^x = \bar{\mu}^x - z_{\bar{p}}\sqrt{x}$, where $z_{\bar{p}}$ is the standard-normal cutoff for tail probability $\bar{p}$. At the balanced-growth point, the manager's first-order condition implies

$$\bar{\mu}^x = \frac{x}{2} + (1 - m)\bar{\lambda}_T^S. \quad (17)$$

Combining these expressions with (16) gives the lognormal capacity equation:

$$m = \max \left\{ 0, \exp \left\{ \frac{x}{2} + (1 - m)\bar{\lambda}_T^S - z_{\bar{p}}\sqrt{x} \right\} - \kappa_C \right\}. \quad (18)$$

How does return variance affect pledgeability? In (18), a higher x raises average excess return through $x/2$, but worsens bad outcomes through $z_{\bar{p}}\sqrt{x}$. When $x < z_{\bar{p}}^2$, the second effect dominates, so lower variance increases pledgeability. Section 4 uses this condition to determine when policy-induced risk reduction expands collateral capacity.

3.5 Financial Conditions and Equilibrium

We now connect asset prices to aggregate demand and define financial conditions.

For the first-order dynamics below, we hold m fixed at its equilibrium value. Consequently, $\lambda_{P,t} = m\lambda_{T,t}^S$ and $\bar{\lambda}_P = m\bar{\lambda}_T^S$, so the two service wedges share the same log deviation:

$$\ell_t \equiv \log \frac{\lambda_{T,t}^S}{\bar{\lambda}_T^S} = \log \frac{\lambda_{P,t}}{\bar{\lambda}_P}.$$

In the linearized pricing equation (15), the final two service terms therefore reduce to $\zeta \ell_t$, where $\zeta \equiv \bar{\lambda}_P - \bar{\lambda}_T^S$. Section 4 allows the equilibrium value of m to adjust with policy and characterizes the resulting change in pledgeability.

Aggregate demand. Following Caballero et al. (2024), an autonomous-expenditure component Δ_t enters goods demand independently of household preferences and asset accumulation. Define $\delta_t \equiv \Delta_t / (\beta \nu \bar{Y})$ and assume it follows an AR(1) process:

$$\delta_t = \varphi_\delta \delta_{t-1} + \varepsilon_{\delta,t}, \quad \varphi_\delta \in [0, 1), \quad \varepsilon_{\delta,t} \sim N(0, \sigma_\delta^2), \quad (19)$$

where the innovation is independent of supply and financial-noise shocks.

Let $C_t^H \equiv C_t$ denote optimizing-household consumption. Household optimization, hand-to-mouth spending, and goods-market clearing imply:

$$\begin{aligned} C_t^H &= (1 - \beta)(\nu Y_t + P_t + \bar{B}) - \lambda_{P,t} P_t - \lambda_{T,t}^S q_{B,t} \bar{B}, \\ C_t^{HM} &= (1 - \nu) Y_t + q_{B,t} \bar{B} - \bar{B}, \\ Y_t &= C_t^H + C_t^{HM} + \Delta_t. \end{aligned} \quad (20)$$

Consolidating these components yields the joint aggregate-demand condition:

$$\beta \nu Y_t = (1 - \beta - \lambda_{P,t}) P_t + \bar{B} \left[\underbrace{q_{B,t}(1 - \lambda_{T,t}^S)}_{=e^{-r_t^{\text{pec}}}} - \beta \right] + \Delta_t. \quad (21)$$

Because the Treasury convenience rent is rebated to households, it cancels out of aggregate demand, leaving the convenience-stripped rate r_t^{pec} . At the deterministic steady state, $\beta \nu \bar{Y} = (1 - \beta - \bar{\lambda}_P) \bar{P}$. We assume $\lambda_{T,t}^S < 1$, which ensures a positive Treasury price, and $\bar{\lambda}_P < 1 - \beta$, which ensures a positive steady-state equity price and dividend weight.

Define $f_t \equiv y_t - \delta_t$ as the component of aggregate demand driven by financial prices. Log-

linearizing (21) yields

$$y_t = f_t + \delta_t, \quad \tilde{y}_t \equiv y_t - y_t^* = f_t - f_t^*, \quad f_t^* \equiv y_t^* - \delta_t, \quad (22)$$

The financial conditions index (FCI) is this component expressed in prices:

$$\begin{aligned} f_t &\equiv p_t - b\hat{r}_t - H\ell_t, \\ \Xi &\equiv \frac{\bar{\lambda}_P}{1 - \beta - \bar{\lambda}_P}, \quad b \equiv \frac{\bar{B}}{\nu\bar{Y}}, \quad b > 0, \quad a_T \equiv \frac{\bar{\lambda}_T^S}{1 - \bar{\lambda}_T^S}, \\ H &\equiv \Xi + ba_T, \end{aligned} \quad (23)$$

A higher f_t denotes looser financial conditions and raises output, through its three components. First, a higher equity price p_t increases optimizing households' wealth relative to current cash flow, which raises their desired consumption. Second, a higher policy rate $\hat{r}_t$ reduces the Treasury rollover proceeds transferred to hand-to-mouth workers, which lowers aggregate demand. Third, a higher common service-wedge deviation ℓ_t raises the service return to saving and tightens conditions by lowering current household spending.

Finally, at fixed mean output, the steady-state aggregate-demand condition implies $d \log \bar{P} = \Xi d \log \bar{\lambda}_P$, as derived in (A.16). Section 4.3 uses this equilibrium price response to solve jointly for usable equity collateral capacity and the Treasury convenience yield.

FCI-targeting regime. Following Caballero et al. (2024); Caballero and Simsek (2026), we capture the “Fed put” by allowing the central bank to supplement its output-gap mandate with a preannounced FCI target. The parameter $\psi \geq 0$ measures the strength of the central bank's commitment to this target. By shaping expectations of future financial conditions, stronger targeting can reduce the return risk faced by portfolio managers. Section 4 characterizes when this reduction in risk raises equity pledgeability and lowers the Treasury convenience yield.

Within period t , macroeconomic innovations $(\varepsilon_{z,t}, \varepsilon_{\delta,t})$ realize first. The central bank then sets the policy rate r_t and announces next period's target f_{t+1}^T . Financial noise $\varepsilon_{\mu,t}$ realizes last, and markets clear. Let $\underline{\mathbb{E}}_t$ denote expectations when the bank acts, and $\mathbb{E}_t$ denote expectations after all date- t shocks realize.

The central bank evaluates policy using the discounted output-gap loss

$$G_t^{\text{mand}} \equiv \underline{\mathbb{E}}_t \left[\sum_{h=0}^{\infty} \beta^h \tilde{y}_{t+h}^2 \right] = \underline{\mathbb{E}}_t \left[\sum_{h=0}^{\infty} \beta^h (f_{t+h} - f_{t+h}^*)^2 \right]. \quad (24)$$

The gap-closing FCI f_t^* therefore expresses the central bank's structural mandate in FCI space.

Given the operational target f_t^T announced at $t - 1$, the bank sets the current rate to minimize:

$$\mathcal{L}_t(\psi; f_t^T) \equiv \min_{\hat{r}_t} \mathbb{E}_t \left[(f_t - f_t^*)^2 + \psi(f_t - f_t^T)^2 \right]. \quad (25)$$

At the same date, it announces the target f_{t+1}^T that minimizes next period's expected operational loss:

$$f_{t+1}^T \in \arg \min_f \mathbb{E}_t[\mathcal{L}_{t+1}(\psi; f)]. \quad (26)$$

The initial target f_0^T is given. Under $\psi = 0$, the central bank operates under pure discretion. For $\psi > 0$, the inherited target operates as a soft commitment: it constrains the current rate choice but does not enter household utility.

The forward equity-price solution satisfies the no-bubble condition:

$$\lim_{j \rightarrow \infty} \omega_p^j \mathbb{E}_t p_{t+j} = 0. \quad (27)$$

The steady-state restriction $\bar{\lambda}_p < 1 - \beta$ ensures $0 < \omega_p < 1$.

Definition 1 (ψ -equilibrium). *Given the shock processes, $\psi \geq 0$, and the initial target f_0^T , a ψ -equilibrium is a collection $\{Y_t, C_t, P_t, q_{B,t}, r_t, f_t, f_t^T, \lambda_t^S, r_t^{\text{pec}}, S_t\}_{t \geq 0}$ and regime-level objects $\{m, x\}$ such that:*

- (i) *households optimize consumption, directly hold and price Treasuries according to (10), and satisfy the budget constraint (7), with $q_{B,t}$ and r_t^{pec} satisfying (6) and (11);*
- (ii) *portfolio managers optimize and the equity market clears, such that p_t satisfies (15) and (27);*
- (iii) *$m_t = m$ at every date, where the scalar m satisfies the total-return tail condition (16), which reduces to (18) under lognormal returns;*
- (iv) *equity and Treasuries supply collateral services with intensities m and one, and the service market clears according to (9) and (10);*
- (v) *optimizing-household consumption, hand-to-mouth spending, autonomous expenditure, and goods-market clearing satisfy (20)–(21);*
- (vi) *the observed Treasury rate and the sequence of FCI targets solve the timing and policy problems in (25) and (26); and*
- (vii) *$x = \text{Var}_t(r_{p,t+1}^x)$ under the resulting equilibrium law of motion.*

Within a fixed ψ -regime, the scalar objects $m(\psi)$, $x(\psi)$, and $\bar{\mu}^x(\psi)$ —along with balanced-growth prices and reduced-form coefficients—are constant. State variables and shocks fluctuate

around this regime-level steady state. Across values of ψ , we jointly solve these regime-level objects. We characterize a differentiable, locally stable linear solution in stationary gaps, returns, and risk measures, but do not establish global existence or uniqueness.

This environment nests two benchmarks. Taking the limit $\xi \rightarrow 0$ and $b \rightarrow 0$ shuts down the convenience and fiscal channels, recovering the financial-conditions model of [Caballero et al. \(2024\)](#). Alternatively, setting $m_t = 0$ eliminates equity's capacity to back safe claims; in that limit, changes in equity risk leave the supply of collateral services unchanged.

4 Financial Conditions Targeting and Service-Wedge Erosion

This section traces the effects of stronger FCI targeting through the closed-economy equilibrium. We first show how policy changes the response of financial conditions to macroeconomic news and identify when this reduces equity-return risk. We then characterize when lower risk raises pledgeability and decreases the Treasury convenience yield.

4.1 Policy Transmission

FCI targeting shifts how the central bank adjusts financial conditions in response to macroeconomic news. Before financial noise is realized, the bank chooses the policy rate by trading off the expected output gap against the deviation of expected financial conditions, $f_t^d \equiv \mathbb{E}_t f_t$, from the preannounced target. Financial noise subsequently moves realized financial conditions.

Define the targeting weight $h_\psi \equiv \psi / (1 + \psi)$. Solving the central bank's sequential rate and target choices yields

$$\begin{aligned} f_{t+1}^T &= \mathbb{E}_t f_{t+1}^*, \\ f_t^T &= \mathbb{E}_{t-1} f_t^* = \mathbb{E}_{t-1} f_t^*, \end{aligned} \tag{28}$$

$$f_t^d \equiv \mathbb{E}_t f_t = f_t^* - h_\psi (\varepsilon_{z,t} - \varepsilon_{\delta,t}). \tag{29}$$

The announced target is the forecast of next period's gap-closing FCI. The implemented conditional index, $f_t^d = (1 - h_\psi) f_t^* + h_\psi f_t^T$, is a weighted average of today's gap-closing FCI and yesterday's announcement. Stronger targeting places greater weight on the preannounced target and less weight on current macroeconomic news. Under pure discretion ($\psi = 0$), the bank adjusts financial conditions to close the current gap.

4.2 Return-Risk Reduction

Stronger targeting changes how macroeconomic news passes through to equity returns. Because the bank adjusts financial conditions less in response to current news, a fraction h_ψ of that news

remains in the output gap:

$$\tilde{y}_t = -h_\psi(\varepsilon_{z,t} - \varepsilon_{\delta,t}) + \mathcal{I}_f(x)\varepsilon_{\mu,t}, \quad \mathcal{I}_f(x) > 0. \quad (30)$$

Under discretion ($h_\psi = 0$), the bank closes the gap following observed macroeconomic news. Stronger targeting leaves more of this news in the gap as the policy rate implements the preannounced FCI target.

This change in the policy response alters equity prices and therefore the equity return's exposure to the underlying shocks. Substituting the induced price response into the total return yields the equilibrium variance map:

$$x = \Phi(x; \psi) = [\gamma_z(x) - \gamma_h(x)h_\psi]^2 \sigma_z^2 + [\gamma_\delta(x) + \gamma_h(x)h_\psi]^2 \sigma_\delta^2 + \mathcal{K}(x)x^2. \quad (31)$$

Here γ_z and γ_δ are the discretionary return loadings on supply and demand news, while $\gamma_h = \omega_D + \omega_P/(1+b) > 0$ is the exposure that targeting shifts from supply toward demand news. The last term is risk generated by financial noise. The return loadings and $\mathcal{K}(x)$ vary with x because pledgeability, steady-state prices, and service provision are solved jointly. Appendix B.4 derives these coefficients and the full price and policy-rate solution.

Define the targeting threshold

$$h_M(x) \equiv \frac{\gamma_z(x)\sigma_z^2 - \gamma_\delta(x)\sigma_\delta^2}{\gamma_h(x)(\sigma_z^2 + \sigma_\delta^2)}. \quad (32)$$

Proposition 1 (Return-risk reduction under FCI targeting). *If*

$$0 \leq h_\psi < h_M(x) \quad \text{and} \quad 1 - \Phi_x(x; \psi) > 0,$$

then

$$\frac{dx}{d\psi} = \frac{\Phi_\psi}{1 - \Phi_x} < 0. \quad (33)$$

The first condition ensures that shifting return exposure from supply news toward demand news directly lowers risk. The second ensures that equilibrium feedback does not overturn that decline. When $0 \leq \Phi_x < 1$, lower risk reduces the price concession required to absorb financial noise, recruiting arbitrageurs and amplifying the decline (Caballero et al., 2024). Appendix B.4 gives the proof and the auxiliary fixed-coefficient benchmark.

To compare policy regimes ex ante, let $G^{\text{mand}}(\psi) \equiv \mathbb{E}[G_t^{\text{mand}}(\psi)]$ denote the unconditional mandate loss under targeting intensity ψ . Factor financial-noise pass-through as $\mathcal{I}_f(x) = g(x)x/\alpha_p$, where x/α_p is the partial-equilibrium price impact of financial noise and $g(x) > 0$ captures its general-equilibrium transmission to the output gap. Define the elasticity of this

transmission multiplier as

$$\epsilon_g(x) \equiv \frac{d \log g(x)}{d \log x}. \quad (34)$$

Proposition 2 (Mandate gain near discretion). *Consider a discretionary equilibrium satisfying the conditions of Proposition 1, with $\sigma_\mu^2 > 0$. If*

$$\epsilon_g(x(0)) > -1, \quad (35)$$

then

$$\left. \frac{dG^{\text{mand}}}{d\psi} \right|_{\psi=0} < 0.$$

Thus, a small departure from discretion strictly lowers the expected output-gap loss.

Near discretion, targeting generates a first-order stabilization benefit that dominates its second-order macroeconomic cost. Because the central bank perfectly closes the output gap against macroeconomic news under pure discretion, allowing a small gap to emerge under soft commitment increases the mandate loss only at second order. Simultaneously, this commitment structurally lowers equity-return risk. Lower risk increases the elasticity of portfolio managers and directly compresses the partial-equilibrium price impact of financial noise, x/α_p . Condition (35) ensures that general-equilibrium feedback through goods and service markets does not overturn this reduction. Financial noise therefore passes through less strongly to the output gap, reducing the mandate loss at first order.

4.3 Convenience-Yield Erosion

In the lognormal benchmark, the direct effect of lower risk raises the lower-tail equity return whenever $x < z_p^2$. This subsection shows that the tail improvement raises pledgeability and decreases the Treasury convenience yield in general equilibrium.

Consider a change in ψ across policy regimes. Mean output is held fixed, and convenience-stripped safe rate remains $\bar{r}^{\text{pec}} = \rho$. The relevant service capacity is usable equity collateral value, $X \equiv mP$. Let $\Gamma > 0$ denote the general-equilibrium inverse-demand elasticity of the Treasury service wedge with respect to X .

Proposition 3 (Collateral expansion and Treasury convenience-yield erosion). *Consider a differentiable comparison across policy regimes holding mean output and $\bar{r}^{\text{pec}} = \rho$ fixed. Suppose $\xi > 0$, the standardized return distribution is fixed across the comparison, pledgeability is interior, $0 < m < 1$, and*

$$x < z_p^2, \quad \frac{dx}{d\psi} < 0$$

as in Proposition 1. Then

$$\begin{aligned}\frac{d \log m}{d\psi} &> 0, & \frac{d \log X}{d\psi} &= \frac{1 + \Xi}{1 + \Gamma \Xi} \frac{d \log m}{d\psi} > 0, \\ \frac{d \log \lambda_T^S}{d\psi} &= -\frac{\Gamma(1 + \Xi)}{1 + \Gamma \Xi} \frac{d \log m}{d\psi} < 0.\end{aligned}\tag{36}$$

The equity-price response is

$$\frac{d \log P}{d\psi} = \frac{\Xi(1 - \Gamma)}{1 + \Gamma \Xi} \frac{d \log m}{d\psi},\tag{37}$$

so the equity price rises if and only if $\Gamma < 1$. Moreover,

$$\frac{d}{d\psi} \log \left(\frac{1 - \lambda_P}{1 - \lambda_T^S} \right) < 0,\tag{38}$$

where $\lambda_P = m\lambda_T^S$.

A stronger financial-conditions target lowers the Treasury convenience yield through three steps. First, lower equilibrium return risk improves equity's collateral tail and raises the pledgeable share m . Second, the increase in m expands usable equity collateral capacity, $X = mP$, and Proposition 3 shows that the increase in m dominates so X increases. Third, the increase in supply of services lowers their marginal value λ_T^S . Equity becomes a closer substitute for Treasuries, and the Treasury–equity service-valuation gap narrows. Because $\bar{r} = \rho + \log(1 - \lambda_T^S)$, the Treasury rate rises toward the convenience-stripped rate ρ . The proof is available in Appendix B.5.

4.4 Collateral Activation

The preceding results evaluate an equilibrium in which equity collateral is active. The capacity condition (16) also allows a corner solution where equity remains nonpledgeable:

$$m = 0 \iff Q_{\bar{p}}^x \leq \log \kappa_C.$$

This lower corner separates policy-induced risk reduction from convenience-yield erosion: targeting can lower equity-return risk even when equity cannot supply collateral services.

Proposition 4 (Dormancy and collateral activation). *Consider the stationary policy comparison used in Proposition 3, with $\xi > 0$ and initially $m = 0$.*

(i) If

$$Q_{\bar{p}}^x < \log \kappa_C,$$

then equity remains nonpledgeable under a marginal increase in targeting. Usable equity collateral remains zero, and the Treasury convenience yield is locally unchanged.

(ii) Suppose instead that

$$Q_p^x = \log \kappa_C.$$

If the standardized return distribution is fixed, $x < z_p^2$, and $dx/d\psi < 0$, then a one-sided increase in targeting raises the lower-tail return above the collateral threshold. Equity becomes pledgeable, usable collateral capacity expands from zero, and the Treasury convenience yield strictly falls.

Corollary 1 (Necessity of usable equity collateral). *Under the stationary policy comparison in Proposition 3, suppose $\xi > 0$ and $m > 0$. Then*

$$\frac{d \log \lambda_T^S}{d\psi} = -\Gamma \frac{d \log(mP)}{d\psi}, \quad \Gamma > 0. \quad (39)$$

Hence the Treasury convenience yield falls if and only if usable equity collateral expands.

When equity cannot be pledged as collateral ($m = 0$), financial-conditions targeting lowers return risk but leaves the Treasury convenience yield unchanged. When stronger targeting moves the lower-tail return across the threshold in Part (ii), creditors' information-insensitivity constraint is satisfied, equity-backed collateral becomes active, and the Treasury convenience yield begins to fall.

Thus how monetary policy affects the Treasury service wedge depends on usable capacity mP . Endogenous pledgeability is a necessary link between the “Fed put” in equity markets and the erosion of the Treasury convenience yield.

5 Foreign Portfolios, Financial Equilibrium, and the Dollar

In this section, we embed the environment of Section 3 into an open economy with exchange rates. Foreign investors hold U.S. equity and Treasuries, delegating their dollar accounts to portfolio managers. These investors value U.S. claims for both pecuniary returns and dollar services, such as liquidity and vehicle-currency functions. As in the closed economy, a dollar of equity provides $m \in [0, 1)$ units of service to Foreigners. A higher pledgeability share m makes equity a closer substitute for Treasuries in Foreign portfolios.

For each targeting regime ψ , the equilibrium jointly determines regime-level return risk, equity pledgeability, Foreign dollar exposure, and claim ownership. These regime-level objects remain constant under a fixed policy rule. Within a regime, Home equity managers absorb equity-market noise, and Foreign currency specialists absorb customer currency-demand noise. Asset prices and the exchange rate adjust to clear these respective markets. We present the

key equilibrium conditions in the text and relegate the complete accounting conventions and derivations to Appendix C.1.

5.1 Open-Economy Financial Conditions

We add two new ingredients. The first is trade in goods: a dollar depreciation and stronger foreign demand increase U.S. net exports. The second is trade in assets: Home households hold N^H equity shares and B^H units of Treasury face value, Foreign investors hold the remaining shares $N^F = 1 - N^H$ and bonds $B^F = \bar{B} - B^H$, and Home holds external assets $A^{H,-}$. Ownership and the external position are determined jointly with asset prices and return risk at the regime level (Section 5.2) and remain constant within a regime. This subsection takes the regime-level allocation as given and derives the transmission of financial prices to aggregate demand.

Let e_t denote the log dollar price of the foreign consumption basket, where a higher e_t represents a dollar depreciation. Let y_t^g be the log deviation of foreign demand for U.S. goods. A weaker dollar and stronger foreign demand both increase U.S. net exports:

$$NX_t \equiv \mathcal{X}(\exp(e_t), Y_t^g) - \mathcal{M}(\exp(e_t)).$$

Following Caballero and Simsek (2026), Home consumption does not alter imports at the margin. We scale foreign demand y_t^g so that both y_t^g and e_t share the net-export sensitivity parameter $\phi > 0$. Log-linearizing goods-market clearing around the regime-level allocation gives

$$y_t = f_t^o + \phi y_t^g + \delta_t, \quad (40)$$

where δ_t is autonomous expenditure, normalized by $\Theta_H \nu \bar{Y}$ as defined in (41).

The open-economy financial conditions index (FCI), f_t^o , measures the degree to which asset prices, exchange rates, and wealth raise aggregate demand:

$$\begin{aligned} f_t^o &\equiv \phi e_t + g_p^o p_t - g_r^o \hat{r}_t - H^o \ell_t + g_a^o a_t^{H,-}, & \Theta_H &\equiv \beta + (1 - \beta)N^F, \\ g_p^o &\equiv \frac{(1 - \beta - \bar{\lambda}_p)\bar{P}N^H}{\Theta_H \nu \bar{Y}} > 0, & g_r^o &\equiv \frac{\beta B^H + \bar{q}_B B^F}{\Theta_H \nu \bar{Y}} > 0, \\ H^o &\equiv \frac{\bar{\lambda}_p \bar{P}N^H + \bar{\lambda}_T^S \bar{q}_B B^H}{\Theta_H \nu \bar{Y}} > 0, & g_a^o &\equiv \frac{(1 - \beta)(1 + \iota_A)\bar{A}^{H,-}}{\Theta_H \nu \bar{Y}} \geq 0. \end{aligned} \quad (41)$$

The regime-level external asset $\bar{A}^{H,-}$ pays yield $\iota_A \geq 0$; its log deviation, $a_t^{H,-}$, is evaluated at current prices before new trading occurs.

A higher f_t^o indicates looser financial conditions and raises output through five signed channels. A dollar depreciation ($e_t > 0$) increases net exports. A higher equity price ($p_t > 0$) raises Home equity wealth and increases consumption. Conversely, a higher policy rate ($\hat{r}_t > 0$)

reduces the rollover proceeds rebated to hand-to-mouth workers and raises the interest paid on Foreign-held Treasuries, lowering demand. A higher service wedge ($\ell_t > 0$) strengthens the household savings motive and lowers current spending. Finally, a larger external asset balance ($a_t^{H,-} > 0$) increases Home external wealth and interest income and raises consumption.

We evaluate the coefficients in (41) holding asset ownership fixed at its regime-level baseline. Home households consume out of dividends and wealth generated by their N^H equity shares. When Foreign investors acquire Home assets, the proceeds accumulate in Home's end-of-period external asset $A_t^{H,+}$ rather than funding current Home consumption. As such, we use the factor $\Theta_H \nu \bar{Y}$ to reflect dividends paid to Foreign investors. When $N^F = 0$, we have $\Theta_H = \beta$, and every coefficient reduces to its closed-economy counterpart in (23).³

Let $f_t^{0,*}$ denote the gap-closing open-economy FCI, defined as the value of f_t^0 that sets $y_t = y_t^*$. Imposing this condition in (40) and subtracting yields the output gap:

$$\tilde{y}_t \equiv y_t - y_t^* = f_t^0 - f_t^{0,*}, \quad f_t^{0,*} \equiv y_t^* - \phi y_t^g - \delta_t. \quad (42)$$

Stronger foreign goods demand y_t^g decreases the gap-closing index $f_t^{0,*}$ because higher net exports absorb productive capacity, requiring tighter domestic financial conditions to prevent overheating.

The composition of Foreign asset demand determines Home's exposure to the service wedge, H^0 . An increase in Foreign holdings of both U.S. assets decreases H^0 because Home households retain fewer total claims, reducing their exposure to service-wedge fluctuations. Conversely, a Foreign portfolio rotation from Treasuries into equity can increase H^0 . Because one dollar of equity provides only $m < 1$ service units, the rotation leaves a larger share of the economy's aggregate service provision with Home households, though the accompanying rise in Foreign equity ownership partially offsets this through Θ_H .

5.2 Foreign Demand, Currency Intermediation, and Ownership

Next, we characterize Foreign demand for U.S. assets and the resulting regime-level ownership. We model Foreign asset demand in two stages: investors first allocate wealth to a U.S. dollar account, and portfolio managers then divide that account between U.S. equity and Treasuries. Fix a targeting regime ψ and suppress the ψ subscript on regime-level objects unless explicitly comparing across regimes.

Let $\mathcal{A}^F > 0$ denote the pool of eligible Foreign wealth and $A^{F,0}$ the amount allocated to dollar assets. The net benefit to a Foreign investor of holding the dollar account rather than a foreign safe asset is $\mathcal{Q}^F$. A higher expected return or greater collateral service provision increases $\mathcal{Q}^F$,

³Appendix C.1 derives these coefficients under the convention of Section 3.1 in which the Treasury rollover flow accrues to hand-to-mouth workers.

while higher equity-return risk or exchange-rate risk decreases it. Because every policy regime is stationary, expected dollar depreciation is zero (Appendix C.3). Investors evaluate Q^F at the equity–Treasury portfolio their account manager chooses below.

The fraction of Foreign wealth allocated to the dollar account follows a logistic function:

$$A^{F,0} = \mathcal{A}^F \frac{\exp\{\chi_A + \kappa_A Q^F\}}{1 + \exp\{\chi_A + \kappa_A Q^F\}}.$$

The parameter χ_A captures baseline dollar share, while $\kappa_A \geq 0$ governs its sensitivity to Q^F .

Portfolio managers then choose the equity share ω^F of the dollar account, placing the remainder in Treasuries. The resulting allocation is:

$$\begin{aligned} \omega^F &= W(\mu^F, x_p, c_{pe}, m), \\ E^F &= \omega^F A^{F,0}, \quad T^F = (1 - \omega^F) A^{F,0}. \end{aligned} \tag{43}$$

Here μ^F is the expected equity excess return, x_p is the equity-return variance, and $c_{pe} \equiv \text{Cov}(r_{p,t+1}^x, e_{t+1} - e_t)$ is the covariance between the equity return and dollar depreciation. A higher expected excess return (μ^F) or greater pledgeability (m) increases the equity share. Conversely, higher equity-return variance (x_p) decreases the equity share. A higher covariance (c_{pe}) increases the equity share because dollar-denominated equity returns offset the loss in purchasing power from a dollar depreciation. Because a dollar of Treasury provides one service unit and a dollar of equity provides m units, services per account dollar are $\vartheta^F \equiv 1 - (1 - m)\omega^F$.

At the regime-level steady state, asset prices convert these dollar demands into claim quantities:

$$\bar{P}N^F = E^F, \quad \bar{q}_B B^F = T^F, \quad N^H + N^F = 1, \quad B^H + B^F = \bar{B}. \tag{44}$$

Home's portfolio managers hold the residual equity supply, N^H . Their optimality condition determines the expected excess return:

$$\mu^F = \frac{x_p}{2} + (1 - m)\lambda_T^S - \frac{x_p}{\alpha_p} N^F. \tag{45}$$

The final term captures structural price pressure: greater Foreign ownership reduces the risk-bearing position of Home managers, which compresses the required traded return. Because Foreign claim quantities (N^F, B^F) remain fixed within a regime, this structural price pressure shifts the regime-level price intercept but preserves the short-run response of prices to financial noise, x_p/α_p . Fluctuating prices alter the marked value of Foreign portfolios but do not change the number of claims beneficially owned.

Let e_t be the log dollar price of the Foreign consumption basket, so an increase in e_t is a dollar depreciation. The Foreign dollar allocation $A^{F,0}$ creates demand for dollar exposure and remains

fixed within a policy regime. Let $Z^e(\bar{e})$, with $Z_e \equiv \partial Z^e / \partial \bar{e} < 0$, be the reduced-form component of Foreign customers' net short-dollar exposure. After accounting for their actual dollar asset allocation and the disturbance μ_t^e , the exposure passed to Foreign currency specialists is

$$S_t^e \equiv Z^e(\bar{e}) + \mu_t^e - A^{F,0}.$$

The endogenous specialist position is $D_t^e = S_t^e$: a positive position is long dollar risk, not ownership of an additional dollar claim. Specialist optimality and currency-risk clearing yield the risk-adjusted uncovered interest parity (UIP) equation:

$$e_t = \mathbb{E}_t e_{t+1} - (r_t - \bar{r}^*) - \frac{x_e}{2} + \frac{x_e}{\alpha_e} \left[Z^e(\bar{e}) + \mu_t^e - A^{F,0} \right]. \quad (46)$$

Here x_e is regime-level exchange-rate variance, α_e is specialist risk-bearing capacity, and μ_t^e is an exogenous disturbance to Foreign customer currency exposure. Following [Caballero and Simsek \(2026\)](#), greater structural Foreign demand shifts the regime-level exchange-rate intercept and exerts appreciation pressure on the dollar; within the regime, specialists absorb high-frequency currency noise.

Appendix [C.1](#) gives the specialist problem and records the opposite Foreign customer position. The two sides have zero initial value and their settlement transfers cancel within Foreign. Thus D_t^e affects asset ownership only indirectly through the equilibrium exchange rate, Foreign participation, and portfolio choice; it is absent from dollar services and the U.S. external balance sheet.

Because collateral rights travel strictly with beneficial ownership, claim ownership determines the allocation of collateral services:

$$S_t^F = q_{B,t} B^F + m P_t N^F, S_t^H = q_{B,t} B^H + m P_t N^H = q_{B,t} \bar{B} + m P_t - S_t^F. \quad (47)$$

5.3 Policy and Joint Risk

We now characterize how financial-conditions targeting operates in the open economy and how this policy jointly determines equity and exchange-rate risk. The central bank retains its domestic output-gap mandate but applies the operational protocol of [Section 4.1](#) to the open-economy index [\(41\)](#). Let $\varepsilon_t^* \equiv f_t^{o,*} - \mathbb{E}_{t-1} f_t^{o,*}$ denote news about the gap-closing FCI. The announced target and the implemented conditional FCI are:

$$f_t^T = \mathbb{E}_{t-1} f_t^{o,*}, \quad \mathbb{E}_t f_t^o = f_t^{o,*} - h_\psi \varepsilon_t^*, \quad h_\psi \equiv \frac{\psi}{1 + \psi}. \quad (48)$$

Equity and currency noise arrive after the central bank sets the current policy rate.

An open-economy equilibrium augments the domestic environment with Foreign portfolio

allocation, currency-market clearing, and the external-wealth state. For each targeting intensity ψ , the equilibrium jointly determines asset prices, ownership shares, dollar allocations, equity pledgeability, and return risk. Within a fixed regime, ownership quantities remain predetermined while fluctuating prices alter their marked values. Appendix C.1 formally defines this equilibrium and the complete market-clearing system.

For analytical tractability, we analyze the symmetric special case adapted from Caballero and Simsek (2026). Under Assumption (S^0) (Appendix C.3), the equity price and exchange rate carry equal weights in the FCI, the two specialist sectors share identical risk-bearing capacity α , and the service-wedge and external-wealth feedbacks enter the risk block solely through regime-level coefficients. Ownership, pledgeability, prices, and service incidence remain jointly determined by their asymmetric equations. Appendix C.2 provides the general asymmetric system.

Let z_t^p and z_t^e denote the innovations to the macroeconomic fundamentals driving the equity and currency markets, respectively. Let $\varepsilon_{\mu^p,t}$ and $\varepsilon_{\mu^e,t}$ denote the financial-noise innovations. Symmetry imposes equal fundamental and noise variances across the two markets:

$$\text{Var}(z_t^p) = \text{Var}(z_t^e) \equiv \sigma_z^2, \quad \text{Var}(\varepsilon_{\mu^p,t}) = \text{Var}(\varepsilon_{\mu^e,t}) \equiv \sigma_\mu^2,$$

with correlations $\rho_z, \rho_\mu \in (-1, 1)$ within the two pairs. Gap-closing news is shared equally, $\varepsilon_t^* = (z_t^p + z_t^e)/2$, and the policy offset passes to each price through the common Jacobian $\gamma_h \equiv 1/(1 + g_r^o)$. Define the mixing coefficient $a_\psi \equiv \gamma_h h_\psi / 2 \in [0, \frac{1}{2})$. Appendix C.3 derives the equity-return and exchange-rate innovations:

$$\begin{aligned} u_{p,t} &= (1 - a_\psi)z_t^p - a_\psi z_t^e + \frac{x}{\alpha} \varepsilon_{\mu^p,t}, \\ u_{e,t} &= (1 - a_\psi)z_t^e - a_\psi z_t^p + \frac{x}{\alpha} \varepsilon_{\mu^e,t}. \end{aligned} \quad (49)$$

Defending the preannounced target against news in one market requires adjusting the policy rate, which simultaneously moves prices in the other market. Policy thus mixes macroeconomic news across the two specialist sectors. Financial noise is not mixed: because it arrives after the policy rate is set and the target resets each period, each noise innovation enters only its own market with price impact x/α . Equity-return and exchange-rate variance coincide ($x_p = x_e \equiv x$), reducing risk consistency to the scalar fixed point:

$$x = v_z(\psi) + \sigma_\mu^2 \left(\frac{x}{\alpha} \right)^2, \quad (50)$$

where

$$v_z(\psi) \equiv \sigma_z^2 \left[(1 - a_\psi)^2 + a_\psi^2 - 2\rho_z a_\psi (1 - a_\psi) \right]. \quad (51)$$

Proposition 5 (Risk reduction under symmetric FCI targeting). *Suppose Assumption (S^0) holds*

and (50) has a stable smaller solution, such that:

$$1 - 2\sigma_\mu^2 \frac{x}{\alpha^2} > 0.$$

Then stronger FCI targeting lowers both equity-return and exchange-rate risk:

$$\frac{dx}{d\psi} = \frac{v'_z(\psi)}{1 - 2\sigma_\mu^2 x / \alpha^2} < 0. \quad (52)$$

In particular, a marginal departure from discretion lowers risk in both markets.

Targeting spreads each market's macroeconomic news across both specialist sectors. Because macroeconomic shocks are not perfectly negatively correlated ($\rho_z > -1$), the reduction in each sector's own-shock exposure outweighs the exposure imported from the other market, lowering the macroeconomic component of variance. The resulting decline in total variance raises the demand elasticity of equity managers and currency specialists, α/x . Consequently, these specialists absorb financial noise with smaller required price concessions. This endogenous increase in market liquidity shrinks the noise component $\sigma_\mu^2(x/\alpha)^2$ and further reduces total variance through the recruitment mechanism.

While stronger targeting unambiguously reduces return variance, its effect on the equity-exchange-rate covariance c_{pe} is ambiguous. Policy mixing lowers the fundamental component of the covariance by the same diversification factor that lowers v_z , while the decline in variance shrinks the noise component $\rho_\mu \sigma_\mu^2(x/\alpha)^2$ toward zero—which raises c_{pe} when noise innovations are negatively correlated. We therefore retain c_{pe} as a separate regime-level object in the Foreign demand system (43) to trace how policy-induced covariance shifts alter Foreign dollar participation.

5.4 Foreign Demand and Convenience-Yield Erosion

This section characterizes how policy-induced risk reduction transmits through Foreign asset demand to erode the Treasury convenience yield. The transmission depends on two endogenous margins: the total supply of usable equity collateral and the share of those collateral services absorbed by Foreign investors. We first derive how lower risk alters equity pledgeability, and then compare total usable capacity against Foreign service absorption to determine the Treasury service wedge.

First, we decompose Foreign demand. The net account value Q^F determines total Foreign dollar exposure $A^{F,0}$. Given this exposure, the portfolio manager chooses the equity share ω^F .

Total dollar equity demand (E^F) and Foreign ownership (N^F) satisfy:

$$\begin{aligned}\frac{d \log E^F}{d\psi} &= \frac{d \log A^{F,0}}{d\psi} + \frac{d \log \omega^F}{d\psi}, \\ \frac{d \log N^F}{d\psi} &= \frac{d \log E^F}{d\psi} - \frac{d \log \bar{P}}{d\psi}.\end{aligned}\tag{53}$$

Holding the expected excess return, pledgeability, and the equity–exchange-rate covariance fixed, a lower return variance x directly increases the Foreign equity share ω^F . General-equilibrium adjustments in these three objects can either reinforce or offset the direct portfolio shift. Ultimately, Foreign ownership N^F rises if and only if total dollar equity demand grows faster than the equilibrium equity price. This ownership response determines the share of collateral services retained by Home households and simultaneously feeds back into the total pledgeable capacity of equity.

Lemma 1 (Open-economy pledgeability). *Assume an active, nonsaturated regime steady state ($0 < m < 1$) with a fixed standardized return distribution. The local response of pledgeability to targeting intensity ψ satisfies:*

$$\begin{aligned}\left[1 + \lambda_T^S(m + \kappa_C)\right] \frac{dm}{d\psi} &= (m + \kappa_C) \left[\left(\frac{1}{2} - \frac{N^F}{\alpha} - \frac{z_{\bar{p}}}{2\sqrt{x}} \right) \frac{dx}{d\psi} \right. \\ &\quad \left. + (1 - m) \frac{d\lambda_T^S}{d\psi} - \frac{x}{\alpha} \frac{dN^F}{d\psi} \right].\end{aligned}\tag{54}$$

Lemma 1 decomposes the general-equilibrium response of pledgeability into three mechanisms, corresponding to the three bracketed terms in (54). First, when $x < z_{\bar{p}}^2$, the policy-induced risk reduction ($dx/d\psi < 0$ from Proposition 5) improves the downside equity payoff that creditors underwrite, directly expanding collateral capacity. Second, an erosion of the Treasury service wedge ($d\lambda_T^S/d\psi < 0$) raises the domestic safe interest rate. This increases the cost of collateralized debt relative to the equity payoff, which reduces the pledgeable share. Third, an increase in Foreign ownership ($dN^F/d\psi > 0$) reduces the risk-bearing position of Home managers, which compresses the required expected equity return and thereby lowers the expected future payoff per dollar of current equity value. Pledgeability rises if and only if the direct risk-reduction benefit dominates the funding-wedge and Foreign-demand feedbacks.

Because the collateralized credit line is denominated in dollars, the exchange rate does not enter the collateral payoff directly. Instead, exchange-rate dynamics alter pledgeability only indirectly through the general-equilibrium responses of return risk x , Foreign ownership N^F , and the Treasury service wedge λ_T^S . The equity–currency covariance c_{pe} operates solely through Foreign portfolio demand.

Let X denote regime-level usable equity collateral capacity and F denote total Foreign service

absorption:

$$X \equiv m\bar{P}, \quad F \equiv \bar{q}_B B^F + m\bar{P}N^F = \vartheta^F A^{F,0}. \quad (55)$$

Home households retain the residual service supply: $S^H = \bar{q}_B \bar{B} + X - F$. Let $dc_\perp^H$ denote the shift in Home consumption demand arising from changes in equity prices, pledgeability, asset ownership, external wealth, and dividend income, holding the service wedge fixed. The Home inverse service-demand curve implies:

$$d \log \lambda_T^S = -\Gamma_X d \log X + \Gamma_F d \log F + \Gamma_C dc_\perp^H, \quad \Gamma_X, \Gamma_F > 0, \quad \Gamma_C > 0, \quad \frac{\Gamma_F}{\Gamma_X} = \frac{F}{X}. \quad (56)$$

An expansion in total usable capacity X increases aggregate service supply and decreases the Home marginal service wedge. Conversely, an increase in Foreign service absorption F drains services from Home households, increasing the Home service wedge. A positive shift in net consumption demand $dc_\perp^H$ raises the marginal utility of services relative to consumption, tightening the Home service wedge. Appendix C.4 derives these coefficients.

Proposition 6 (Foreign service absorption). *At an interior regime steady state, Foreign service absorption and the capacity retained by Home satisfy:*

$$\frac{dF}{d\psi} = \vartheta^F \frac{dA^{F,0}}{d\psi} + A^{F,0} \omega^F \frac{dm}{d\psi} - A^{F,0} (1 - m) \frac{d\omega^F}{d\psi}, \quad (57)$$

$$\frac{dX}{d\psi} - \frac{dF}{d\psi} = \bar{P} (1 - N^F) \frac{dm}{d\psi} + m \frac{d\bar{P}}{d\psi} + A^{F,0} (1 - m) \frac{d\omega^F}{d\psi} - \vartheta^F \frac{dA^{F,0}}{d\psi}. \quad (58)$$

Proposition 6 decomposes the response of Foreign service absorption into three distinct economic margins, corresponding to the terms in (57). First, an expansion in total Foreign dollar exposure ($dA^{F,0} > 0$) directly scales up Foreign service absorption. By increasing Foreign demand for U.S. assets, this expansion can compress the required equity return, reduce pledgeability per equity dollar, and transfer collateral services to Foreign beneficial owners. Second, holding the portfolio fixed, higher pledgeability ($dm > 0$) increases the service units provided by the existing Foreign equity position. Third, holding total dollar exposure fixed, a portfolio rotation from Treasuries into equity ($d\omega^F > 0$) strictly reduces total Foreign service absorption. Because each acquired equity dollar provides only a fraction $m < 1$ of the service unit surrendered by selling a Treasury dollar, this portfolio rotation increases Foreign equity services at the direct expense of total service absorption. Foreign equity ownership and total Foreign service absorption are therefore distinct margins that need not move in tandem. Appendix C.4 provides the exact derivatives for $A^{F,0}$ and ω^F .

The regime-level equity price capitalizes future collateral services and responds to Foreign order flow. Because usable capacity is the product of pledgeability and price ($X = m\bar{P}$), an increase in pledgeability expands usable capacity as long as it dominates any offsetting decline

in the equity price. Appendix C.5 derives the joint price and capacity responses.

Corollary 2 (Open service-market incidence). *At a differentiable interior equilibrium, the Treasury service wedge, and hence the Treasury convenience yield, falls if and only if*

$$\Gamma_X \frac{d \log X}{d\psi} > \Gamma_F \frac{d \log F}{d\psi} + \Gamma_C \frac{dc_{\perp}^H}{d\psi}. \quad (59)$$

When Home consumption demand is unchanged, $dc_{\perp}^H/d\psi = 0$, and the condition reduces to

$$\frac{dX}{d\psi} > \frac{dF}{d\psi}.$$

When Home consumption demand is unchanged, usable equity capacity grows faster than Foreign service absorption if and only if the Treasury service wedge falls. More generally, an increase in net consumption demand ($dc_{\perp}^H > 0$) raises the marginal utility of services relative to consumption, which tightens the Home service wedge and raises the threshold required for convenience-yield erosion.

Proposition 7 (Dollar and Foreign participation responses). *At a differentiable interior regime equilibrium, expected depreciation is zero within every regime (Lemma C.1), and regime-level currency-market clearing implies*

$$Z_e \frac{d\bar{e}}{d\psi} = \frac{dA^{F,0}}{d\psi} + \frac{\alpha}{x} \frac{d\bar{r}}{d\psi} - \frac{\alpha(\bar{r} - \bar{r}^*)}{x^2} \frac{dx}{d\psi}. \quad (60)$$

Because $Z_e < 0$, the dollar appreciates if and only if the right-hand side of (60) is positive. Foreign participation satisfies

$$\frac{d \log A^{F,0}}{d\psi} = \kappa_A \left(1 - \frac{A^{F,0}}{\mathcal{A}^F} \right) \frac{d\mathcal{Q}^F}{d\psi}, \quad \mathcal{Q}^F = \mathcal{V}^F. \quad (61)$$

At an interior portfolio, the optimized noncurrency account value obeys

$$\begin{aligned} \frac{d\mathcal{V}^F}{d\psi} &= \frac{d\bar{r}}{d\psi} + \omega^F \frac{d\mu^F}{d\psi} - \frac{\alpha_F}{2} \left[1 + (\omega^F)^2 \right] \frac{dx}{d\psi} \\ &\quad + \alpha_F \omega^F \frac{dc_{pe}}{d\psi} + \zeta_F v'_F(\vartheta^F) \omega^F \frac{dm}{d\psi}. \end{aligned} \quad (62)$$

Equation (60) governs the regime-level exchange rate through three distinct economic channels. First, an expansion of Foreign dollar exposure ($dA^{F,0} > 0$) exerts direct appreciation pressure on the dollar. Second, a higher domestic policy rate ($d\bar{r} > 0$) appreciates the dollar by widening the interest differential against the foreign safe asset. Third, when the domestic interest rate exceeds the foreign rate ($\bar{r} > \bar{r}^*$), a reduction in exchange-rate risk ($dx < 0$) makes the positive differential more attractive to currency specialists, further appreciating the

dollar. Because each policy regime is stationary, within-regime expected depreciation is zero; policy shifts the steady-state exchange rate $\bar{e}$ across regimes without inducing transitional drift (Appendix C.3).

Foreign dollar participation responds to the optimized noncurrency account value $\mathcal{V}^F$. Equation (62) shows that a reduction in common return variance ($dx < 0$) or an increase in equity pledgeability ($dm > 0$) directly increases this account value. General equilibrium shifts in the expected excess return and the equity–currency covariance can either amplify or dampen this increase. If the participation elasticity κ_A is strictly positive, an overall increase in account value expands Foreign dollar exposure; if $\kappa_A = 0$, exposure remains fixed across policy regimes.

5.5 External Accounts and Valuation

We now characterize how policy-induced asset appreciation and portfolio flows alter the U.S. net foreign asset (NFA) position. Following Atkeson et al. (2025), U.S. net foreign assets immediately after date- t transactions are:

$$\text{NFA}_t \equiv A_t^{H,+} - P_t N_t^F - q_{B,t} B_t^F. \quad (63)$$

Only beneficially owned claims enter this position. The currency-derivative position D_t^e is excluded because its cash flows settle entirely within the Foreign sector.

The stock–flow identity links this position to the current account, $\text{CA}_t = \text{NX}_t + \text{NPI}_t$, and the ex-cash revaluation of inherited positions, VAL_t :

$$\text{NFA}_t - \text{NFA}_{t-1} = \text{CA}_t + \text{VAL}_t. \quad (64)$$

For foreign-owned equity, price fluctuations and quantity flows separate exactly at the period boundary:

$$P_t N_t^F - P_{t-1} N_{t-1}^F = N_{t-1}^F (P_t - P_{t-1}) + P_t (N_t^F - N_{t-1}^F). \quad (65)$$

The first term isolates valuation changes on inherited shares. The second term captures financial-account flows at current prices. Appendix C.7 provides the complete income and transaction accounting.

Let d^{acq} denote the acquisition component of a policy-regime comparison, evaluated at fixed prices and current-account flows. To satisfy balance-of-payments clearing, Home’s post-transaction external asset $A^{H,+}$ absorbs the proceeds from Foreign acquisitions of U.S. claims:

$$d^{\text{acq}} A^{H,+} = \bar{P} d^{\text{acq}} N^F + \bar{q}_B d^{\text{acq}} B^F. \quad (66)$$

Corollary 3 (External position and valuation exposure). *The acquisition component in (66) preserves*

net foreign assets:

$$d^{\text{acq}}\text{NFA}_t = 0.$$

The contribution of equity-price appreciation to U.S. valuation is determined by inherited Foreign ownership:

$$d\text{VAL}_{P,t} = -N_{t-1}^F dP_t. \quad (67)$$

Holding inherited ownership fixed, this equity-liability valuation contribution obeys across policy regimes:

$$\frac{d\text{VAL}_{P,t}}{d\psi} = -\bar{P}_\psi N_{t-1}^F \frac{d \log \bar{P}_\psi}{d\psi}. \quad (68)$$

Corollary 3 establishes that cross-border financial transactions swap equal-value claims, leaving the net external position unchanged on impact. Because Foreign-held U.S. equity is an external liability, an increase in the steady-state equity price ($\bar{P}_\psi$) generates a strictly negative valuation contribution to U.S. net foreign assets whenever $N_{t-1}^F > 0$. Whether the appreciation reflects greater service capitalization or an outward shift in Foreign demand does not alter this accounting result. New Foreign purchases enter the financial account at the current price and do not alter the valuation of inherited positions.

An erosion of the Treasury service wedge creates a distinct income effect on the external balance. Because the observed Treasury rate satisfies $r = \rho + \log(1 - \lambda_T^S)$, a lower service wedge raises the quoted rate and lowers the issue price $q_{B,t} = e^{-r_t}$. Holding the rolled-over Foreign-held face value B_t^F fixed, the interest paid at $t + 1$ is $I_{T,t+1}^F = (1 - q_{B,t})B_t^F$. The policy-induced change is:

$$dI_{T,t+1}^F = q_{B,t} B_t^F dr_t > 0. \quad (69)$$

This higher interest payment strictly lowers U.S. net investment income when the newly issued debt matures, whereas inherited debt continues to pay its previously contracted return.

The return differential is defined exclusively over beneficially owned external claims. Let r_A^{dol} denote the expected dollar return on Home's foreign asset, incorporating both cash income and valuation changes. Because ω^F is the equity share of the Foreign dollar account and μ^F is the expected equity excess return, the expected pecuniary return paid on U.S. liabilities is:

$$r_L^F = \bar{r} + \omega^F \mu^F. \quad (70)$$

Define the U.S. external return differential as $\mathcal{RD} \equiv r_A^{\text{dol}} - r_L^F$. Its local policy response is:

$$\frac{d\mathcal{RD}}{d\psi} = \frac{dr_A^{\text{dol}}}{d\psi} - \frac{d\bar{r}}{d\psi} - \omega^F \frac{d\mu^F}{d\psi} - \mu^F \frac{d\omega^F}{d\psi}. \quad (71)$$

Stronger targeting lowers the return differential if and only if the response of the U.S. liability

return exceeds that of the foreign-asset return. An erosion of the Treasury convenience yield raises the domestic safe rate $\bar{r}$, which increases liability costs and lowers the return differential. Conversely, a lower expected equity excess return (μ^F) reduces the cost of U.S. equity liabilities, which raises the return differential. Because the return process on Home's foreign asset is an exogenous primitive, the overall direction of the differential remains analytically ambiguous and depends strictly on the balance of forces in (71). The currency-derivative position D^e does not affect this differential because it carries zero initial value and generates no cross-border cash flows.

In summary, when stronger targeting raises the equity price and erodes the Treasury service wedge, it generates two distinct reductions in the U.S. external balance: a negative valuation effect on inherited Foreign-owned equity and higher interest payments on newly issued Foreign-held Treasury debt. The total NFA response also depends on the remaining current-account and valuation adjustments.

5.6 Home Asset-Holder and Foreign Participation Incidence

Home asset-holding households' lifetime utility is:

$$\mathcal{W}_H \equiv \mathbb{E}_0 \sum_{t \geq 0} \beta^t \left[\log C_t^H + \zeta v \left(\frac{S_t^H}{C_t^H} \right) \right]. \quad (72)$$

Because the marginal rate of substitution between services and consumption equals the Treasury service wedge ($U_{S,t}^H / U_{C,t}^H = \lambda_{T,t}^S$), the total policy derivative of Home utility is:

$$\frac{d\mathcal{W}_H}{d\psi} = \mathbb{E}_0 \sum_{t \geq 0} \beta^t U_{C,t}^H \left(\frac{dC_t^H}{d\psi} + \lambda_{T,t}^S \frac{dS_t^H}{d\psi} \right). \quad (73)$$

At the regime-level steady state, Home retained services equal usable equity capacity net of Foreign absorption: $S^H = \bar{q}_B \bar{B} + X - F$.

Foreign welfare refers strictly to the participation surplus of customers in the delegated dollar-account block; it excludes the rents earned by currency specialists. This customer participation surplus obeys the envelope condition:

$$\frac{d\mathcal{W}_\psi^F}{d\psi} = A_{\psi}^{F,0} \frac{dQ_\psi^F}{d\psi}. \quad (74)$$

Proposition 8 (Incidence for Home asset holders and Foreign participation). *For a differentiable policy comparison, Home asset-holder utility rises if and only if the present value of consumption and service contributions in (73) is positive. Holding Home consumption fixed, the contemporaneous service contribution is strictly positive if and only if the Treasury service wedge falls. Foreign participation*

surplus strictly rises if and only if $dQ_\psi^F/d\psi > 0$.

Equation (73) isolates two distinct welfare channels for Home households. First, because inverse service demand slopes downward, a lower Treasury service wedge implies an expansion of services retained by Home at fixed consumption ($dS_t^H > 0$), which directly increases welfare. Second, cross-border investment income, valuation effects on inherited equity liabilities, and current-account flows alter Home consumption (dC_t^H). The total change in Home welfare is the sum of these service and consumption responses.

For Foreign dollar-account customers, the envelope theorem implies that policy-induced changes in the optimal equity share ω^F have no first-order effect on their welfare. Consequently, their participation surplus strictly increases with Q^F , the maximized net benefit of the dollar account relative to the foreign safe asset.

6 Conclusion

This paper shows that monetary policy frameworks designed to stabilize financial conditions also determine the supply and pricing of safe assets. A stronger financial-conditions target lowers equilibrium equity-return risk. When this return risk lies below the information-insensitivity threshold, a safer lower-tail payoff expands equity's capacity to back safe-asset claims. The resulting increase in private service supply reduces the scarcity value of safety, which lowers both the Treasury convenience yield and the Treasury–equity convenience spread. This transmission channel requires private pledgeability: if equity cannot supply collateral services, reductions in equity risk leave the Treasury convenience yield unchanged.

Internationally, lower return risk and higher pledgeability draw foreign portfolios toward dollar equity, while a compressed equity premium pushes them toward Treasuries. Because foreign ownership drains domestic convenience services, the domestic Treasury convenience yield falls—and domestic service welfare rises—only if the expansion in collateral capacity outpaces this foreign absorption. Finally, when the policy strictly raises the net valuation of dollar assets, the dollar appreciates, aggregate foreign dollar exposure expands, and foreign participation surplus rises.

We leave several natural extensions for future research. Empirically, the framework suggests that aggressive monetary backstops should induce structural breaks in bond–equity correlations and persistently compress the equity risk premium. Future work could explore the systemic fragility inherent in this regime: while quasi-safe equity stabilizes output in normal times, the induced comovement between stocks and bonds risks severely amplifying crises if the central bank exhausts its policy space.

References

- ATKESON, A., J. HEATHCOTE, AND F. PERRI (2025): "The End of Privilege: A Reexamination of the Net Foreign Asset Position of the United States," *American Economic Review*, 115, 2151–2206.
- BAUER, M. D., B. S. BERNANKE, AND E. MILSTEIN (2023): "Risk Appetite and the Risk-Taking Channel of Monetary Policy," *Journal of Economic Perspectives*, 37, 77–100.
- BAUER, M. D. AND E. T. SWANSON (2023): "An Alternative Explanation for the "Fed Information Effect"," *American Economic Review*, 113, 664–700.
- BEKAERT, G., M. HOEROVA, AND M. LO DUCA (2013): "Risk, uncertainty and monetary policy," *Journal of Monetary Economics*, 60, 771–788.
- BERNANKE, B. S. AND K. N. KUTTNER (2005): "What Explains the Stock Market's Reaction to Federal Reserve Policy?" *The Journal of Finance*, 60, 1221–1257, _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1540-6261.2005.00760.x>.
- BERTAUT, C. C., S. E. CURCURU, E. FAIA, AND P.-O. GOURINCHAS (2024): "New Evidence on the US Excess Return on Foreign Portfolios," *IMF Working Paper*.
- BJØRNLAND, H. C. AND K. LEITEMO (2009): "Identifying the interdependence between US monetary policy and the stock market," *Journal of Monetary Economics*, 56, 275–282.
- CABALLERO, R. J., T. E. CARAVELLO, AND A. SIMSEK (2024): "Financial Conditions Targeting," *NBER Working Paper*.
- CABALLERO, R. J. AND A. SIMSEK (2026): "Financial Conditions Targeting in a Multi-Asset Open Economy," *Working Paper*.
- CAMANHO, N., H. HAU, AND H. REY (2022): "Global Portfolio Rebalancing and Exchange Rates," *The Review of Financial Studies*, 35, 5228–5274.
- CHODOROW-REICH, G., P. T. NENOV, AND A. SIMSEK (2021): "Stock Market Wealth and the Real Economy: A Local Labor Market Approach," *American Economic Review*, 111, 1613–1657.
- CIESLAK, A., A. MORSE, AND A. VISSING-JORGENSEN (2019): "Stock Returns over the FOMC Cycle," *The Journal of Finance*, 74, 2201–2248, _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/jofi.12818>.
- CIESLAK, A. AND H. PANG (2021): "Common Shocks in Stocks and Bonds," *Journal of Financial Economics*, 142, 880–904.

- CIESLAK, A. AND A. VISSING-JORGENSEN (2021): “The Economics of the Fed Put,” *The Review of Financial Studies*, 34, 4045–4089.
- DAHIYA, S., B. KAMRAD, V. POTI, AND A. SIDDIQUE (2026): “Fed put in the equity options markets,” *Journal of Banking & Finance*, 188, 107697.
- DANG, T. V., G. GORTON, AND B. HOLMSTRÖM (2015): “The Information Sensitivity of a Security,” *Working Paper*.
- DI MAGGIO, M., A. KERMANI, AND K. MAJLESI (2020): “Stock Market Returns and Consumption,” *The Journal of Finance*, 75, 3175–3219, *_eprint*: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/jofi.12968>.
- DU, W., J. IM, AND J. SCHREGER (2018): “The U.S. Treasury Premium,” *Journal of International Economics*, 112, 167–181.
- DU, W., R. KEERATI, AND J. SCHREGER (2025): “Decoupling Dollar and Treasury Privilege,” *International Finance Discussion Papers*.
- FUHRER, J. AND G. TOOTELL (2008): “Eyes on the prize: How did the fed respond to the stock market?” *Journal of Monetary Economics*, 55, 796–805.
- GABAIX, X. AND R. S. J. KOIJEN (2021): “In Search of the Origins of Financial Fluctuations: The Inelastic Markets Hypothesis,” *NBER Working Paper*.
- GERTLER, M. AND P. KARADI (2015): “Monetary Policy Surprises, Credit Costs, and Economic Activity,” *American Economic Journal: Macroeconomics*, 7, 44–76.
- GORTON, G. AND G. PENNACCHI (1990): “Financial Intermediaries and Liquidity Creation,” *The Journal of Finance*, 45, 49–71.
- GOURINCHAS, P.-O. AND H. REY (2007): “From World Banker to World Venture Capitalist: US External Adjustment and the Exorbitant Privilege,” *NBER Chapters*, 11–66.
- GÜRKAYNAK, R. S., B. SACK, AND E. T. SWANSON (2005): “Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements,” *International Journal of Central Banking*, 1, 55–93.
- HADDAD, V., A. MOREIRA, AND T. MUIR (2025): “Whatever It Takes? The Impact of Conditional Policy Promises,” *American Economic Review*, 115, 295–329.
- JAROCIŃSKI, M. AND P. KARADI (2020): “Deconstructing Monetary Policy Surprises—The Role of Information Shocks,” *American Economic Journal: Macroeconomics*, 12, 1–43.

- JIANG, Z. (2024): “Exorbitant Privilege: A Safe-Asset View,” *NBER Working Paper*.
- JIANG, Z., R. J. RICHMOND, AND T. ZHANG (2025): “Convenience Lost,” .
- KOIJEN, R. S. J. AND M. YOGO (2026): “Exchange Rates and Asset Prices in a Global Demand System,” *Journal of Political Economy*.
- KRISHNAMURTHY, A. AND A. VISSING-JORGENSEN (2012): “The Aggregate Demand for Treasury Debt,” *Journal of Political Economy*, 120, 233–267.
- KUROV, A., E. OLSON, AND G. R. ZAYNUTDINOVA (2022): “When does the fed care about stock prices?” *Journal of Banking & Finance*, 142, 106556.
- MAGGIORI, M. (2017): “Financial Intermediation, International Risk Sharing, and Reserve Currencies,” *American Economic Review*, 107, 3038–3071.
- MAGGIORI, M., B. NEIMAN, AND J. SCHREGER (2020): “International Currencies and Capital Allocation,” *Journal of Political Economy*, 128, 2019–2066.
- MIRANDA-AGRIPPINO, S. AND G. RICCO (2021): “The Transmission of Monetary Policy Shocks,” *American Economic Journal: Macroeconomics*, 13, 74–107.
- NAGEL, S. (2016): “The Liquidity Premium of Near-Money Assets*,” *The Quarterly Journal of Economics*, 131, 1927–1971.
- NAKAMURA, E. AND J. STEINSSON (2018): “High-Frequency Identification of Monetary Non-Neutrality: The Information Effect,” *The Quarterly Journal of Economics*, 133, 1283–1330.
- RIGOBON, R. AND B. SACK (2003): “Measuring The Reaction of Monetary Policy to the Stock Market,” *The Quarterly Journal of Economics*, 118, 639–669.
- SHAPIRO, A. H. AND D. J. WILSON (2022): “Taking the Fed at its Word: A New Approach to Estimating Central Bank Objectives using Text Analysis,” *The Review of Economic Studies*, 89, 2768–2805.
- SHLEIFER, A. AND R. W. VISHNY (1997): “The Limits of Arbitrage,” *The Journal of Finance*, 52, 35–55.
- TABOVA, A. M. AND F. E. WARNOCK (2025): “Exorbitant Changes in Three Parts,” NBER Working Paper 34372, National Bureau of Economic Research.

Appendix

A Omitted Derivations

A.1 The Household Consumption Rule

Let N_t^H and B_t^H denote the household's economically owned equity and Treasury positions after date- t trade. The primitive accounting constraint is

$$C_t + P_t N_t^H + q_{B,t} B_t^H = (P_t + d_{P,t}) N_{t-1}^H + B_{t-1}^H. \quad (\text{A.1})$$

Cash on hand and saved wealth are, respectively,

$$W_t \equiv (P_t + d_{P,t}) N_{t-1}^H + B_{t-1}^H, \quad W_t - C_t = P_t N_t^H + q_{B,t} B_t^H.$$

The services attached to those positions and the household balance-sheet service intensity are

$$S_t = q_{B,t} B_t^H + m_t P_t N_t^H, \quad h_t = \frac{q_{B,t} B_t^H + m_t P_t N_t^H}{q_{B,t} B_t^H + P_t N_t^H}, \quad S_t = h_t (W_t - C_t). \quad (\text{A.2})$$

The term $m_t P_t N_t^H$ is the usable collateral value attached to the household's beneficial equity position. Households choose consumption taking h_t and their delegated portfolio return as given. If R_{t+1}^{port} denotes that return, then $W_{t+1} = R_{t+1}^{\text{port}} (W_t - C_t)$. In the closed economy, $N_t^H = 1$, $B_t^H = B_{t-1}^H = \bar{B}$, and $d_{P,t} = \nu Y_t$, so (A.1) reduces to (7) and $W_t = \nu Y_t + P_t + \bar{B}$.

Guess $V(W) = A \log W + B$. The envelope equation gives

$$A = 1 + \beta A = \frac{1}{1 - \beta}.$$

Because

$$z_t = \frac{h_t (W_t - C_t)}{C_t},$$

the consumption first-order condition is

$$\frac{1}{C_t} - \frac{\xi h_t W_t v'(z_t)}{C_t^2} - \frac{\beta A}{W_t - C_t} = 0.$$

Combining this condition with the envelope equation gives

$$\frac{C_t}{W_t} = (1 - \beta) [1 - \xi z_t v'(z_t)]. \quad (\text{A.3})$$

The maintained condition $U_C > 0$ is equivalent to $\xi z_t v'(z_t) < 1$. Since $\lambda_t^S = \xi v'(z_t) / [1 - \xi z_t v'(z_t)]$ and $S_t = z_t C_t$,

$$C_t = (1 - \beta)W_t - \lambda_t^S S_t, \quad W_t = vY_t + P_t + \bar{B}. \quad (\text{A.4})$$

This rule follows from household optimization. The autonomous-expenditure component Δ_t enters goods demand below, outside the household problem, as in Caballero et al. (2024). Setting $\xi = 0$ recovers the log-utility consumption rule in Caballero et al. (2024).

Direct Treasury pricing. An additional directly held Treasury unit costs $q_{B,t}$, raises date- t services by $q_{B,t}$, and pays one unit at date $t + 1$. Its household first-order condition is

$$q_{B,t} \left(1 - \frac{U_S(C_t, S_t)}{U_C(C_t, S_t)} \right) = \mathbb{E}_t[\Lambda_{t+1}] = Q_{B,t},$$

which is exactly (10). Thus the positive Treasury supply is priced by direct household holdings, not by the managers' relative-exposure condition.

Inverse service demand. Write

$$\gamma_S \equiv - \left. \frac{d \log \lambda_t^S}{d \log z_t} \right|_{\bar{z}}.$$

If $\gamma_v \equiv -zv''(z)/v'(z)$, differentiation of the service price gives

$$\frac{d\lambda^S(z)}{dz} = \frac{\xi \{v''(z) + \xi[v'(z)]^2\}}{[1 - \xi z v'(z)]^2}.$$

For $\xi > 0$, strict joint concavity is equivalent to $v''(z) + \xi[v'(z)]^2 < 0$. Hence $d\lambda^S(z)/dz < 0$ and $\gamma_S > 0$. Given $U_C > 0$, the separate Treasury-pricing restriction $\lambda^S(z) < 1$ is equivalent to $\xi(1 + z)v'(z) < 1$. The elasticity is

$$\gamma_S = \frac{\gamma_v - \xi z v'(z)}{1 - \xi z v'(z)}. \quad (\text{A.5})$$

Because $z_t = S_t/C_t$, its log deviation is $s_t - c_t$. A first-order expansion of inverse service demand around $\bar{z}$ therefore gives

$$\ell_t = \log \frac{\lambda_{T,t}^S}{\bar{\lambda}_T^S} = -\gamma_S(s_t - c_t) = \gamma_S(c_t - s_t), \quad (\text{A.6})$$

where ℓ_t is a current level deviation, not a time difference. Under $v = \log$, $\gamma_S = 1$.

A.2 First-Order Equilibrium

This subsection log-linearizes the equilibrium conditions around an interior steady state.

Aggregate demand. Let $C_t^H \equiv C_t$. The household rule (A.4), hand-to-mouth spending, and goods-market clearing are

$$\begin{aligned} C_t^H &= (1 - \beta)(\nu Y_t + P_t + \bar{B}) - \lambda_{P,t} P_t - \lambda_{T,t}^S q_{B,t} \bar{B}, \\ C_t^{HM} &= (1 - \nu)Y_t + T_t, \quad T_t = q_{B,t} \bar{B} - \bar{B}, \\ Y_t &= C_t^H + C_t^{HM} + \Delta_t. \end{aligned} \quad (\text{A.7})$$

Combining these equations gives

$$\beta \nu Y_t = (1 - \beta - \lambda_{P,t}) P_t + \bar{B} \left[q_{B,t} (1 - \lambda_{T,t}^S) - \beta \right] + \Delta_t. \quad (\text{A.8})$$

The bond-pricing condition makes the term in square brackets $e^{-r_t^{\text{pec}}} - \beta$. At $\bar{r}^{\text{pec}} = \rho$ and $\Delta_t = 0$,

$$\beta \nu \bar{Y} = (1 - \beta - \bar{\lambda}_P) \bar{P}, \quad \bar{\lambda}_P < 1 - \beta. \quad (\text{A.9})$$

Log-linearization gives

$$y_t = p_t - b \hat{r}_t - (\Xi + b a_T) \ell_t + \delta_t, \quad (\text{A.10})$$

where

$$a_T \equiv \frac{\bar{\lambda}_T^S}{1 - \bar{\lambda}_T^S}, \quad \hat{r}_t^{\text{pec}} = \hat{r}_t + a_T \ell_t, \quad (\text{A.11})$$

and

$$\Xi \equiv \frac{\bar{\lambda}_P}{1 - \beta - \bar{\lambda}_P}, \quad b \equiv \frac{\bar{B}}{\nu \bar{Y}}. \quad (\text{A.12})$$

Equation (A.10) gives (22)–(23).

Services and household consumption. Because m is fixed and $d \log q_{B,t} = -\hat{r}_t$, log-linearizing (9) gives

$$s_t = \bar{\omega}_P^S p_t - (1 - \bar{\omega}_P^S) \hat{r}_t. \quad (\text{A.13})$$

The closed-economy household identity (7) reduces to $C_t = \nu Y_t + (1 - q_{B,t}) \bar{B}$. Therefore

$$c_t = \chi_y y_t + \chi_r \hat{r}_t, \quad (\text{A.14})$$

$$\chi_y \equiv \frac{\nu \bar{Y}}{\bar{C}}, \quad \chi_r \equiv \frac{\bar{q}_B \bar{B}}{\bar{C}}, \quad \bar{C} = \nu \bar{Y} + (1 - \bar{q}_B) \bar{B}. \quad (\text{A.15})$$

At fixed mean output, differentiating (A.9) gives the convenience component of equity

valuation,

$$d \log \bar{P} = \Xi d \log \bar{\lambda}_P. \quad (\text{A.16})$$

Equity payoff and pricing. The traded equity return satisfies

$$\begin{aligned} r_{p,t+1}^P &= \bar{r}^P + \omega_D y_{t+1} + \omega_P p_{t+1} - p_t, \\ \omega_D &= \frac{\nu \bar{Y}}{\bar{P} + \nu \bar{Y}} = \frac{1 - \beta - \bar{\lambda}_P}{1 - \bar{\lambda}_P}, \\ \omega_P &= \frac{\bar{P}}{\bar{P} + \nu \bar{Y}} = \frac{\beta}{1 - \bar{\lambda}_P}. \end{aligned} \quad (\text{A.17})$$

where the second equalities use the balanced-growth relation (A.9). The payoff weights satisfy $\omega_D + \omega_P = 1$.

Return convention. For otherwise identical cash flows, the exact rate-equivalent difference between Treasury and equity services is

$$\log \left(\frac{1 - \bar{\lambda}_P}{1 - \bar{\lambda}_T^S} \right) = \bar{\lambda}_T^S - \bar{\lambda}_P + O \left(\max \{ \bar{\lambda}_T^S, \bar{\lambda}_P \}^2 \right).$$

The delegated lognormal mandate adds the risk correction $x/2$, so its balanced-growth condition is

$$\bar{\mu}^x = \frac{x}{2} + \bar{\lambda}_T^S - \bar{\lambda}_P,$$

as in (17). We use this condition to locate the conditional return distribution in the pledgeability block. In the within-environment first-order system, the risk correction and the balanced-growth service spread enter the regime-specific intercept. Subtracting the balanced-growth manager condition therefore removes both from the equity-price recursion.

At fixed m , the first-order deviation of the marginal service-wedge difference is

$$(\lambda_{T,t}^S - \lambda_{P,t}) - (\bar{\lambda}_T^S - \bar{\lambda}_P) = (\bar{\lambda}_T^S - \bar{\lambda}_P) \ell_t.$$

An arbitrageur chooses relative equity exposure ω_t^A against the Treasury benchmark. Following Caballero et al. (2024), we use the local lognormal representation of an expected-log-assets-under-management mandate:

$$\max_{\omega_t^A} \left\{ \omega_t^A \left[\mathbb{E}_t r_{p,t+1}^x + \lambda_{P,t} - \lambda_{T,t}^S + \frac{x}{2} \right] - \frac{x}{2} (\omega_t^A)^2 \right\}.$$

The additive service spread is the first-order marginal-utility adjustment in the delegated

mandate. The first-order condition is

$$\mathbb{E}_t r_{p,t+1}^x + \lambda_{p,t} - \lambda_{T,t}^S = \left(\omega_t^A - \frac{1}{2} \right) x,$$

the manager condition used in (14). Across all manager types, equity-market clearing makes average equity exposure one, so their positions relative to the safe benchmark sum to zero. The manager condition therefore does not price the positive Treasury supply.

Subtracting the balanced-growth manager condition from its date- t counterpart and substituting (A.17) yields

$$p_t = \omega_P \mathbb{E}_t p_{t+1} + \omega_D \mathbb{E}_t y_{t+1} - \hat{r}_t + \mathcal{I}_P \mu_t^p + (\lambda_{p,t} - \bar{\lambda}_P) - (\lambda_{T,t}^S - \bar{\lambda}_T^S).$$

This gives (15). On the active branch, the final two terms reduce to $(\bar{\lambda}_P - \bar{\lambda}_T^S) \ell_t = \zeta \ell_t$. Thus the equity-pricing equation contains the current service-wedge level. It contains no service-wedge growth term.

Return risk. Conditional on date- t information, p_t and r_t are known. Equation (A.17) therefore implies

$$x = \text{Var}_t(\omega_D y_{t+1} + \omega_P p_{t+1}). \quad (\text{A.18})$$

This is the total-return variance relevant for manager risk bearing. The pledged account in Appendix A.3 retains dividends and therefore uses the same risk measure when dividends have positive weight ω_D .

A.3 Tail Microfoundation for Pledgeability

This subsection derives (16) and (18) and states the regularity conditions used in Lemma A.1.

As a microfoundation for m_t , place one share of equity in a segregated account that supports an undrawn one-period collateralized credit line. If the line is drawn, the account retains dividends until repayment and then pays $P_{t+1} + d_{p,t+1}$. The date- t credit limit is $m_t P_t$, and the corresponding promised date- $t + 1$ repayment is $R_t^{\text{mar}} m_t P_t$, where $R_t^{\text{mar}} = e^{r_t}$. Liquidation and operating costs absorb κ_C units of Treasury-value payoff per dollar of collateral. Repayment fails when

$$\frac{P_{t+1} + d_{p,t+1}}{R_t^{\text{mar}} P_t} - \kappa_C < m_t. \quad (\text{A.19})$$

Define the total excess return

$$r_{p,t+1}^x \equiv \log\left(\frac{P_{t+1} + d_{p,t+1}}{P_t}\right) - r_t, \quad x \equiv \text{Var}_t(r_{p,t+1}^x). \quad (\text{A.20})$$

Let $Q_{\bar{p}}^x$ denote the lower $\bar{p}$ -quantile of the one-period conditional distribution of this return at

the balanced-growth point. Assume that this distribution is continuous and strictly increasing in a neighborhood of $Q_{\bar{p}}^x$. The requirement that the shortfall probability in (A.19) not exceed $\bar{p}$ gives

$$m = \max \left\{ 0, \exp\{Q_{\bar{p}}^x\} - \kappa_C \right\}, \quad (\text{A.21})$$

We assume that the implied advance rate remains below one:

$$Q_{\bar{p}}^x < \log(1 + \kappa_C). \quad (\text{A.22})$$

This condition ensures $m < 1$. The lower corner binds when $Q_{\bar{p}}^x \leq \log \kappa_C$, so equity has zero service intensity in sufficiently adverse return distributions.

To separate scale from distributional shape, write

$$r_{p,t+1}^x = \bar{\mu}^x + \sqrt{x} \varepsilon_{t+1},$$

where the standardized innovation has mean zero, variance one, and conditional distribution F_ε . Its lower standardized quantile is $q_{\bar{p}} = F_\varepsilon^{-1}(\bar{p}) < 0$, so

$$Q_{\bar{p}}^x = \bar{\mu}^x + q_{\bar{p}} \sqrt{x}. \quad (\text{A.23})$$

Variance alone summarizes lower-tail changes along branches on which F_ε is fixed. This condition holds in the lognormal equilibrium.

Under lognormal returns, the log excess return is normal. Its lower $\bar{p}$ -quantile is $Q_{\bar{p}}^x = \bar{\mu}^x - z_{\bar{p}} \sqrt{x}$, where $z_{\bar{p}}$ is the standard-normal cutoff for tail probability $\bar{p}$. The manager condition at the balanced-growth point gives

$$\bar{\mu}^x = \frac{x}{2} + \bar{\lambda}_T^S - \bar{\lambda}_P = \frac{x}{2} + (1 - m) \bar{\lambda}_T^S. \quad (\text{A.24})$$

Substitution into (A.21) yields

$$m = \max \left\{ 0, \exp \left\{ \frac{x}{2} + (1 - m) \bar{\lambda}_T^S - z_{\bar{p}} \sqrt{x} \right\} - \kappa_C \right\}. \quad (\text{A.25})$$

For a given x , this equation is solved jointly with steady-state equity pricing and service clearing. At the balanced-growth point, x is the same one-period payoff variance that enters manager risk bearing. The probability bound is imposed at that point rather than state by state.

Lemma A.1 (Lower-tail returns and pledgeability). *Consider a differentiable change along a branch with $0 < m < 1$, $x > 0$, and fixed F_ε . Pledgeability rises if and only if the lower-tail total excess return rises:*

$$d\bar{\mu}^x + \frac{q_{\bar{p}}}{2\sqrt{x}} dx > 0. \quad (\text{A.26})$$

In that case,

$$d \log m = \left(1 + \frac{\kappa_C}{m}\right) \left[d\bar{\mu}^x + \frac{q_{\bar{p}}}{2\sqrt{x}} dx\right] > 0. \quad (\text{A.27})$$

Proof. On the active branch, $m + \kappa_C = \exp\{\bar{\mu}^x + q_{\bar{p}}\sqrt{x}\}$. Differentiating gives (A.27); its multiplicative factor is positive, so the sign of $d \log m$ is the sign of the change in the lower-tail return. $\square$

At $m = 0$, comparative statics use level or one-sided changes because the interior log derivative does not apply; the nonsaturation condition excludes an upper corner. Under lognormal returns, $q_{\bar{p}} = -z_{\bar{p}}$. Lower dispersion raises the tail payoff for a fixed mean, while a lower conditional mean reduces it. Pledgeability rises precisely when the net lower-tail response is positive.

A.4 Collected Equilibrium Conditions

For reference, we collect separately the within-regime first-order system and the regime-level consistency conditions. Tags report the economic source and the equation from which each condition is obtained. We use AD for the joint aggregate-demand condition, TS for Treasury pricing, SS for service clearing, PM for portfolio management and equity-market clearing, HH for household accounting, EP for the equity-payoff identity, PC for pledgeability, and VC for variance consistency. Here $\bar{C}$ denotes balanced-growth household consumption.

Within-regime first-order system. Conditional on the exogenous shock laws and the policy rule, and for given regime-level objects m and x , the first-order equilibrium satisfies

$$y_t = f_t + \delta_t \quad (\text{AD [22]})$$

$$f_t = p_t - b\hat{r}_t - (\Xi + ba_T)\ell_t \quad (\text{AD+TS [23]})$$

$$\ell_t = \gamma_S(c_t - s_t) \quad (\text{HH [A.6]})$$

$$s_t = \bar{\omega}_p^S p_t - (1 - \bar{\omega}_p^S)\hat{r}_t \quad (\text{SS [A.13]})$$

$$c_t = \frac{v\bar{Y}}{\bar{C}}y_t + \frac{\bar{q}_B\bar{B}}{\bar{C}}\hat{r}_t \quad (\text{HH [A.14, A.15]})$$

$$p_t = \omega_P \mathbb{E}_t p_{t+1} + \omega_D \mathbb{E}_t y_{t+1} - \hat{r}_t + \frac{x}{\alpha_p} \mu_t^p + \zeta \ell_t \quad (\text{PM+EP [15]})$$

$$r_{p,t+1}^P = \bar{r}^P + \omega_D y_{t+1} + \omega_P p_{t+1} - p_t. \quad (\text{EP [A.17]})$$

The no-bubble condition (27) completes the forward equity-price equation. The system uses the Treasury service-wedge deviation ℓ_t ; when $m > 0$, it is also the log deviation of the equity service wedge.

Regime-level consistency. The stochastic law generated by the first-order system determines return risk, while portfolio management and the collateral constraint determine the balanced-growth expected return and pledgeability:

$$x = \text{Var}_t(\omega_D y_{t+1} + \omega_P p_{t+1}) \quad (\text{VC [A.18]})$$

$$\bar{\mu}^x = \frac{x}{2} + (1 - m)\bar{\lambda}_T^S \quad (\text{PM [17]})$$

$$m = \max\{0, \exp\{\bar{\mu}^x - z_{\bar{p}}\sqrt{x}\} - \kappa_C\}. \quad (\text{PC [16, 18]})$$

These conditions determine x , $\bar{\mu}^x$, and m jointly. The maximum operator includes the lower collateral corner $m = 0$; on an interior branch it is inactive.

B Derivations and Proofs for Section 3

B.1 Eliminating the Endogenous Service Wedge

The FCI is given by (23). Inverse service demand is given by (A.6); service supply and household consumption are given by (A.13) and (A.14).

Let

$$H \equiv \Xi + ba_T, \quad \mathcal{D}_S \equiv 1 + H\gamma_S\chi_y.$$

Combining the service-demand, service-supply, and consumption equations with $f_t = p_t - b\hat{r}_t - H\ell_t$ and $y_t = f_t + \delta_t$ gives the intermediate service-market equation

$$\mathcal{D}_S \ell_t = \gamma_S \left[(\chi_y - \bar{\omega}_P^S) p_t + (\chi_r + 1 - \bar{\omega}_P^S - b\chi_y) \hat{r}_t + \chi_y \delta_t \right]. \quad (\text{B.1})$$

Thus

$$\ell_t = \Lambda_p p_t + \Lambda_r \hat{r}_t + \Lambda_\delta \delta_t, \quad (\text{B.2})$$

where

$$\begin{aligned} \Lambda_p &= \frac{\gamma_S(\chi_y - \bar{\omega}_P^S)}{\mathcal{D}_S}, & \Lambda_r &= \frac{\gamma_S(\chi_r + 1 - \bar{\omega}_P^S - b\chi_y)}{\mathcal{D}_S}, \\ \Lambda_\delta &= \frac{\gamma_S\chi_y}{\mathcal{D}_S}. \end{aligned} \quad (\text{B.3})$$

Substitution into the FCI gives

$$f_t = (1 - H\Lambda_p)p_t - (b + H\Lambda_r)\hat{r}_t - H\Lambda_\delta\delta_t. \quad (\text{B.4})$$

The demand shifter enters this reduced representation because it changes service demand.

The household budget identity

$$\chi_r + 1 = (1 + b)\chi_y \quad (\text{B.5})$$

implies

$$\Lambda_r - \Lambda_p = \frac{\gamma_S[\chi_r + 1 - (1 + b)\chi_y]}{\mathcal{D}_S} = 0.$$

Write their common value as Λ . Equation (B.4) then becomes

$$f_t = (1 - H\Lambda)p_t - (b + H\Lambda)\hat{r}_t - H\Lambda_\delta\delta_t. \quad (\text{B.6})$$

The price and rate coefficients sum to $1 + b$. Moreover,

$$1 - H\Lambda = \frac{1 + H\gamma_S\bar{\omega}_P^S}{\mathcal{D}_S} > 0, \quad b + H\Lambda = \frac{b + H\gamma_S(\chi_r + 1 - \bar{\omega}_P^S)}{\mathcal{D}_S} > 0,$$

and $H\Lambda_\delta = H\gamma_S\chi_y/\mathcal{D}_S \in [0, 1)$ under the maintained interior restrictions.

B.2 The Reduced Pricing Recursion

This subsection eliminates ℓ_t , y_t , and $\hat{r}_t$ from the equity-pricing equation. It contains the coefficient algebra suppressed in the main text. Let

$$A_0 \equiv 1 - \zeta\Lambda, \quad Q_0 \equiv A_0 \left[1 + \frac{1 - H\Lambda}{b + H\Lambda} \right] = A_0 \frac{1 + b}{b + H\Lambda}, \quad \zeta \equiv \bar{\lambda}_P - \bar{\lambda}_T^S. \quad (\text{B.7})$$

We assume $A_0 > 0$. Because $b + H\Lambda > 0$, the expression in (B.7) then implies $Q_0 > 0$. Substituting (B.2) into (15) and using $y_{t+1} = f_{t+1} + \delta_{t+1}$ gives the intermediate pricing equation

$$A_0 p_t = \omega_P \mathbb{E}_t p_{t+1} + \omega_D \mathbb{E}_t (f_{t+1} + \delta_{t+1}) - A_0 \hat{r}_t + \zeta \Lambda_\delta \delta_t + \mathcal{I}_P \mu_t^P. \quad (\text{B.8})$$

Eliminating the current rate with (B.6) gives

$$p_t = d_P \mathbb{E}_t p_{t+1} + d_f \mathbb{E}_t f_{t+1} + d_\delta \mathbb{E}_t \delta_{t+1} + c_f f_t + c_\delta \delta_t + j \mathcal{I}_P \mu_t^P, \quad (\text{B.9})$$

where

$$d_p = \frac{\omega_P}{Q_0}, \quad d_f = \frac{\omega_D}{Q_0}, \quad (\text{B.10})$$

$$d_\delta = \frac{\omega_D}{Q_0}, \quad c_f = \frac{A_0}{(b + H\Lambda)Q_0} = \frac{1}{1 + b}, \quad (\text{B.11})$$

$$c_\delta = \frac{A_0 H \Lambda_\delta / (b + H\Lambda) + \zeta \Lambda_\delta}{Q_0}, \quad j = \frac{1}{Q_0}. \quad (\text{B.12})$$

B.3 Monetary Policy

Rate-to-FCI mapping. Hold continuation values, the current demand state, and current financial noise fixed while current goods and service markets clear. Differentiating (B.8) gives

$$A_0 \frac{dp_t}{d\hat{r}_t} = -A_0, \quad \frac{dp_t}{d\hat{r}_t} = -1. \quad (\text{B.13})$$

The service wedge is endogenous in this derivative. Equations (B.2) and (B.5) imply

$$\frac{d\ell_t}{d\hat{r}_t} = \Lambda \left(\frac{dp_t}{d\hat{r}_t} + 1 \right) = 0.$$

Differentiating $f_t = p_t - b\hat{r}_t - H\ell_t$ therefore yields

$$\frac{df_t}{d\hat{r}_t} = \frac{d\tilde{y}_t}{d\hat{r}_t} = -(1 + b), \quad c_f = \frac{1}{1 + b} > 0. \quad (\text{B.14})$$

These are total conditional-equilibrium derivatives at the rate-setting stage.

Derivation of the desired financial conditions. Let $f_t^d \equiv \mathbb{E}_t f_t$ and $\nu_t \equiv f_t - f_t^d$, with $\mathbb{E}_t \nu_t = 0$. Equations (B.13) and (B.14), together with the regularity conditions in Appendix B.2, imply that the current observed rate can choose f_t^d without changing the loading on the subsequently realized financial-noise innovation. Conditional on the candidate risk x and the associated linear coefficients, the current rate does not change the exogenous state or the future feasible policy set. Given the predetermined target f_t^T , the current-rate problem in (25) can therefore be written as

$$\mathbb{E}_t[(f_t - f_t^*)^2 + \psi(f_t - f_t^T)^2] = (f_t^d - f_t^*)^2 + \psi(f_t^d - f_t^T)^2 + (1 + \psi)\mathbb{E}_t \nu_t^2.$$

The first-order condition gives

$$f_t^d = \frac{f_t^* + \psi f_t^T}{1 + \psi}.$$

The target problem in (26) applies this minimized loss one period ahead. For a candidate

announcement f , its target-dependent component is

$$\frac{\psi}{1+\psi} \mathbb{E}_t \left[(f_{t+1}^* - f)^2 \right].$$

For $\psi > 0$, the new target is therefore $f_{t+1}^T = \mathbb{E}_t f_{t+1}^*$. Shifting the expression back one period gives $f_t^T = \mathbb{E}_{t-1} f_t^* = \mathbb{E}_{t-1} f_t^*$ because date- $(t-1)$ financial noise does not forecast f_t^* . At $\psi = 0$ the announcement is irrelevant, and we use the same expression as the continuous limiting convention. Because $f_t^* - E_{t-1} f_t^* = \varepsilon_{z,t} - \varepsilon_{\delta,t}$, equations (28)–(29) follow.

B.4 Equilibrium and Risk Reduction under FCI Targeting

Derivation of the first-order equilibrium. For each candidate x , we jointly solve for pledgeability m , steady-state asset prices, and the marginal value and aggregate supply of convenience services. All coefficients below are evaluated at the associated collateral-and-service solution. Because policy deviations vanish at the balanced-growth point, these coefficients have no direct dependence on ψ conditional on x ; total derivatives account for their response as equilibrium risk changes.

Start from (B.9). Define

$$A_y \equiv \frac{d_f + c_f}{1 - d_p}, \quad A_\delta \equiv \frac{\varphi_\delta(d_\delta - d_f) + c_\delta - c_f}{1 - d_p \varphi_\delta}, \quad (B.15)$$

$$F_p \equiv 1 - c_f(1 - H\Lambda), \quad \chi_p \equiv \frac{j}{F_p(1 - d_p \varphi_\mu)}, \quad \ell_\mu \equiv \varphi_\mu F_p \chi_p. \quad (B.16)$$

The definitions in Appendix B.2 imply $F_p = A_0/Q_0 > 0$. We restrict attention to the bounded fundamental solution satisfying the no-bubble condition in (27).

The rate-to-FCI mapping gives $\Lambda_r = \Lambda_p = \Lambda$ and $c_f = 1/(1 + b)$. Equivalently,

$$F_p = 1 - \frac{1 - H\Lambda}{1 + b} = \frac{b + H\Lambda}{1 + b} = \frac{A_0}{Q_0} > 0. \quad (B.17)$$

Price and output gap. First suppress financial noise. Since policy observes the macroeconomic innovations, $f_t = f_t^d$. Conjecture

$$p_t^M = A_y y_t^* + A_\delta \delta_t - c_f h_\psi (\varepsilon_{z,t} - \varepsilon_{\delta,t}).$$

Then

$$\mathbb{E}_t p_{t+1}^M = A_y y_t^* + A_\delta \varphi_\delta \delta_t, \quad \mathbb{E}_t f_{t+1}^M = y_t^* - \varphi_\delta \delta_t.$$

Substitution in (B.9) gives the two coefficient restrictions in (B.15).

Now suppress macroeconomic shocks. Policy removes the predictable component of the FCI but cannot react to $\varepsilon_{\mu,t}$. Write

$$p_t^N = u_\mu \mu_{t-1}^p + q_\mu \varepsilon_{\mu,t}, \quad f_t^N = (1 - H\Lambda) q_\mu \varepsilon_{\mu,t}.$$

The second equality follows from (B.6): the current rate has no loading on the unobserved innovation. Since $\mathbb{E}_t p_{t+1}^N = u_\mu \mu_t^p$ and $\mathbb{E}_t f_{t+1}^N = 0$, coefficient matching in (B.9) gives

$$u_\mu = \varphi_\mu (d_p u_\mu + j \mathcal{I}_P), \quad q_\mu = d_p u_\mu + j \mathcal{I}_P + c_f (1 - H\Lambda) q_\mu.$$

Therefore

$$u_\mu = \ell_\mu \mathcal{I}_P, \quad q_\mu = \chi_p \mathcal{I}_P.$$

Adding the macroeconomic and financial-noise components gives

$$p_t = A_y y_t^* + A_\delta \delta_t - \frac{h_\psi}{1+b} (\varepsilon_{z,t} - \varepsilon_{\delta,t}) + \ell_\mu \frac{x}{\alpha_p} \mu_{t-1}^p + \chi_p \frac{x}{\alpha_p} \varepsilon_{\mu,t}. \quad (\text{B.18})$$

Using $\tilde{y}_t = f_t - f_t^*$ gives (30).

Policy rule. Inverting (B.6) expresses the instrument as

$$\hat{r}_t = \frac{(1 - H\Lambda) p_t - H\Lambda_\delta \delta_t - f_t}{b + H\Lambda}.$$

For the macroeconomic component, substitute p_t^M and $f_t^M = f_t^d$ from (29). Matching coefficients gives loadings $[(1 - H\Lambda) A_y - 1] / (b + H\Lambda) = b_y$ on y_t^* , $[1 + (1 - H\Lambda) A_\delta - H\Lambda_\delta] / (b + H\Lambda) = b_\delta$ on δ_t , and, on the news $\varepsilon_{z,t} - \varepsilon_{\delta,t}$,

$$\frac{h_\psi F_p}{b + H\Lambda} = \frac{h_\psi}{1+b},$$

using (B.17). For the noise component, $f_t^N = (1 - H\Lambda) q_\mu \varepsilon_{\mu,t}$ and $p_t^N = u_\mu \mu_{t-1}^p + q_\mu \varepsilon_{\mu,t}$ give

$$\hat{r}_t^N = \frac{1 - H\Lambda}{b + H\Lambda} u_\mu \mu_{t-1}^p = \frac{1 - H\Lambda}{b + H\Lambda} \varphi_\mu F_p \chi_p \mathcal{I}_P \mu_{t-1}^p = \frac{\varphi_\mu \mathcal{I}_f(x)}{1+b} \mu_{t-1}^p,$$

again by (B.17) and $\mathcal{I}_f(x) = (1 - H\Lambda) \chi_p \mathcal{I}_P$. Define

$$b_y \equiv \frac{(1 - H\Lambda) A_y - 1}{b + H\Lambda}, \quad b_\delta \equiv \frac{1 + (1 - H\Lambda) A_\delta - H\Lambda_\delta}{b + H\Lambda}.$$

Adding the components gives the implementing policy-rate process

$$\hat{r}_t = b_y y_t^* + b_\delta \delta_t + \frac{h_\psi}{1+b} (\varepsilon_{z,t} - \varepsilon_{\delta,t}) + \frac{\varphi_\mu \mathcal{I}_f(x)}{1+b} \mu_{t-1}^p. \quad (\text{B.19})$$

Advancing one period and conditioning on full date- t information gives

$$\mathbb{E}_t \hat{r}_{t+1} = b_y y_t^* + b_\delta \varphi_\delta \delta_t + \frac{\varphi_\mu \mathcal{I}_f(x)}{1+b} \mu_t^p. \quad (\text{B.20})$$

Regime-level return variance. The conditional innovation loadings of the equity price are

$$A_{p,z} = A_y - c_f h_\psi, \quad A_{p,\delta} = A_\delta + c_f h_\psi, \quad A_{p,\mu} = \chi_p \frac{x}{\alpha_p}. \quad (\text{B.21})$$

The output innovations have loadings $1 - h_\psi$ on supply news, h_ψ on demand news, and $\mathcal{I}_f(x) = (1 - H\Lambda)\chi_p x / \alpha_p$ on financial noise. Substituting these loadings into (A.17) and using $c_f = 1/(1+b)$ gives

$$\gamma_z \equiv \omega_D + \omega_P A_y, \quad \gamma_\delta \equiv \omega_P A_\delta, \quad \gamma_h \equiv \omega_D + \frac{\omega_P}{1+b} > 0. \quad (\text{B.22})$$

Thus

$$\begin{aligned} r_{p,t+1}^x - \mathbb{E}_t r_{p,t+1}^x &= (\gamma_z - \gamma_h h_\psi) \varepsilon_{z,t+1} + (\gamma_\delta + \gamma_h h_\psi) \varepsilon_{\delta,t+1} \\ &\quad + [\omega_D(1 - H\Lambda) + \omega_P] \chi_p \frac{x}{\alpha_p} \varepsilon_{\mu,t+1}. \end{aligned} \quad (\text{B.23})$$

Here and below, all coefficients are evaluated at the balanced-growth solution associated with candidate risk x , and their x arguments are suppressed where no ambiguity arises. The primitive innovations are mutually independent, and their variances are fixed across ψ -regimes. Their squared loadings yield the scalar fixed-point map

$$\Phi(x; \psi) = \mathcal{V}(\psi, x) + \mathcal{K}(x)x^2, \quad (\text{B.24})$$

where

$$\mathcal{V}(\psi, x) = (\gamma_z - \gamma_h h_\psi)^2 \sigma_z^2 + (\gamma_\delta + \gamma_h h_\psi)^2 \sigma_\delta^2, \quad \mathcal{K}(x) = [\omega_D(1 - H\Lambda) + \omega_P]^2 \frac{\chi_p^2 \sigma_\mu^2}{\alpha_p^2}. \quad (\text{B.25})$$

This establishes (31). □

Auxiliary fixed-coefficient benchmark. If the coefficients are frozen at a given balanced-growth solution while the explicit risk argument is varied, the variance map becomes

$$\Phi^{\text{fix}}(x; \psi) = \mathcal{V}^{\text{fix}}(\psi) + \mathcal{K}^{\text{fix}} x^2, \quad \mathcal{V}^{\text{fix}}(\psi) = (\gamma_z - \gamma_h h_\psi)^2 \sigma_z^2 + (\gamma_\delta + \gamma_h h_\psi)^2 \sigma_\delta^2. \quad (\text{B.26})$$

If $\mathcal{K}^{\text{fix}} = 0$, the auxiliary equation $x = \Phi^{\text{fix}}(x; \psi)$ has the unique solution $x = \mathcal{V}^{\text{fix}}(\psi)$. If $\mathcal{K}^{\text{fix}} > 0$, real roots require $4\mathcal{K}^{\text{fix}}\mathcal{V}^{\text{fix}}(\psi) \leq 1$; under the strict inequality they are

$$x_\pm = \frac{1 \pm \sqrt{1 - 4\mathcal{K}^{\text{fix}}\mathcal{V}^{\text{fix}}(\psi)}}{2\mathcal{K}^{\text{fix}}}, \quad 2\mathcal{K}^{\text{fix}}x_- < 1 < 2\mathcal{K}^{\text{fix}}x_+.$$

The slope of the auxiliary map at a root is $2\mathcal{K}^{\text{fix}}x$, so x_- is locally stable. This benchmark illustrates root selection and feedback with frozen coefficients; it does not establish existence or uniqueness for the full endogenous-coefficient map.

Direct risk effects. At fixed candidate risk x , the equilibrium coefficients have no direct dependence on ψ . Differentiating the supply-and-demand component of (B.25) gives

$$\frac{\partial \mathcal{V}(\psi, x)}{\partial \psi} = 2\gamma_h^2 h'_\psi (\sigma_z^2 + \sigma_\delta^2) (h_\psi - h_M), \quad (\text{B.27})$$

with $h_M(x)$ as defined in (32). Equation (B.14) gives $\gamma_h = \omega_D + \omega_P / (1 + b) > 0$. It follows that $h_M(x) > 0$ if and only if $\gamma_z \sigma_z^2 > \gamma_\delta \sigma_\delta^2$. Since $h'_\psi > 0$, the direct risk effect is negative precisely when $h_\psi < h_M(x)$. Using (B.15), the risk-reduction condition can be written entirely in the model coefficients as

$$\begin{aligned} & \left[\omega_D + \omega_P \frac{d_f + (1 + b)^{-1}}{1 - d_p} \right] \sigma_z^2 \\ & > \omega_P \frac{\varphi_\delta (d_\delta - d_f) + c_\delta - (1 + b)^{-1}}{1 - d_p \varphi_\delta} \sigma_\delta^2. \end{aligned} \quad (\text{B.28})$$

Appendix B.2 gives each coefficient as a function of the structural parameters and steady service shares.

Proof of Proposition 1. At a smooth point away from the collateral-activation boundary, the risk fixed point solves $x = \Phi(x; \psi)$. Implicit differentiation gives

$$(1 - \Phi_x) \frac{dx}{d\psi} = \Phi_\psi.$$

At fixed x , the coefficients and primitive shock variances have no direct dependence on ψ , so equation (B.24) gives $\Phi_\psi = \mathcal{V}_\psi$. Equation (B.27) makes this direct effect negative whenever

$0 \leq h_\psi < h_M(x)$. The condition $1 - \Phi_x > 0$ then proves (33). If $0 \leq \Phi_x < 1$, the denominator is between zero and one, so feedback amplifies the direct decline. If $\Phi_x < 0$, the denominator exceeds one, so feedback dampens it. At the activation boundary, the same argument applies one-sided whenever the corresponding one-sided derivatives satisfy these inequalities. $\square$

Tail condition. Suppose $x(0) < z_{\bar{p}}^2$. Proposition 1 implies that $x(\psi)$ is decreasing. Thus $x(\psi) \leq x(0) < z_{\bar{p}}^2$ for $\psi \in [0, \bar{\psi}]$, and

$$\frac{d}{d\psi} \left(\frac{x}{2} - z_{\bar{p}} \sqrt{x} \right) = \left(\frac{1}{2} - \frac{z_{\bar{p}}}{2\sqrt{x}} \right) \frac{dx}{d\psi} > 0.$$

This condition signs the direct lognormal risk component of the lower-tail log return. Positive collateral capacity separately requires $Q_{\bar{p}}^x > \log \kappa_C$. On an active branch, the full pledgeability response also includes the endogenous service wedge through (17).

For the auxiliary benchmark in (B.26), a sufficient bound is

$$\mathcal{V}^{\text{fix}}(0) + \mathcal{K}^{\text{fix}} z_{\bar{p}}^4 < z_{\bar{p}}^2. \quad (\text{B.29})$$

To see this, let $F(x) = \mathcal{V}^{\text{fix}}(0) + \mathcal{K}^{\text{fix}} x^2 - x$. Nondegenerate macroeconomic risk gives $F(0) > 0$, while (B.29) gives $F(z_{\bar{p}}^2) < 0$. Continuity therefore places the lower fixed point in $(0, z_{\bar{p}}^2)$. At this first crossing $F'(x) < 0$, or equivalently $\Phi_x^{\text{fix}} < 1$, so it is the lower stable equilibrium.

Proof of Proposition 2. Equation (30) and orthogonality imply

$$G^{\text{mand}}(\psi) = \frac{1}{1 - \beta} \left[h_\psi^2 (\sigma_z^2 + \sigma_\delta^2) + \mathcal{I}_f(x(\psi))^2 \sigma_\mu^2 \right]. \quad (\text{B.30})$$

At discretion, $h_0 = 0$, so the derivative of the macroeconomic component is zero. Moreover, $\mathcal{I}_f(x) = g(x)x/\alpha_p$ implies

$$\frac{d\mathcal{I}_f(x(\psi))}{d\psi} = \frac{g(x)}{\alpha_p} [1 + \epsilon_g(x)] \frac{dx}{d\psi} < 0$$

under (35) and Proposition 1. Hence

$$\left. \frac{dG^{\text{mand}}}{d\psi} \right|_{\psi=0} = \frac{2\sigma_\mu^2}{1 - \beta} \mathcal{I}_f(x) \left. \frac{d\mathcal{I}_f(x(\psi))}{d\psi} \right|_{\psi=0} < 0$$

because $\mathcal{I}_f(x) > 0$. Continuity gives a right neighborhood of zero on which the mandate loss is below its discretionary value. In the fixed-coefficient benchmark, $g'(x) = 0$ and $\epsilon_g(x) = 0$, so (35) holds immediately. $\square$

B.5 Convenience-Yield Erosion

Proof of Proposition 3. Let

$$u \equiv d \log m, \quad L \equiv d \log \lambda_T^S, \quad E \equiv d \log \lambda_P, \quad \Pi \equiv d \log P, \quad \hat{X} \equiv d \log X.$$

Holding $\bar{r}^{\text{pec}}$ fixed gives $d \log q_B = a_T L$. At fixed mean output, (A.14) gives $d \log C = -\chi_r a_T L$. Since $S = q_B \bar{B} + X$, service supply and inverse demand imply

$$L = -\gamma_S[(\chi_r + 1 - \omega_P^S)a_T L + \omega_P^S \hat{X}] = -\Gamma \hat{X}, \quad \Gamma \equiv \frac{\gamma_S \omega_P^S}{1 + \gamma_S(\chi_r + 1 - \omega_P^S)a_T} > 0. \quad (\text{B.31})$$

The usable-collateral identity, the equity service wedge, and the valuation decomposition give

$$\hat{X} = u + \Pi, \quad E = u + L, \quad \Pi = \Xi E. \quad (\text{B.32})$$

Substitute $L = -\Gamma \hat{X}$ into this system:

$$\hat{X} = (1 + \Xi)u - \Xi \Gamma \hat{X}.$$

Therefore

$$\hat{X} = \mathcal{A}u, \quad L = -\Gamma \mathcal{A}u, \quad (\text{B.33})$$

$$\Pi = \frac{\Xi(1 - \Gamma)}{1 + \Gamma \Xi}u, \quad \mathcal{A} \equiv \frac{1 + \Xi}{1 + \Gamma \Xi} > 0. \quad (\text{B.34})$$

The balanced-growth-point mean in (17) satisfies

$$d\bar{\mu}^x = \frac{1}{2}dx + (1 - m)\lambda_T^S L - m\lambda_T^S u = \frac{1}{2}dx - \mathcal{B}u,$$

where

$$\mathcal{B} \equiv \lambda_T^S [m + (1 - m)\Gamma \mathcal{A}] > 0.$$

On the interior branch, Lemma A.1 gives

$$u = \left(1 + \frac{\kappa_C}{m}\right) \left[d\bar{\mu}^x - \frac{z_{\bar{p}}}{2\sqrt{x}} dx \right] = \left(1 + \frac{\kappa_C}{m}\right) \left[\left(\frac{1}{2} - \frac{z_{\bar{p}}}{2\sqrt{x}} \right) dx - \mathcal{B}u \right].$$

Solving for u and dividing by $d\psi$ gives

$$\frac{d \log m}{d\psi} = \frac{1 + \kappa_C/m}{1 + (1 + \kappa_C/m)\mathcal{B}} \left(\frac{1}{2} - \frac{z_{\bar{p}}}{2\sqrt{x}} \right) \frac{dx}{d\psi} > 0.$$

Risk reduction and $\sqrt{x} < z_{\bar{p}}$ give the inequality. Equations (B.33)–(B.34) then establish the responses of usable collateral, the Treasury service wedge, and the equity price in (36)–(37).

Since $d \log \lambda_P = u + L$,

$$\begin{aligned} d(r^{\text{pec}} - r) &= \frac{\lambda_T^S}{1 - \lambda_T^S} L < 0, \\ d \log \left(\frac{1 - \lambda_P}{1 - \lambda_T^S} \right) &= \left[\frac{\lambda_T^S}{1 - \lambda_T^S} - \frac{\lambda_P}{1 - \lambda_P} \right] L - \frac{\lambda_P}{1 - \lambda_P} u \\ &< 0. \end{aligned} \tag{B.35}$$

Both signs follow from $0 < \lambda_P = m \lambda_T^S < \lambda_T^S < 1$, $L < 0$, and $u > 0$. $\square$

Proof of Corollary 1. At $m > 0$, $X = mP > 0$. Equation (B.31) gives

$$d \log \lambda_T^S = -\Gamma d \log(mP), \quad \Gamma > 0.$$

Dividing by $d\psi$ proves (39). Since $-\log(1 - \lambda_T^S)$ is strictly increasing in λ_T^S , its policy derivative is negative if and only if $d \log(mP)/d\psi > 0$. $\square$

Proof of Proposition 4. Dormancy. Suppose $Q_{\bar{p}}^x < \log \kappa_C$. By continuity, the inequality remains strict under sufficiently small policy changes, so (16) gives $m = X = \lambda_P = 0$. Their log changes are undefined, and their level changes are zero. With $S = q_B \bar{B}$ and $L = d \log \lambda_T^S$, holding $\bar{r}^{\text{pec}} = \rho$ and mean output fixed gives

$$\begin{aligned} d \log q_B &= a_T L, \quad d \log C = -\chi_r a_T L, \\ L &= \gamma_S (d \log C - d \log S) = -\gamma_S (\chi_r + 1) a_T L. \end{aligned}$$

Because $1 + \gamma_S (\chi_r + 1) a_T > 0$, it follows that $L = 0$. The policy intensity ψ does not enter this service-market equation, so $d \log \lambda_T^S/d\psi = 0$, regardless of the response of x .

Activation. Now suppose $Q_{\bar{p}}^x = \log \kappa_C$. Let $dm \geq 0$ denote a one-sided change onto the active branch. At $m = 0$, $dX = P dm$. Writing the inverse-demand and service-market-clearing condition in levels gives

$$L = -\mathcal{G}_0 dm, \quad \mathcal{G}_0 \equiv \frac{\gamma_S P/S}{1 + \gamma_S (\chi_r + 1) a_T} > 0.$$

The balanced-growth mean condition then implies

$$dQ_{\bar{p}}^x = \left(\frac{1}{2} - \frac{z_{\bar{p}}}{2\sqrt{x}} \right) dx - \lambda_T^S (1 + \mathcal{G}_0) dm.$$

At the threshold, differentiating the positive branch of (16) gives $dm = \kappa_C dQ_{\bar{p}}^x$. Hence

$$\left[1 + \kappa_C \lambda_T^S (1 + \mathcal{G}_0)\right] dm = \kappa_C \left(\frac{1}{2} - \frac{z_{\bar{p}}}{2\sqrt{x}}\right) dx.$$

The bracketed term is positive. When $x < z_{\bar{p}}^2$ and $dx/d\psi < 0$, the right-hand side is positive, so $dm/d\psi > 0$ one-sidedly. Because $P > 0$, usable collateral capacity expands from zero. The level service-market condition gives $L < 0$, so the Treasury service wedge and its convenience yield begin to fall. Condition (A.22) rules out an upper corner on the branch under consideration. $\square$

C Proofs for Section 4

C.1 Open-Economy Microfoundation, Timing, and Equilibrium

Fix the targeting intensity ψ , and let bars denote the associated regime-level allocation. Prices, risk, pledgeability, Foreign dollar allocation, portfolios, and ownership are determined jointly. Within the regime, $(m_\psi, N_\psi^F, B_\psi^F)$ and the regime-level dollar exposures are predetermined with respect to calendar-time shocks. Delegated managers trade on behalf of Home and Foreign investors, while cash flows and collateral services accrue to the investors for whom they act. The beneficial owner holds the exclusive collateral right attached to equity; the right cannot be rented or rehypothecated.

Three conventions are maintained throughout. The Section 2 fiscal convention is unchanged: the Treasury rollover flow $q_{B,t}\bar{B} - \bar{B}$ accrues to hand-to-mouth workers, and asset-holding households receive no separate fiscal transfer. Import spending is independent of Home consumption at the margin, as in Caballero and Simsek (2026), so the expenditure-switching margin operates through the exchange rate alone. Home households own the delegated equity book, and Foreign investors own their dollar accounts. Acquisitions of actual U.S. claims are external transactions. Under the maintained external-saving convention, the resulting net external financing accumulates in Home's external account rather than current household cash on hand. Foreign currency specialists and Foreign customers take opposite sides of a zero-value currency-risk contract; it transfers neither beneficial ownership nor dollar services and cancels upon consolidation of the Foreign currency sector.

Delegated Foreign portfolio choice. A foreign dollar account assigns the share $\omega_\psi^F \in [0, 1]$ to U.S. equity and the remainder to Treasuries. Let $x_{p,\psi} > 0$ denote conditional equity-return variance and μ_ψ^F the expected equity return in excess of the Treasury return. Let $x_{e,\psi} > 0$ be exchange-rate variance and let

$$c_{pe,\psi} \equiv \text{Cov}_\psi(r_{p,t+1}^x, e_{t+1} - e_t)$$

be the equity–exchange-rate covariance generated by the joint equilibrium. Because one dollar of Treasuries supplies one unit of dollar services and one dollar of foreign-owned equity carries m_ψ units of usable collateral value, services per account dollar are

$$\vartheta_\psi^F \equiv 1 - (1 - m_\psi)\omega_\psi^F. \quad (\text{C.1})$$

The manager solves

$$J_\psi^F \equiv \max_{\omega \in [0,1]} \left\{ \omega \mu_\psi^F - \frac{\alpha_F}{2} \left(\omega^2 x_{p,\psi} + x_{e,\psi} - 2\omega c_{pe,\psi} \right) + \xi_F v_F(1 - (1 - m_\psi)\omega) \right\}, \quad (\text{C.2})$$

where $\alpha_F > 0$, $\xi_F > 0$, and v_F is increasing and concave. The objective is strictly concave when $x_{p,\psi} > 0$. At an interior solution its first-order condition is

$$\mu_\psi^F - \alpha_F(x_{p,\psi}\omega_\psi^F - c_{pe,\psi}) - \xi_F(1 - m_\psi)v_F'(\vartheta_\psi^F) = 0. \quad (\text{C.3})$$

In the regime-level allocation, foreign-owned shares displace shares held by Home delegated managers. Under the unit-supply and unit-exposure normalization inherited from Section 3.3, equity clearing is

$$\alpha_p \omega_{t|\psi}^A + \eta \left(1 + \frac{\mu_t^p}{\eta} \right) + (1 - \eta - \alpha_p) + N_\psi^F = 1, \quad \omega_{t|\psi}^A = 1 - \frac{N_\psi^F + \mu_t^p}{\alpha_p}. \quad (\text{C.4})$$

Combining this equation with the Section 3.3 manager first-order condition and evaluating at the zero-noise equilibrium gives (45) when Foreign and Home managers share beliefs. The term involving N_ψ^F is structural order flow. Because N_ψ^F is fixed within the regime, only μ_t^p moves the local manager position at a fixed ψ ; the high-frequency price-impact slope therefore remains $x_{p,\psi}/\alpha_p$. The first-order condition and the appropriate endpoint inequalities define the reduced portfolio map W in (43). Exchange-rate variance is common to every value of ω , so it affects the optimized account value and dollar allocation but not the interior equity share. Covariance affects both.

Bounded dollar allocation and ownership. Define regime-level expected dollar depreciation by

$$\overline{\Delta e}_\psi \equiv \mathbb{E}_t(e_{t+1} - e_t) |_{\text{regime equilibrium } \psi}.$$

The optimized noncurrency value of a dollar in the U.S. account, relative to the foreign safe asset, and its currency-adjusted value are

$$\mathcal{V}_\psi^F \equiv \bar{r}_\psi - \bar{r}^* + J_\psi^F, \quad (\text{C.5})$$

$$\mathcal{Q}_\psi^F \equiv \mathcal{V}_\psi^F - \bar{\Delta}e_\psi. \quad (\text{C.6})$$

The foreign safe rate $\bar{r}^*$ is held fixed across ψ . Currency risk enters J_ψ^F through $x_{e,\psi}$ and $c_{pe,\psi}$, while expected depreciation enters $\mathcal{Q}_\psi^F$ through $\bar{\Delta}e_\psi$. Lemma C.1 below establishes that $\bar{\Delta}e_\psi = 0$ within every fixed regime, so $\mathcal{Q}_\psi^F = \mathcal{V}_\psi^F$; we keep the depreciation term explicit in the definitions to make the currency adjustment auditable.

Let $\mathcal{A}^F > 0$ be the eligible pool of Foreign investable wealth. Each unit of the pool draws a fixed cost of entering the U.S. dollar account. Normalize the cost distribution so its CDF is the logistic function below. A unit enters when its optimized account value exceeds its cost, which gives the bounded dollar-allocation schedule

$$A^{F,0}(q) \equiv \mathcal{A}^F \frac{\exp\{\chi_A + \kappa_A q\}}{1 + \exp\{\chi_A + \kappa_A q\}}, \quad \kappa_A \geq 0. \quad (\text{C.7})$$

Thus $0 < A^{F,0} < \mathcal{A}^F$. The intercept χ_A fixes the location of the entry-cost distribution and κ_A its sensitivity to the optimized currency-adjusted account value; the local semi-elasticity is $\kappa_A(1 - A^{F,0}/\mathcal{A}^F)$, while the level slope satisfies

$$0 \leq \frac{\partial A^{F,0}}{\partial q} = \kappa_A A^{F,0} \left(1 - \frac{A^{F,0}}{\mathcal{A}^F}\right) \leq \frac{\mathcal{A}^F \kappa_A}{4}.$$

The case $\kappa_A = 0$ is the locally inelastic dollar-allocation limit; it does not fix portfolio composition, prices, or ownership across policy regimes. Hence

$$A_\psi^{F,0} = A^{F,0}(\mathcal{Q}_\psi^F), \quad E_\psi^F = \omega_\psi^F A_\psi^{F,0}, \quad T_\psi^F = (1 - \omega_\psi^F) A_\psi^{F,0}. \quad (\text{C.8})$$

Equation (44) converts these dollar demands into foreign-owned equity shares and Treasury face value at equilibrium prices. Within the regime the quantities remain fixed, while the marked portfolio value and its value equity share are

$$V_{t|\psi}^F \equiv P_t N_\psi^F + q_{B,t} B_\psi^F, \quad \tilde{\omega}_{t|\psi}^F \equiv \frac{P_t N_\psi^F}{V_{t|\psi}^F}. \quad (\text{C.9})$$

Acquisition value, current marked value, and ownership are therefore distinct.

For later linearization, let q_ψ^F , v_ψ^F , and $\hat{A}_\psi^{F,0}$ denote local deviations of $\mathcal{Q}_\psi^F$, $\mathcal{V}_\psi^F$, and acquisition

exposure. Define

$$\eta_A \equiv \frac{\partial \log A^{F,0}}{\partial Q^F} = \kappa_A \left(1 - \frac{A^{F,0}}{\mathcal{A}^F} \right) \geq 0. \quad (\text{C.10})$$

Then

$$\hat{A}_\psi^{F,0} = \eta_A q_\psi^F. \quad (\text{C.11})$$

Together with zero-noise currency clearing and the regime-level supply schedule, this condition determines the equilibrium exchange rate, Foreign dollar allocation, and regime-level dollar exposure. We require equilibrium excess demand to have a nonzero derivative with respect to the equilibrium exchange rate.

At an interior portfolio, the envelope theorem gives

$$d\mathcal{V}^F = d\bar{r} + \omega^F d\mu^F - \frac{\alpha_F}{2}(\omega^F)^2 dx_p - \frac{\alpha_F}{2} dx_e + \alpha_F \omega^F dc_{pe} + \zeta_F v'_F(\vartheta^F) \omega^F dm. \quad (\text{C.12})$$

Consequently,

$$dQ^F = d\mathcal{V}^F - d\bar{\Delta e} = d\mathcal{V}^F, \quad d \log A^{F,0} = \kappa_A \left(1 - \frac{A^{F,0}}{\mathcal{A}^F} \right) dQ^F, \quad (\text{C.13})$$

where the second equality uses $\bar{\Delta e}_\psi = 0$ from Lemma C.1.

Currency specialists and currency-risk clearing. The Foreign dollar allocation $A_\psi^{F,0}$ creates demand for dollar exposure and remains fixed within the regime. It is therefore distinct from the account's fluctuating marked value $V_{t|\psi}^F$. At regime prices, (C.8) and (44) imply

$$A_\psi^{F,0} = \bar{P}_\psi N_\psi^F + \bar{q}_{B,\psi} B_\psi^F.$$

Let $Z^e(\bar{e}_\psi)$, with $Z_e \equiv \partial Z^e / \partial \bar{e}_\psi < 0$, be the reduced-form component of Foreign customers' net short-dollar currency exposure. The disturbance μ_t^e shifts this exposure. After accounting for actual dollar assets held by Foreign investors, the net exposure offered to specialists is

$$\mathcal{S}_t^e \equiv Z^e(\bar{e}_\psi) + \mu_t^e - A_\psi^{F,0}. \quad (\text{C.14})$$

This is a customer hedging schedule, not a supply of claims generated by a Home budget.

Let e_t be the log dollar price of the Foreign consumption basket, so an increase in e_t is a dollar depreciation. A unit long-dollar currency contract has zero value at date t and a date- $t+1$ payoff, in Foreign consumption units and per unit of dollar-equivalent notional, proportional to

$$\Pi_{t+1}^e \equiv \exp\{r_t - \bar{r}^* + e_t - e_{t+1}\} - 1. \quad (\text{C.15})$$

No separate dollar funding rate is introduced: r_t and $\bar{r}^*$ are the existing pecuniary returns in the two currencies. To second order, the conditional mean payoff is $r_t - \bar{r}^* + e_t - \mathbb{E}_t e_{t+1} + x_{e,\psi}/2$, and its conditional variance is locally $x_{e,\psi}$.

Let D_t^e be the specialists' endogenous notional, positive when they are long dollar risk. Given risk-bearing capacity $\alpha_e > 0$, their conditional mean–variance objective is, up to an additive constant,

$$D_t^e \left[r_t - \bar{r}^* + e_t - \mathbb{E}_t e_{t+1} + \frac{x_{e,\psi}}{2} \right] - \frac{x_{e,\psi}}{2\alpha_e} (D_t^e)^2.$$

Its first-order condition gives

$$D_t^e = \frac{\alpha_e}{x_{e,\psi}} \left[r_t - \bar{r}^* + e_t - \mathbb{E}_t e_{t+1} + \frac{x_{e,\psi}}{2} \right]. \quad (\text{C.16})$$

Foreign customers take the opposite contract position $D_t^{e,c} = -D_t^e$. Currency-risk clearing requires specialists to absorb the net customer exposure:

$$D_t^e = S_t^e = Z^e(\bar{e}_\psi) + \mu_t^e - A_\psi^{F,0}, \quad D_t^e + D_t^{e,c} = 0. \quad (\text{C.17})$$

A positive realization of μ_t^e raises the short-dollar exposure that specialists absorb and puts depreciation pressure on the dollar. Substitution gives (46). The specialist and customer settlements are $D_t^e \Pi_{t+1}^e$ and $-D_t^e \Pi_{t+1}^e$, respectively, and therefore cancel exactly. The contract neither changes the beneficial ownership recorded in (N^F, B^F) nor creates an additional service-bearing asset. Its effect on ownership is only indirect, through e_t , Q^F , $A^{F,0}$, and the portfolio allocation.

Absorbing the regime-level Jensen term and structural imbalance into the linearization intercept gives

$$\mu_t^e = \frac{\alpha_e}{x_e} [\hat{r}_t + e_t - \mathbb{E}_t e_{t+1}], \quad (\text{C.18})$$

or, equivalently,

$$e_t = \mathbb{E}_t e_{t+1} - \hat{r}_t + \frac{x_e}{\alpha_e} \mu_t^e. \quad (\text{C.19})$$

The endogenous FX price impact is x_e/α_e .

Home's external asset, budget, and retained services. Let $P_{A,t}$ be the dollar ex-cash price of Home's foreign asset and $Q_{A,t}^H$ its post-transaction quantity. The inherited marked position and end-of-period position are

$$A_t^{H,-} \equiv P_{A,t} Q_{A,t-1}^H, \quad A_t^{H,+} \equiv P_{A,t} Q_{A,t}^H. \quad (\text{C.20})$$

Lowercase $a_t^{H,-}$ and $a_t^{H,+}$ denote log deviations from the stationary position. The acquisition component in (66) holds prices and current-account flows fixed. It is an accounting decomposi-

tion of a comparison across policy regimes, not a calendar-time shock.

The consolidated Home resource constraint records only beneficially owned claims:

$$C_t + P_t N_t^H + q_{B,t} B_t^H + A_t^{H,+} = (P_t + d_{P,t}) N_{t-1}^H + B_{t-1}^H + A_t^{H,-} + I_{A,t}^H, \quad (\text{C.21})$$

where $I_{A,t}^H = \iota_A A_t^{H,-}$ is cash income on the inherited foreign asset, with deterministic yield $\iota_A \geq 0$. No fiscal transfer appears: under the maintained fiscal convention, the Treasury rollover flow accrues to hand-to-mouth workers as in Section 2. Acquisitions of equity and Treasuries are routed through $A_t^{H,+}$ by the financial account and do not augment household cash on hand available for current consumption, which is

$$W_t^H \equiv (P_t + d_{P,t}) N_{t-1}^H + B_{t-1}^H + (1 + \iota_A) A_t^{H,-}, \quad (\text{C.22})$$

and the consumption rule (A.4) of Appendix A.1 applies to W_t^H :

$$C_t^H = (1 - \beta) W_t^H - \lambda_{T,t}^S S_t^H. \quad (\text{C.23})$$

Within a fixed regime, ownership is stationary: $N_{t-1}^H = N_t^H = N^H$ and $B_{t-1}^H = B_t^H = B^H$. Current cash on hand values dividends, principal, and equity wealth on inherited Home-owned holdings, while current services accrue on post-trade holdings; under stationary ownership both are evaluated at (N^H, B^H) . Foreign acquisition proceeds enter $A_t^{H,+}$ and never appear in W_t^H .

Because the collateral right travels with beneficial equity ownership, ownership clearing implies

$$P_t N^H = P_t - P_t N^F, \quad q_{B,t} B^H = q_{B,t} \bar{B} - q_{B,t} B^F. \quad (\text{C.24})$$

It follows that

$$\begin{aligned} S_t^H &= q_{B,t} B^H + m P_t N^H \\ &= q_{B,t} \bar{B} + m P_t - (q_{B,t} B^F + m P_t N^F) = q_{B,t} \bar{B} + X_t - F_t, \end{aligned} \quad (\text{C.25})$$

where $X_t \equiv m P_t$ is aggregate usable equity collateral value and $F_t \equiv q_{B,t} B^F + m P_t N^F$. This proves (47). Regime-level acquisition value is distinct from the marked portfolio value within the regime.

Define retained Home service shares in the regime-level allocation by

$$\varrho_P^H \equiv \frac{m \bar{P} N^H}{\bar{S}^H}, \quad \varrho_T^H \equiv \frac{\bar{q}_B B^H}{\bar{S}^H}, \quad \varrho_P^H + \varrho_T^H = 1. \quad (\text{C.26})$$

Because claim quantities and m are fixed within the regime,

$$s_t^H \equiv \log(S_t^H / \bar{S}^H) = \varrho_P^H p_t - \varrho_T^H \hat{r}_t. \quad (\text{C.27})$$

Consumption, inverse service demand, and the open FCI. Substituting stationary ownership, the retained-service allocation (C.25), and $d_{P,t} = \nu Y_t$ into the consumption rule (C.23) gives the open consumption level

$$\begin{aligned} C_t^H &= (1 - \beta)\nu Y_t N^H + (1 - \beta - \lambda_{P,t})P_t N^H \\ &\quad + (1 - \beta)B^H - \lambda_{T,t}^S q_{B,t} B^H + (1 - \beta)(1 + \iota_A)A_t^{H,-}. \end{aligned} \quad (\text{C.28})$$

Pecuniary cash on hand is valued at inherited holdings and services at post-trade holdings; within the regime both equal (N^H, B^H) . Linearizing around the regime-level allocation gives

$$c_t^H = \chi_y^o y_t + \chi_p^o p_t + \chi_r^o \hat{r}_t - \chi_\ell^o \ell_t + \chi_a^o a_t^{H,-}, \quad (\text{C.29})$$

with

$$\begin{aligned} \chi_y^o &\equiv \frac{(1 - \beta)\nu \bar{Y} N^H}{\bar{C}^H}, & \chi_p^o &\equiv \frac{(1 - \beta - \bar{\lambda}_P)\bar{P} N^H}{\bar{C}^H}, & \chi_r^o &\equiv \frac{\bar{\lambda}_T^S \bar{q}_B B^H}{\bar{C}^H}, \\ \chi_\ell^o &\equiv \frac{\bar{\lambda}_T^S \bar{S}^H}{\bar{C}^H}, & \chi_a^o &\equiv \frac{(1 - \beta)(1 + \iota_A)\bar{A}^{H,-}}{\bar{C}^H}. \end{aligned} \quad (\text{C.30})$$

This is the primitive optimized consumption response implied by the open budget. The closed-economy identity $c_t = \chi_y y_t + \chi_r \hat{r}_t$ of Appendix A.1 emerges only after closed ownership and the closed accounting identities eliminate the remaining variables; it cannot be imported into the open system directly. The acquisition component creates no net wealth by construction, while subsequent income and relative valuation changes affect spending through $a_t^{H,-}$.

Inverse service demand depends on S_t^H / C_t^H , so $\ell_t = \gamma_S(c_t^H - s_t^H)$ with s_t^H given by (C.27). Because (C.29) itself carries the service-cost term $-\chi_\ell^o \ell_t$, solving for ℓ_t gives

$$\ell_t = \Lambda_y^o y_t + \Lambda_p^o p_t + \Lambda_r^o \hat{r}_t + \Lambda_a^o a_t^{H,-}, \quad (\text{C.31})$$

where

$$\begin{aligned} \mathcal{D}_C^o &\equiv 1 + \gamma_S \chi_\ell^o > 0, & \Lambda_y^o &\equiv \frac{\gamma_S \chi_y^o}{\mathcal{D}_C^o}, & \Lambda_p^o &\equiv \frac{\gamma_S(\chi_p^o - \varrho_P^H)}{\mathcal{D}_C^o}, \\ \Lambda_r^o &\equiv \frac{\gamma_S(\chi_r^o + \varrho_T^H)}{\mathcal{D}_C^o}, & \Lambda_a^o &\equiv \frac{\gamma_S \chi_a^o}{\mathcal{D}_C^o}. \end{aligned} \quad (\text{C.32})$$

Unlike the closed-economy row, the corrected inverse-demand row carries both the direct

equity-wealth response χ_p^o and the direct service-cost response χ_ℓ^o .

Let $\mathcal{X}(\mathcal{E}_t, Y_t^s)$ denote foreign expenditure on U.S. goods and $\mathcal{M}(\mathcal{E}_t)$ Home imports, with $\mathcal{E}_t = \exp(e_t)$; under the maintained import convention, imports carry no C_t^H margin. Normalize the direct net-export contribution to $\phi(e_t + y_t^s)$. Hand-to-mouth consumption keeps its Section 2 form, $C_t^{HM} = (1 - \nu)Y_t + q_{B,t}\bar{B} - \bar{B}$. Combining (C.28), hand-to-mouth spending, net exports, and goods-market clearing $Y_t = C_t^H + C_t^{HM} + NX_t + \Delta_t$, and using $\bar{B} = B^H + B^F$, yields the exact identity

$$\begin{aligned} \Theta_H \nu Y_t &= (1 - \beta - \lambda_{P,t})P_t N^H + \left[q_{B,t}(1 - \lambda_{T,t}^S) - \beta \right] B^H + (q_{B,t} - 1)B^F \\ &\quad + (1 - \beta)(1 + \iota_A)A_t^{H,-} + NX_t + \Delta_t, \quad \Theta_H \equiv \beta + (1 - \beta)N^F. \end{aligned} \quad (\text{C.33})$$

Home receives dividends and price value only on its N^H shares, retains services only on Home-owned holdings, and the interest on Foreign-held Treasuries leaves the domestic private sector through the $(q_{B,t} - 1)B^F$ term. At $N^F = 0$, $\Theta_H = \beta$ and (C.33) reduces to the closed-economy condition (21). Linearizing and dividing by $\Theta_H \nu \bar{Y}$ gives

$$y_t = \phi(e_t + y_t^s) + g_p^o p_t - g_r^o \hat{r}_t - H^o \ell_t + g_a^o a_t^{H,-} + \delta_t, \quad (\text{C.34})$$

where $\delta_t \equiv \Delta_t / (\Theta_H \nu \bar{Y})$ and the level-accounting coefficients are

$$\begin{aligned} g_p^o &\equiv \frac{(1 - \beta - \bar{\lambda}_P) \bar{P} N^H}{\Theta_H \nu \bar{Y}} > 0, & g_r^o &\equiv \frac{\beta B^H + \bar{q}_B B^F}{\Theta_H \nu \bar{Y}} > 0, \\ H^o &\equiv \frac{\bar{\lambda}_T^S \bar{S}^H}{\Theta_H \nu \bar{Y}} > 0, & g_a^o &\equiv \frac{(1 - \beta)(1 + \iota_A) \bar{A}^{H,-}}{\Theta_H \nu \bar{Y}} \geq 0. \end{aligned} \quad (\text{C.35})$$

These are the ownership-accounting coefficients under stationary inherited ownership: only Home-owned equity $\bar{P} N^H$ enters the contemporaneous equity-wealth coefficient, and the normalization $\Theta_H \nu \bar{Y}$ reflects the dividend flow on Foreign-owned shares that leaves Home demand. No residual cash-flow state remains: the fiscal transfer is zero under the maintained convention, and the currency-risk contract cancels within Foreign.

Eliminating ℓ_t with (C.31) gives

$$y_t = \bar{\phi}(e_t + y_t^s) + \bar{g}_p^o p_t - \bar{g}_r^o \hat{r}_t + \bar{g}_a^o a_t^{H,-} + \bar{\delta}_t, \quad (\text{C.36})$$

where

$$\begin{aligned} \mathcal{D}_G^o &\equiv 1 + H^o \Lambda_y^o > 0, & \bar{\phi} &\equiv \frac{\phi}{\mathcal{D}_G^o}, & \bar{g}_p^o &\equiv \frac{g_p^o - H^o \Lambda_p^o}{\mathcal{D}_G^o}, \\ \bar{g}_r^o &\equiv \frac{g_r^o + H^o \Lambda_r^o}{\mathcal{D}_G^o} > 0, & \bar{g}_a^o &\equiv \frac{g_a^o - H^o \Lambda_a^o}{\mathcal{D}_G^o}, & \bar{\delta}_t &\equiv \frac{\delta_t}{\mathcal{D}_G^o}. \end{aligned} \quad (\text{C.37})$$

The raw coefficients in (C.35) have the stated signs; the reduced price and external-wealth coefficients should be signed only after evaluating (C.37), because service-demand feedback can partially offset the direct spending effects. Equations (C.34) and (C.31) deliver the open FCI summarized in (40)–(41); raw and reduced representations should not be mixed.

External-account state and actual claims. Let $\mathcal{T}_{P,t}$ and $\mathcal{T}_{T,t}$ denote current-price Foreign acquisitions of U.S. equity and Treasuries. The external asset obeys

$$A_t^{H,+} - A_t^{H,-} = CA_t + \mathcal{T}_{P,t} + \mathcal{T}_{T,t}. \quad (\text{C.38})$$

Within a fixed regime, N^F is fixed and $\mathcal{T}_{P,t} = 0$; Treasury rollover remains a financial transaction and records new issuance net of principal repayment. Appendix C.7 derives this identity from the balance of payments.

For the benchmark, the foreign-currency ex-cash price of Home's external asset has no primitive innovation. Let $\omega_A^e \in [0, 1]$ be its foreign-currency share and define the post-transaction, currency-stripped state

$$\mathbf{a}_t \equiv a_t^{H,+} - \omega_A^e e_t. \quad (\text{C.39})$$

Then, up to a deterministic return intercept,

$$a_t^{H,-} = \mathbf{a}_{t-1} + \omega_A^e e_t, \quad (\text{C.40})$$

and

$$a_{t+1}^{H,-} - \mathbb{E}_t a_{t+1}^{H,-} = \omega_A^e (e_{t+1} - \mathbb{E}_t e_{t+1}). \quad (\text{C.41})$$

Normalizing current-account and transaction flows by steady-state external assets gives

$$\mathbf{a}_t = \mathbf{a}_{t-1} + \widehat{CA}_t + \widehat{\mathcal{T}}_{P,t} + \widehat{\mathcal{T}}_{T,t}. \quad (\text{C.42})$$

The date- t income and transaction components of the one-period Treasury claim are

$$I_{T,t}^F \equiv (1 - q_{B,t-1}) B_{t-1}^F, \quad \mathcal{T}_{T,t} \equiv q_{B,t} B_t^F - q_{B,t-1} B_{t-1}^F. \quad (\text{C.43})$$

Combining interest and rollover gives

$$-I_{T,t}^F + \mathcal{T}_{T,t} = q_{B,t} B_t^F - B_{t-1}^F. \quad (\text{C.44})$$

Thus lagged Treasury prices cancel. The currency-risk contract contributes no Home income or financial transaction because its two Foreign-side settlements sum to zero. The current account is a linear function of current equilibrium variables and inherited actual positions, as made

explicit in Appendix C.7.

Formal equilibrium definition.

Definition 2 (Open financial equilibrium). *Given the Section 2 domestic primitives, the Section 3 policy protocol, foreign demand state y_t^g , constant foreign safe rate $\bar{r}^*$, external-asset yield ι_A , Foreign dollar-demand parameters $(\mathcal{A}^F, \chi_A, \kappa_A)$, Foreign customer currency-exposure schedule Z^e , and inherited external positions, an open financial equilibrium at targeting intensity ψ consists of a regime-level allocation and a stochastic allocation within the regime such that:*

- (i) *domestic households, managers, firms, and the fiscal authority satisfy Definition 1, with the Home resource constraint, household cash on hand, and services given by (C.21), (C.22)–(C.23), and (47), and with (13) replaced by (C.4);*
- (ii) *goods demand and policy satisfy (40)–(42) and (48);*
- (iii) *Foreign portfolios solve (C.2), dollar allocation satisfies (C.7), and ownership clears according to (C.8) and (44);*
- (iv) *Foreign currency specialists satisfy (C.16), customers take the opposite position, and currency risk clears according to (C.17);*
- (v) *external saving satisfies (C.38); and*
- (vi) *$x_{p,\psi}$, $x_{e,\psi}$, and $c_{pe,\psi}$ equal the two variances and covariance generated by the equilibrium laws of motion.*

All regime-level objects are solved jointly as functions of ψ . Within a fixed regime, foreign-owned quantities and regime-level dollar exposures remain fixed while marked values vary.

C.2 The Joint Linear Equilibrium and Risk Map

The main text imposes local symmetry to obtain a sharp analytical risk result. This section presents the general asymmetric system with distinct equity-return and exchange-rate variances and an endogenous covariance.

Equity pricing. Equation (C.4) adds $(x_{p,\psi}/\alpha_p)N_\psi^F$ to the regime-level equity-price equation. Since $1 - \omega_{P,\psi}^o = \omega_{D,\psi}^o$, its direct structural-order component can be written

$$\log \bar{P}_\psi = p_\psi^0 + \Lambda_\psi N_\psi^F, \quad \Lambda_\psi = \frac{x_{p,\psi}}{\alpha_p \omega_{D,\psi}^o} > 0. \quad (\text{C.45})$$

Here p_ψ^0 is the contribution of all other equilibrium-price equations at the same allocation; it is not the counterfactual price obtained by setting foreign ownership to zero. In particular, it retains the effects of foreign ownership on services, external wealth, expected payoffs, and the regime-level coefficients.

Foreign-owned quantities are predetermined within a fixed ψ -regime, so Home managers accommodate high-frequency equity noise around their residual regime-level position. Demeaning around the regime-level allocation absorbs the structural term in (C.45). It therefore leaves the local noise-clearing slope x_p/α_p unchanged, and the within-regime equity-price equation remains

$$p_t = \omega_D^0 \mathbb{E}_t p_{t+1} + \omega_P^0 \mathbb{E}_t y_{t+1} - \hat{r}_t + \frac{x_p}{\alpha_p} \mu_t^p + \zeta^0 \ell_t, \quad (\text{C.46})$$

where $\omega_D^0, \omega_P^0 > 0$, $\omega_D^0 + \omega_P^0 = 1$, and

$$\zeta^0 \equiv \bar{\lambda}_P - \bar{\lambda}_T^S.$$

The regime-specific $x_p/2$ term is part of the balanced-growth mean return and disappears when the price equation is demeaned around the regime-level allocation. The regime-level payoff weights are the primitive dividend and price shares

$$\omega_D^0 \equiv \frac{\nu \bar{Y}}{\bar{P} + \nu \bar{Y}}, \quad \omega_P^0 \equiv \frac{\bar{P}}{\bar{P} + \nu \bar{Y}}. \quad (\text{C.47})$$

Closed-form substitutions of the closed-economy steady-state relation, as in the second equalities of (A.17), are not available here: the stationary open goods-market level (C.33) ties $\nu \bar{Y}/\bar{P}$ to ownership and the external balance, so the weights are kept in primitive form. At $N^F = 0$ with a zero external balance they reduce to (A.17).

Primitive states and policy. Let

$$\begin{aligned} y_t^* &= y_{t-1}^* + \varepsilon_{z,t}, & \delta_t &= \varphi_\delta \delta_{t-1} + \varepsilon_{\delta,t}, \\ y_t^g &= \varphi_g y_{t-1}^g + \varepsilon_{g,t}, & \mu_t^j &= \varphi_j \mu_{t-1}^j + \varepsilon_{\mu^j,t}, \quad j \in \{p, e\}. \end{aligned} \quad (\text{C.48})$$

The three macroeconomic innovations are mutually independent and independent of the financial-noise pair. The two financial-noise innovations may be correlated; Σ_ε denotes the resulting covariance matrix. The foreign safe rate is constant at fixed ψ and therefore adds no state or shock. Gap-closing-FCI news is

$$\varepsilon_t^* = \varepsilon_{z,t} - \phi \varepsilon_{g,t} - \varepsilon_{\delta,t}. \quad (\text{C.49})$$

The policy row is (48). The maintained spanning condition is

$$\kappa_f^o \equiv -\frac{\partial \mathbb{E}_t f_t^o}{\partial \hat{r}_t} > 0, \quad (\text{C.50})$$

allowing the contemporaneous exchange rate, asset price, and service wedge to clear. Holding expected future variables fixed and allowing all current markets to clear, define the remaining total policy derivatives by

$$\kappa_p^o \equiv -\frac{\partial p_t}{\partial \hat{r}_t}, \quad \kappa_e^o \equiv -\frac{\partial e_t}{\partial \hat{r}_t}. \quad (\text{C.51})$$

These derivatives are generated by the joint open-economy system; they are not copied from the closed economy. When the currency and external-demand blocks are shut down, $\kappa_p^o = 1$ and $\kappa_f^o = 1 + b$, recovering (B.13) and (B.14).

Augmented coefficient system. For a candidate $\mathbf{x} = (x_p, x_e)'$, collect current prices and output in

$$\mathbf{q}_t \equiv (p_t, e_t, \hat{r}_t, \ell_t, y_t)'$$

and the predetermined states in

$$\mathbf{s}_t \equiv (y_t^*, \delta_t, y_t^g, \mu_{t-1}^p, \mu_{t-1}^e, \mathbf{a}_{t-1})'.$$

Substitute (C.40) into reduced goods demand and inverse service demand. The five current equations are:

- (i) reduced goods demand (C.36);
- (ii) inverse service demand (C.31);
- (iii) equity pricing (C.46);
- (iv) currency clearing (C.19); and
- (v) policy (48).

Together they have the form

$$\mathbf{M}_0(\mathbf{x})\mathbf{q}_t = \mathbf{M}_1(\mathbf{x})\mathbb{E}_t\mathbf{q}_{t+1} + \mathbf{M}_s(\mathbf{x}; \psi)\mathbf{s}_t + \mathbf{M}_\mu(\mathbf{x}) \begin{pmatrix} (x_p/\alpha_p)\varepsilon_{\mu^p,t} \\ (x_e/\alpha_e)\varepsilon_{\mu^e,t} \end{pmatrix}. \quad (\text{C.52})$$

The external-account law supplies the state-transition row

$$\mathbf{a}_t = \mathbf{R}_q\mathbf{q}_t + \mathbf{R}_s\mathbf{s}_t + \mathbf{R}_\mu \begin{pmatrix} \varepsilon_{\mu^p,t} \\ \varepsilon_{\mu^e,t} \end{pmatrix}. \quad (\text{C.53})$$

The row in (C.53) is obtained from net exports, net investment income, the Treasury identity in (C.44), and (C.42). No saving or external-wealth state equation is left implicit, and currency-risk clearing creates no repayment state.

Stacking (C.52), (C.53), and the primitive state processes gives a standard linear rational-expectations system. We maintain that its current block is nonsingular, its stable generalized solution is unique, and $\kappa_f^o > 0$:

$$\det M_0 \neq 0, \quad \text{the augmented system has a unique stable solution,} \quad \kappa_f^o > 0. \quad (\text{C.54})$$

These are local regularity conditions on the maintained joint equilibrium branch; they do not assert global existence or uniqueness.

Local equilibrium construction. The stable solution can be written

$$\mathbf{q}_t = \mathbf{A}(\mathbf{x}; \psi) \mathbf{s}_t + \mathbf{C}(\mathbf{x}; \psi) \begin{pmatrix} \varepsilon_{\mu^p, t} \\ \varepsilon_{\mu^e, t} \end{pmatrix}, \quad (\text{C.55})$$

with $\mathbf{a}_t$ following from (C.53). Let a superscript $+$ denote the next-date innovation row generated by this augmented solution. The return and exchange-rate loadings used in (C.57) are

$$\begin{aligned} B_p &\equiv \omega_D^o B_y^+ + \omega_p^o B_p^+, \\ B_e &\equiv B_e^+. \end{aligned} \quad (\text{C.56})$$

Let B_y denote the two financial-noise columns of the output-gap row and define

$$\boldsymbol{\varepsilon}_t \equiv (\varepsilon_{z, t}, \varepsilon_{\delta, t}, \varepsilon_{g, t}, \varepsilon_{\mu^p, t}, \varepsilon_{\mu^e, t})'.$$

The asymmetric return and output-gap system is

$$\begin{pmatrix} r_{p, t+1}^x - \mathbb{E}_t r_{p, t+1}^x \\ e_{t+1} - \mathbb{E}_t e_{t+1} \end{pmatrix} = \begin{pmatrix} B_p \\ B_e \end{pmatrix} \boldsymbol{\varepsilon}_{t+1}, \quad (\text{C.57})$$

$$\tilde{y}_t = -h_\psi \varepsilon_t^* + B_y \begin{pmatrix} \varepsilon_{\mu^p, t} \\ \varepsilon_{\mu^e, t} \end{pmatrix}. \quad (\text{C.58})$$

Let $\mathbf{x} \equiv (x_p, x_e)'$. Risk consistency requires

$$\mathbf{x} = \boldsymbol{\Phi}^o(\mathbf{x}; \psi) \equiv \begin{pmatrix} B_p \Sigma_\varepsilon B_p' \\ B_e \Sigma_\varepsilon B_e' \end{pmatrix}, \quad (\text{C.59})$$

while the same loadings determine the covariance:

$$c_{pe} = B_p \Sigma_e B_e'. \quad (\text{C.60})$$

These formulas define the reduced risk map after the regime-level dollar-allocation, ownership, pledgeability, and pricing equations have been substituted. For each candidate (x_p, x_e) , covariance consistency in (C.60) is imposed jointly with those regime-level equations, and the resulting c_{pe} is passed into (C.2). Covariance is therefore a computed equilibrium moment rather than an omitted portfolio input. The reduced two-dimensional map incorporates its feedback through Foreign demand. Implicit differentiation gives

$$\frac{dx}{d\psi} = (I - \Phi_x^o)^{-1} \Phi_\psi^o.$$

Unlike the symmetric benchmark, the general system does not impose a common sign on the two variance responses.

Under (C.54), the augmented system has the stable local solution (C.55) for a candidate x . The external-account transition introduces no primitive shock, so every innovation is a loading on ε_t . Policy (48) leaves the fraction h_ψ of gap-closing-FCI news in the output gap. The current rate cannot respond to the last-moving innovations $(\varepsilon_{\mu^p,t}, \varepsilon_{\mu^e,t})$; their output loadings are the corresponding columns of the stable solution. This yields (C.58). Equation (C.56) gives the return system, and the three covariance calculations give (C.59) and (C.60).

Primitive policy loadings. At fixed candidate regime-level risk and regime-level coefficients, the policy rule moves the gap-closing-FCI innovation $\varepsilon_z - \phi\varepsilon_g - \varepsilon_\delta$ with intensity h_ψ . The total Jacobian in (C.51) maps this adjustment into the equity and currency returns. For $j \in \{p, e\}$, define $\gamma_{h,j} > 0$ as the resulting return response to one unit of that policy adjustment. Let $(\gamma_{z,j}, \gamma_{\delta,j}, \gamma_{g,j})$ be the return loadings under discretion and let $n_{j,t}$ collect the two last-moving financial-noise innovations. Then

$$u_{j,t} = (\gamma_{z,j} - \gamma_{h,j}h_\psi)\varepsilon_{z,t} + (\gamma_{\delta,j} + \gamma_{h,j}h_\psi)\varepsilon_{\delta,t} + (\gamma_{g,j} + \phi\gamma_{h,j}h_\psi)\varepsilon_{g,t} + n_{j,t}, \quad \gamma_{h,j} > 0. \quad (\text{C.61})$$

Since macroeconomic innovations are independent of financial noise,

$$\mathcal{N}_j(x) \equiv \text{Var}(n_{j,t})$$

is additively separable from the three macroeconomic variance terms. Holding regime-level risk and coefficients fixed gives the direct variance map

$$V_j(h) = (\gamma_{z,j} - \gamma_{h,j}h)^2\sigma_z^2 + (\gamma_{\delta,j} + \gamma_{h,j}h)^2\sigma_\delta^2 + (\gamma_{g,j} + \phi\gamma_{h,j}h)^2\sigma_g^2 + \mathcal{N}_j(x). \quad (\text{C.62})$$

This construction carries the total current-market policy response into both return rows. In the closed-economy limit, $\kappa_p^o = 1$, $\kappa_f^o = 1 + b$, and $\gamma_{h,p} = \omega_D + \omega_P / (1 + b)$. Appendix C.3 evaluates these loadings in closed form under the symmetric benchmark and proves Proposition 5.

Nested cases. If $\sigma_{\mu^e}^2 = 0$, the current FX-noise column disappears. If in addition $\phi = 0$, $\omega_A^e = 0$, and Foreign dollar exposure is locally inelastic ($\kappa_A = 0$), the currency block is triangular. Setting foreign ownership to zero then makes service shares, spending coefficients, and payoff weights converge to the closed-economy objects in Sections 3–4.

C.3 Regime-Level Stationarity and the Symmetric Benchmark

This subsection establishes two results. First, regime-level expected depreciation is zero within every fixed regime; this closes the dollar-participation and currency-clearing block. Second, the symmetric risk system (49)–(51) is derived, rather than imposed, from the primitive shocks, the policy protocol, and an explicit set of symmetry restrictions. The construction follows the random-walk-anchor equilibrium of Caballero and Simsek (2026), adapted to the FCI protocol of Section 4.1 and to the presence of dividends, Treasury wealth, and the collateral-service block.

Regime-level stationarity.

Lemma C.1 (No expected depreciation within a regime). *Fix ψ and let the within-regime equilibrium be the stable solution (C.55). Then regime-level expected depreciation is zero:*

$$\overline{\Delta e_\psi} = 0 \quad \text{for every } \psi. \quad (\text{C.63})$$

Proof. Evaluate $\mathbb{E}_t(e_{t+1} - e_t)$ at the regime-level allocation, where every deviation state and every current innovation is zero. Because the date- t noise term of (C.55) vanishes at that point, the e -row A_e of the stable solution gives $\mathbb{E}_t(e_{t+1} - e_t) = A_e(\mathbb{E}_t \mathbf{s}_{t+1} - \mathbf{s}_t)$. Potential output is a driftless random walk, so its component of $\mathbb{E}_t \mathbf{s}_{t+1} - \mathbf{s}_t$ is zero at every point. Each stationary component $s \in \{\delta_t, y_t^g, \mu_{t-1}^p, \mu_{t-1}^e\}$ satisfies $\mathbb{E}_t s_{t+1} - s_t = (\varphi - 1)s_t = 0$ at the regime-level point. The external state obeys (C.53), whose right-hand side is evaluated at the stationary external position and therefore contributes no expected change. Hence $\mathbb{E}_t \mathbf{s}_{t+1} - \mathbf{s}_t = \mathbf{0}$, which proves (C.63). $\square$

Lemma C.1 closes the block left open by (C.6) in three steps. First, the currency-adjusted and noncurrency account values coincide:

$$Q_\psi^F = V_\psi^F, \quad \frac{dQ^F}{d\psi} = \frac{dV^F}{d\psi}, \quad (\text{C.64})$$

so Foreign participation responds only to returns, services, risk, and pledgeability. Second, regime-level currency clearing—the zero-noise value of (C.16) equated with $Z^e(\bar{e}_\psi) - A_\psi^{F,0}$ —becomes

$$Z^e(\bar{e}_\psi) - A_\psi^{F,0} = \alpha_e \left[\frac{\bar{r}_\psi - \bar{r}^*}{x_{e,\psi}} + \frac{1}{2} \right], \quad (\text{C.65})$$

which determines the regime-level exchange rate: given $(x_\psi, \bar{r}_\psi, J_\psi^F)$, the account value $\mathcal{V}_\psi^F$ determines $A_\psi^{F,0}$ through (C.7), and $Z_e < 0$ then pins $\bar{e}_\psi$ uniquely. The regime-level dollar block is therefore recursive rather than underdetermined. Third, comparisons across ψ are comparisons of stationary allocations: they shift the level $\bar{e}_\psi$ but introduce no within-regime drift. An anticipated transition between regimes would add a one-time expected revaluation; that experiment lies outside the maintained stationary comparison.

The symmetric benchmark. Assumption (S^0) collects the restrictions under which the general system (C.52) reduces to the two-market symmetric block of Section 5.3. It adapts Assumption (S) of Caballero and Simsek (2026) to the present model.

- (i) *Symmetric intermediation and noise.* Risk-bearing capacities are equal, $\alpha_p = \alpha_e \equiv \alpha$, and financial noise is serially uncorrelated ($\varphi_p = \varphi_e = 0$) with equal innovation variances σ_μ^2 and correlation $\rho_\mu \in (-1, 1)$.
- (ii) *Random-walk anchors.* Potential output and foreign demand are driftless random walks ($\varphi_g \rightarrow 1, \sigma_\delta = 0$), so the two market anchors

$$p_t^* \equiv y_t^*, \quad c_t^* \equiv y_t^* - y_t^g$$

are random walks. Their innovations $z_t^p \equiv \varepsilon_{z,t}$ and $z_t^e \equiv \varepsilon_{z,t} - \varepsilon_{g,t}$ have equal variances σ_z^2 and correlation $\rho_z \in (-1, 1)$; this replaces the mutual independence of ε_z and ε_g maintained in (C.48) with a free correlation.

- (iii) *Equal demand weights.* $\phi = g_p^o = \frac{1}{2}$, so the financial-price block of the FCI is $(e_t + p_t)/2$ and gap-closing news is shared equally across markets, $\varepsilon_t^* = (z_t^p + z_t^e)/2$, consistent with (C.49).
- (iv) *Within-regime neutrality of the wedge and external blocks.* The service-wedge deviation ℓ_t and the external loadings (ω_A^e, g_a^o) are set to zero in the within-regime innovation block, so the wedge and the external account affect the risk system only through regime-level coefficients, which continue to determine $(m_\psi, N_\psi^F, \bar{\lambda}_{T,\psi}^S, \bar{P}_\psi, x_\psi)$ across regimes. This formalizes the statement in Section 5.3 that the nonpolicy feedback is exchangeable at the comparison point.

(v) *Price-dominant payoffs.* Return innovations are evaluated in the limit $\omega_D^o \rightarrow 0$, the open-economy counterpart of the $\beta \rightarrow 1$ normalization in [Caballero and Simsek \(2026\)](#); the exact loadings before the limit are displayed in (C.70), with the dividend contribution of order $\omega_D^o = \nu \bar{Y} / (\bar{P} + \nu \bar{Y})$, the dividend yield.

Anchored equilibrium and the policy response. Conjecture, and verify below, that within the regime expectations are centered on the anchors:

$$\mathbb{E}_t p_{t+1} = \mathbb{E}_t y_{t+1} = y_t^*, \quad \mathbb{E}_t e_{t+1} = e_t^*. \quad (\text{C.66})$$

Date- t macroeconomic news (z_t^p, z_t^e) arrives first, the policy rate is set next, and financial noise arrives last. Substituting (C.66) into the demeaned equity-price equation (C.46) under (S^o) (iv), and into the currency row (C.19), the pre-noise (conditional) innovations satisfy

$$\tilde{p}_t = z_t^p - \tilde{r}_t, \quad \tilde{e}_t = z_t^e - \tilde{r}_t. \quad (\text{C.67})$$

The identity $\omega_D^o + \omega_P^o = 1$ delivers the equity row for any ω_D^o : expected dividends and expected prices load on the same anchor, so the payoff weights sum out of the conditional response. Under (S^o) (iii)–(iv) the conditional FCI innovation is

$$\tilde{f}_t = \frac{\tilde{e}_t + \tilde{p}_t}{2} - g_r^o \tilde{r}_t = \varepsilon_t^* - (1 + g_r^o) \tilde{r}_t.$$

The protocol (48) sets $\tilde{f}_t = (1 - h_\psi) \varepsilon_t^*$, so

$$\tilde{r}_t = \gamma_h h_\psi \varepsilon_t^*, \quad \gamma_h \equiv \frac{1}{1 + g_r^o} \in (0, 1], \quad (\text{C.68})$$

and the conditional prices are $\tilde{p}_t = z_t^p - \gamma_h h_\psi \varepsilon_t^*$ and $\tilde{e}_t = z_t^e - \gamma_h h_\psi \varepsilon_t^*$. Under discretion ($h_\psi = 0$) the rate does not move: fundamental news passes through prices one for one and closes the gap by itself. Targeting leans against current news to defend the preannounced target, and the direct tightening of the FCI by the rate itself (the g_r^o term) scales down the required adjustment.

Financial noise then moves each market along its own demand slope and moves output through the FCI:

$$\hat{p}_t = \tilde{p}_t + \frac{x_p}{\alpha} \varepsilon_{\mu^p, t}, \quad \hat{e}_t = \tilde{e}_t + \frac{x_e}{\alpha} \varepsilon_{\mu^e, t}, \quad \tilde{y}_t = -h_\psi \varepsilon_t^* + \frac{1}{2} \left(\frac{x_p}{\alpha} \varepsilon_{\mu^p, t} + \frac{x_e}{\alpha} \varepsilon_{\mu^e, t} \right). \quad (\text{C.69})$$

Because noise arrives after the rate is set, noise is serially uncorrelated under (S^o) (i), and the target resets to $\mathbb{E}_t f_{t+1}^{o,*}$ each period, neither the current nor any future rate responds to financial noise. This is the same timing as the closed economy: the noise term $\mathcal{K}(x)x^2$ in (31) likewise

carries no h_ψ .

Taking date- t expectations of the date- $t + 1$ counterparts of (C.67)–(C.69) verifies (C.66): every date- $t + 1$ innovation has zero date- t mean, and no predetermined state enters under (S^o) . Evaluating $\mathbb{E}_t e_{t+1} - e_t$ at the regime-level point reproduces Lemma C.1 inside the benchmark.

Return innovations and the mixing coefficient. Substituting the solution into the loading definitions (C.56) gives the exact date- t return innovations

$$\begin{aligned} u_{p,t} &= z_t^p - \gamma_h^p h_\psi \varepsilon_t^* + \left(1 - \frac{\omega_D^o}{2}\right) \frac{x_p}{\alpha} \varepsilon_{\mu^p,t} + \frac{\omega_D^o}{2} \frac{x_e}{\alpha} \varepsilon_{\mu^e,t}, & \gamma_h^p &\equiv \omega_D^o + \frac{\omega_P^o}{1 + g_r^o}, \\ u_{e,t} &= z_t^e - \gamma_h^e h_\psi \varepsilon_t^* + \frac{x_e}{\alpha} \varepsilon_{\mu^e,t}. \end{aligned} \quad (\text{C.70})$$

The equity return mixes price and dividend news, so its policy pass-through γ_h^p and its noise loadings differ from the currency row by terms of order ω_D^o ; in the closed-economy limit γ_h^p reduces to the loading $\gamma_h = \omega_D + \omega_P/(1 + b)$ of (31). Exact exchange symmetry therefore requires the price-dominant limit $(S^o)(v)$. Setting $\omega_D^o \rightarrow 0$, defining the mixing coefficient

$$a_\psi \equiv \frac{\gamma_h h_\psi}{2} \in \left[0, \frac{1}{2}\right), \quad (\text{C.71})$$

and writing $\varepsilon_t^* = (z_t^p + z_t^e)/2$ delivers exactly (49). The two rows are exchangeable under the market swap $(z^p, \varepsilon_{\mu^p}, p) \leftrightarrow (z^e, \varepsilon_{\mu^e}, e)$, so the conditional variances coincide, $x_p = x_e \equiv x$. Taking conditional variances and the conditional covariance of (49) gives the scalar fixed point (50) with $\nu_z(\psi)$ as in (51), the constant noise weight σ_μ^2 , and

$$c_{pe} = \sigma_z^2 \left[\rho_z \left((1 - a_\psi)^2 + a_\psi^2 \right) - 2a_\psi(1 - a_\psi) \right] + \rho_\mu \sigma_\mu^2 \left(\frac{x}{\alpha} \right)^2. \quad (\text{C.72})$$

Policy mixes macroeconomic news across the two markets but does not mix financial noise; the noise contribution $\sigma_\mu^2 (x/\alpha)^2$ responds to policy only through the equilibrium price impact x/α —the recruitment channel.

Proof of Proposition 5. Hold the regime-level coefficient γ_h fixed at the local comparison, as in (C.62). Since $a_\psi = \gamma_h h_\psi/2$ and $h_\psi = \psi/(1 + \psi)$,

$$a'_\psi = \frac{\gamma_h}{2(1 + \psi)^2} > 0, \quad 0 \leq a_\psi < \frac{1}{2}.$$

Differentiating (51) gives

$$\nu'_z(\psi) = -2\sigma_z^2(1 + \rho_z)(1 - 2a_\psi)a'_\psi < 0,$$

where the strict inequality uses $\rho_z > -1$ and $a_\psi < 1/2$. Differentiating (50) and collecting the terms multiplying $dx/d\psi$ gives (52): the numerator $v'_z(\psi)$ is strictly negative, and the stable-smaller-root condition makes the denominator $1 - 2\sigma_\mu^2 x/\alpha^2$ positive. $\square$

Relation to Caballero and Simsek (2026). Assumption (S^o) parallels their Assumption (S): equal FCI weights, equal shock variances with free correlations, and a payoff normalization ($\omega_D^o \rightarrow 0$ here, $\beta \rightarrow 1$ there). Two differences reflect the present environment. First, the Treasury-wealth term $g_r^o \hat{r}_t$ in the open FCI scales the policy pass-through to $\gamma_h = 1/(1 + g_r^o)$; in their model the gradualism parameter θ plays the corresponding role. Second, their central bank adjusts the rate after observing noise-driven output gaps, so financial noise is partially offset and enters their variance map with the policy mixing weight. Under the protocol of Section 4.1, the rate is set before the noise realization and the target resets each period, so noise is unmixed and enters (50) with the constant weight σ_μ^2 , exactly as in the closed-economy map (31).

C.4 Foreign Demand and Service Comparative Statics

Delegated portfolio choice. Let the left side of (C.3) be

$$G(\omega; \mu, x, c, m) \equiv \mu - \alpha_F(x\omega - c) - \xi_F(1 - m)v'_F(\vartheta^F).$$

Its derivatives are

$$\begin{aligned} G_\omega &= -\alpha_F x + \xi_F(1 - m)^2 v''_F(\vartheta^F) < 0, \\ G_\mu &= 1, & G_x &= -\alpha_F \omega, & G_c &= \alpha_F, & (C.73) \\ G_m &= \xi_F \left[v'_F(\vartheta^F) - (1 - m)\omega v''_F(\vartheta^F) \right] > 0. \end{aligned}$$

Strict concavity makes the optimizer unique and the reduced map W single-valued. Increasing differences in (ω, μ) , (ω, c) , and (ω, m) , and decreasing differences in (ω, x) , imply that W is increasing in μ , c , and m and decreasing in x at an interior solution. Define

$$\mathcal{D}_\omega \equiv \frac{d\mu^F}{d\psi} - \alpha_F \omega^F \frac{dx_p}{d\psi} + \alpha_F \frac{dc_{pe}}{d\psi} + G_m \frac{dm}{d\psi}. \quad (C.74)$$

Implicit differentiation gives

$$\frac{d\omega^F}{d\psi} = -\frac{1}{G_\omega} \left[\frac{d\mu^F}{d\psi} - \alpha_F \omega^F \frac{dx_p}{d\psi} + \alpha_F \frac{dc_{pe}}{d\psi} + G_m \frac{dm}{d\psi} \right] = -\frac{\mathcal{D}_\omega}{G_\omega}. \quad (C.75)$$

Using (45),

$$\frac{d\mu^F}{d\psi} = \left(\frac{1}{2} - \frac{N^F}{\alpha_p}\right) \frac{dx_p}{d\psi} + (1-m)\lambda_T^S \frac{d\log \lambda_T^S}{d\psi} - m\lambda_T^S \frac{d\log m}{d\psi} - \frac{x_p N^F}{\alpha_p} \frac{d\log N^F}{d\psi}. \quad (\text{C.76})$$

Because $-1/G_\omega > 0$, the sign of the portfolio-share response is the sign of $\mathcal{D}_\omega$. Log differentiation of $E^F = \omega^F A^{F,0}$ and $N^F = E^F / \bar{P}$ gives (53), while (C.13) determines the bounded dollar-allocation term. These identities establish the three response margins reported in the main text; they do not impose a common equilibrium sign on portfolio share, dollar demand, and ownership.

Home service wedge and the external account. At the regime-level allocation, define

$$\omega_T^H \equiv \frac{\bar{q}_B \bar{B}}{S^H}, \quad \omega_X^H \equiv \frac{X}{S^H}, \quad \omega_F^H \equiv \frac{F}{S^H}, \quad (\text{C.77})$$

so $\omega_T^H + \omega_X^H - \omega_F^H = 1$. For comparisons across policy regimes at fixed mean output and $\bar{r}^{\text{pec}} = \rho$, let

$$L \equiv d\log \lambda_T^S, \quad \hat{X} \equiv d\log X, \quad \hat{F} \equiv d\log F.$$

Let $a_T > 0$ denote the regime-level Treasury-price elasticity, so Treasury pricing gives $d\log \bar{q}_B = a_T L$. Under the corrected consumption level (C.28), hold P , m , ownership, mean output, and the external position fixed while λ_T^S and q_B move with L . The direct consumption response and its complement are

$$d\log C^H = \kappa_L^H L + dc_\perp^H, \quad \kappa_L^H = -\frac{\bar{\lambda}_P \bar{P} N^H + \bar{\lambda}_T^S \bar{q}_B B^H (1 + a_T)}{\bar{C}^H} < 0, \quad (\text{C.78})$$

where the net consumption-demand component $dc_\perp^H$ collects the price, pledgeability, ownership, external-wealth, and income responses outside the direct L term. Under the maintained consumption rule, the acquisition component in (66) contributes no direct pecuniary term to $dc_\perp^H$ by construction; income, current-account flows, and relative valuation can contribute. This component is determined by the household problem, ownership clearing, and the external-account transition, not supplied as an additional shock. If ownership is fixed and the equity price and inherited external wealth are the only additional margins, then $dc_\perp^H = \chi_p^o d\log \bar{P} + \chi_a^o d\log A^H$, by (C.29). Since

$$d\log S^H = \omega_T^H a_T L + \omega_X^H \hat{X} - \omega_F^H \hat{F},$$

inverse service demand $L = \gamma_S (d\log C^H - d\log S^H)$ implies

$$L = \gamma_S \left[\kappa_L^H L + dc_\perp^H - \omega_T^H a_T L - \omega_X^H \hat{X} + \omega_F^H \hat{F} \right].$$

Solving gives (56), with

$$\begin{aligned}\Gamma_X &\equiv \frac{\gamma_S \omega_X^H}{1 - \gamma_S \kappa_L^H + \gamma_S \omega_T^H a_T} > 0, \\ \Gamma_F &\equiv \frac{\gamma_S \omega_F^H}{1 - \gamma_S \kappa_L^H + \gamma_S \omega_T^H a_T} > 0, \\ \Gamma_C &\equiv \frac{\gamma_S}{1 - \gamma_S \kappa_L^H + \gamma_S \omega_T^H a_T} > 0,\end{aligned}\tag{C.79}$$

where positivity uses $\kappa_L^H < 0$. Since $\Gamma_F/\Gamma_X = F/X$, holding the net consumption-demand margin fixed gives

$$L < 0 \iff X\hat{X} > F\hat{F} \iff dX > dF.$$

Pledgeability and foreign absorption. At an active, nonsaturated tail solution with fixed standardized return distribution,

$$m + \kappa_C = \exp \left\{ \frac{x_p}{2} + (1 - m)\lambda_T^S - \frac{x_p}{\alpha_p} N^F - z_{\bar{p}} \sqrt{x_p} \right\}.$$

Differentiating this equation and collecting dm gives

$$\left[1 + \lambda_T^S(m + \kappa_C) \right] dm = (m + \kappa_C) \left[\left(\frac{1}{2} - \frac{N^F}{\alpha_p} - \frac{z_{\bar{p}}}{2\sqrt{x_p}} \right) dx_p + (1 - m)d\lambda_T^S - \frac{x_p}{\alpha_p} dN^F \right]. \tag{C.80}$$

The coefficient on dm is positive. Because $N^F \geq 0$, $x_p < z_{\bar{p}}^2$ is sufficient for the coefficient on dx_p to be negative. A decline in risk then improves the lower tail; a decline in the Treasury service wedge or an increase in Foreign ownership lowers the expected excess return and works in the opposite direction.

For the policy comparison, write

$$\pi_\psi \equiv \frac{d \log \bar{P}}{d\psi}, \quad u_\psi \equiv \frac{d \log m}{d\psi}, \quad \hat{F}_\psi \equiv \frac{d \log F}{d\psi}, \quad c_\psi \equiv \frac{dc_\perp^H}{d\psi}.$$

The pledgeability and service-market responses can also be displayed as one joint equation.

Substituting (56) and collecting u_ψ gives

$$\begin{aligned} & \left\{ m \left[1 + \lambda_T^S(m + \kappa_C) \right] + (m + \kappa_C)(1 - m)\lambda_T^S \Gamma_X \right\} u_\psi \\ &= (m + \kappa_C) \left[\left(\frac{1}{2} - \frac{N^F}{\alpha_p} - \frac{z_{\bar{p}}}{2\sqrt{x_p}} \right) \frac{dx_p}{d\psi} - (1 - m)\lambda_T^S \Gamma_X \pi_\psi \right. \\ & \quad \left. + (1 - m)\lambda_T^S (\Gamma_F \hat{F}_\psi + \Gamma_C c_\psi) - \frac{x_p}{\alpha_p} \frac{dN^F}{d\psi} \right]. \end{aligned} \quad (\text{C.81})$$

The positive coefficient on u_ψ shows that service-wedge erosion attenuates rather than reverses the direct tail mechanism mechanically. The sign still depends on the valuation, absorption, and consumption responses on the right-hand side.

Proof of Lemma 1. Evaluating (C.80) along the ψ -indexed equilibrium branch and imposing $x_p = x$ and $\alpha_p = \alpha$ gives (54). Under $x < z_{\bar{p}}^2$, its risk coefficient is negative. Proposition 5 gives $dx/d\psi < 0$, so their product is positive. Since the coefficient multiplying $dm/d\psi$ is positive, the sign of pledgeability is the sign of the three terms on the right-hand side. $\square$

Foreign service absorption obeys

$$d \log F = d \log A^{F,0} + \frac{\omega^F m}{\vartheta^F} d \log m - \frac{1 - m}{\vartheta^F} d \omega^F. \quad (\text{C.82})$$

Thus an equity rotation can raise equity demand while lowering foreign service absorption, because a Treasury dollar is more service intensive when $m < 1$. In levels, $F = \vartheta^F A^{F,0}$ and $A^{F,0} \omega^F = \bar{P} N^F$. Therefore

$$\begin{aligned} dF &= \vartheta^F dA^{F,0} + A^{F,0} \omega^F dm - A^{F,0} (1 - m) d\omega^F, \\ dX - dF &= \bar{P} (1 - N^F) dm + m d\bar{P} + A^{F,0} (1 - m) d\omega^F - \vartheta^F dA^{F,0}. \end{aligned} \quad (\text{C.83})$$

The bounded logit schedule implies

$$\frac{dA^{F,0}}{d\psi} = A^{F,0} \kappa_A \left(1 - \frac{A^{F,0}}{\mathcal{A}^F} \right) \frac{dQ^F}{d\psi}. \quad (\text{C.84})$$

Thus greater pledgeability on Home-owned shares, equity-price appreciation, and a Treasury-to-equity rotation all raise the capacity retained by Home relative to Foreign absorption. Expansion of the Foreign dollar share works in the opposite direction. Its marginal contribution is governed by κ_A and the slope bound following (C.7).

Proof of Proposition 6. Differentiating $F = \vartheta^F A^{F,0}$, where $\vartheta^F = 1 - (1 - m)\omega^F$, gives the first line of (57). Since $X = m\bar{P}$ and $A^{F,0} \omega^F = \bar{P} N^F$, subtracting dF from dX gives (58). These are the two level identities in (C.83). $\square$

Proof of Corollary 2. Equation (56) gives $d \log \lambda_T^S < 0$ exactly when

$$\Gamma_X d \log X > \Gamma_F d \log F + \Gamma_C dc_{\perp}^H,$$

which is (59). When $dc_{\perp}^H = 0$, the share ratio $\Gamma_F/\Gamma_X = F/X$ reduces the condition to $dX > dF$. Finally, the change in the Treasury convenience yield has the same sign as the service-wedge response. $\square$

C.5 Equilibrium Equity Price and Foreign Order Flow

Write the reduced Foreign ownership schedule as

$$\frac{d \log N^F}{d\psi} = g_{N,\psi}^d - \eta_{N,\psi} \pi_{\psi}, \quad \eta_{N,\psi} \geq 0, \quad (\text{C.85})$$

where $g_{N,\psi}^d$ is the outward shift in Foreign share demand holding the equity price fixed and $\eta_{N,\psi}$ is its price semi-elasticity. Define the inverse price impact of structural Foreign demand and its policy response by

$$\Lambda_{\psi} \equiv \frac{x_{p,\psi}}{\alpha_p \omega_{D,\psi}^o} > 0, \quad g_{\Lambda,\psi} \equiv \frac{d \log \Lambda_{\psi}}{d\psi}, \quad (\text{C.86})$$

where $\omega_{D,\psi}^o > 0$ is the regime-level payoff loading. Let $\Xi_P^o > 0$ denote the capitalization elasticity of the equity service wedge, while $\pi_{\perp,\psi}$ collects all nonservice price shifts.

Proposition C.1 (Equilibrium equity price and usable collateral capacity). *At a differentiable interior regime equilibrium satisfying (C.85), the equilibrium price and capacity responses obey*

$$\begin{aligned} \mathcal{D}_P \pi_{\psi} &= \Xi_P^o \left[(1 - \Gamma_X) u_{\psi} + \Gamma_F \hat{F}_{\psi} + \Gamma_C c_{\psi} \right] + \pi_{\perp,\psi} \\ &\quad + \Lambda_{\psi} N^F \left(g_{N,\psi}^d + g_{\Lambda,\psi} \right). \end{aligned} \quad (\text{C.87})$$

$$\begin{aligned} \mathcal{D}_P \frac{d \log X}{d\psi} &= \left(1 + \Xi_P^o + \Lambda_{\psi} N^F \eta_{N,\psi} \right) u_{\psi} + \Xi_P^o \left(\Gamma_F \hat{F}_{\psi} + \Gamma_C c_{\psi} \right) + \pi_{\perp,\psi} \\ &\quad + \Lambda_{\psi} N^F \left(g_{N,\psi}^d + g_{\Lambda,\psi} \right). \end{aligned} \quad (\text{C.88})$$

Here the positive price denominator is

$$\mathcal{D}_P \equiv 1 + \Xi_P^o \Gamma_X + \Lambda_{\psi} N^F \eta_{N,\psi} > 0.$$

Consequently, the price and usable-capacity responses have the signs of the right-hand sides of (C.87) and (C.88), respectively.

Proof of Proposition C.1. At a differentiable point on the maintained local equilibrium branch,

write

$$\pi_\psi^0 \equiv \frac{d\mathbf{p}^0}{d\psi}.$$

Differentiating (C.45) and using (C.85) gives

$$\begin{aligned} \pi_\psi &= \pi_\psi^0 + \Lambda N^F \left(g_{N,\psi}^d - \eta_{N,\psi} \pi_\psi + g_{\Lambda,\psi} \right), \\ (1 + \Lambda N^F \eta_{N,\psi}) \pi_\psi &= \pi_\psi^0 + \Lambda N^F \left(g_{N,\psi}^d + g_{\Lambda,\psi} \right). \end{aligned} \quad (\text{C.89})$$

The term $g_{N,\psi}^d$ is the shift of the reduced ownership schedule at a fixed equity price, after portfolio choice, dollar allocation, and the expected- return feedback in (45) have been solved. The semi-elasticity

$$\eta_{N,\psi} \equiv - \left. \frac{\partial \log N^F}{\partial \log \bar{P}} \right|_\psi$$

is nonnegative when that reduced ownership schedule slopes downward.

Define $\Xi_P^o > 0$ as the local capitalization elasticity of the non-order- flow component $\mathbf{p}^0$ with respect to the equity service wedge. Since $\lambda_P = m\lambda_T^S$, its local differential can be written

$$\pi_\psi^0 = \Xi_P^o \left(\frac{d \log m}{d\psi} + \frac{d \log \lambda_T^S}{d\psi} \right) + \pi_{\perp,\psi}. \quad (\text{C.90})$$

Here $\pi_{\perp,\psi}$ is the response of $\mathbf{p}^0$ holding the equity service wedge fixed; it collects changes in nonservice regime-level fundamentals, including any independent movement in mean pay-offs, pecuniary discounting, and the external-demand block. In the closed-economy limit, Ξ_P^o converges to Ξ and $\pi_{\perp,\psi} = 0$ under the fixed- mean-output comparison of Section 4.3.

Set

$$u_\psi \equiv \frac{d \log m}{d\psi}, \quad \hat{F}_\psi \equiv \frac{d \log F}{d\psi}, \quad c_\psi \equiv \frac{dc_\perp^H}{d\psi}.$$

Because $X = m\bar{P}$, equation (56) becomes

$$\frac{d \log \lambda_T^S}{d\psi} = -\Gamma_X(u_\psi + \pi_\psi) + \Gamma_F \hat{F}_\psi + \Gamma_C c_\psi. \quad (\text{C.91})$$

Substituting (C.90) and (C.91) into (C.89) gives (C.87). Every term in $\mathcal{D}_P$ is nonnegative and its constant term is one, so $\mathcal{D}_P > 0$. The sign of π_ψ , and therefore of $d\bar{P}/d\psi$, is exactly the sign of the right-hand side. Under the corrected consumption rule, c_ψ itself contains the equity-wealth response $\chi_p^o \pi_\psi$, so (C.87) determines π_ψ jointly with the consumption margin rather than in isolation; the display treats c_ψ as a collected input.

Because $d \log X / d\psi = u_\psi + \pi_\psi$, adding $\mathcal{D}_P u_\psi$ to the two sides of (C.87) and using

$$\mathcal{D}_P + \Xi_P^o(1 - \Gamma_X) = 1 + \Xi_P^o + \Lambda N^F \eta_{N,\psi}$$

gives (C.88). The capacity equation therefore does not impose the sign of the equilibrium price. Higher pledgeability contributes positively to $d \log X / d\psi$ even when service-market or nonservice forces make $\pi_\psi < 0$.

For completeness, the fixed-price foreign-demand shift can be recovered from the portfolio and bounded dollar-allocation block. Let a superscript d denote the reduced response holding $\bar{P}$ fixed, after the ownership feedback has been solved. Since $N^F = \omega^F A^{F,0} / \bar{P}$,

$$g_{N,\psi}^d = \kappa_A \left(1 - \frac{A^{F,0}}{\mathcal{A}^F} \right) \frac{dQ^{F,d}}{d\psi} - \frac{\mathcal{D}_{\omega,\psi}^d}{\omega^F G_\omega}. \quad (\text{C.92})$$

Because $G_\omega < 0$, the portfolio contribution is positive when $\mathcal{D}_{\omega,\psi}^d > 0$, and the dollar-allocation contribution is positive when $dQ^{F,d}/d\psi > 0$. A decline in x_p contributes positively to both conditions: its portfolio term and account-value envelope term are

$$-\alpha_F \omega^F \frac{dx_p}{d\psi} > 0, \quad -\frac{\alpha_F}{2} (\omega^F)^2 \frac{dx_p}{d\psi} > 0.$$

A decline in x_e raises account value but does not change the interior portfolio share, while a rise in c_{pe} raises both:

$$-\frac{\alpha_F}{2} \frac{dx_e}{d\psi} > 0, \quad \alpha_F \frac{dc_{pe}}{d\psi} > 0, \quad \alpha_F \omega^F \frac{dc_{pe}}{d\psi} > 0.$$

These last two inequalities reverse when covariance falls. Expected-return, service, covariance, and expected-depreciation responses can therefore offset the direct variance effects.

Finally, the asymmetric price-impact definition in (C.45) implies

$$g_{\Lambda,\psi} = \frac{d \log x_p}{d\psi} - \frac{d \log \omega_D^o}{d\psi}. \quad (\text{C.93})$$

When the regime-level payoff loading is locally fixed, the direct order-flow contribution is positive exactly when

$$g_{N,\psi}^d > -\frac{d \log x_p}{d\psi}. \quad (\text{C.94})$$

Under the symmetric benchmark, $x_p = x$ in these two expressions. This condition does not sign the service, nonservice, or total price response. $\square$

C.6 Equilibrium Dollar and Foreign Participation

Proof of Proposition 7. At the zero-noise equilibrium, the currency-specialist position in (C.16) is

$$\bar{D}^e = \frac{\alpha_e}{x_e} \left[\bar{r} - \bar{r}^* - \bar{\Delta}e + \frac{x_e}{2} \right] = \frac{\alpha_e}{x_e} \left[\bar{r} - \bar{r}^* + \frac{x_e}{2} \right],$$

using $\bar{\Delta}e_\psi = 0$ from Lemma C.1. Currency-risk clearing, $\bar{D}^e = Z^e(\bar{e}) - A^{F,0}$, gives the regime-level closure (C.65). Differentiating that equation and imposing $x_e = x$ and $\alpha_e = \alpha$ yields (60). Since an increase in e is dollar depreciation and $Z_e < 0$, $d\bar{e}/d\psi < 0$ holds exactly when the right-hand side of (60) is positive.

The bounded exposure schedule in (C.7) gives

$$d \log A^{F,0} = \kappa_A \left(1 - \frac{A^{F,0}}{\mathcal{A}^F} \right) dQ^F, \quad dQ^F = dV^F,$$

by (C.64). This proves (61). Finally, applying the envelope theorem to (C.2) gives (C.12). Setting $dx_p = dx_e = dx$ combines its two variance terms and gives (62) along the ψ -indexed equilibrium allocation. Because $0 < A^{F,0}/\mathcal{A}^F < 1$, the participation response has the sign of $dQ^F/d\psi$ when $\kappa_A > 0$; participation is locally fixed when $\kappa_A = 0$. $\square$

C.7 External Stock–Flow Accounting

We now derive (C.38) and (64) with transactions and valuation recorded separately. Let the beginning-of-date inherited long-lived quantities be $(Q_{A,t-1}^H, N_{t-1}^F)$. Immediately after current prices are observed but before transactions, their marked gross positions are

$$\mathcal{A}_t^- = P_{A,t} Q_{A,t-1}^H, \quad \mathcal{L}_{P,t}^- = P_t N_{t-1}^F. \quad (\text{C.95})$$

The Treasury position inherited from $t-1$ instead matures at date t . Its interest component enters net investment income, while principal repayment and new issuance enter financial transactions.

Net investment income is

$$\text{NPI}_t = I_{A,t}^H - d_{P,t} N_{t-1}^F - I_{T,t}^F. \quad (\text{C.96})$$

Treasury interest is defined in (C.43). Cash income is excluded from the ex-cash marked positions above. Define transactions in the two long-lived claims by

$$\mathcal{T}_{A,t} \equiv A_t^{H,+} - A_t^{H,-}, \quad \mathcal{T}_{P,t} \equiv P_t (N_t^F - N_{t-1}^F). \quad (\text{C.97})$$

Together with the Treasury transaction in (C.43), these definitions separate every current trans-

action from cash income and inherited-position valuation.

Let FA_t be Home acquisition of foreign assets minus Foreign acquisition of U.S. liabilities. Then

$$FA_t = \mathcal{T}_{A,t} - \mathcal{T}_{P,t} - \mathcal{T}_{T,t}. \quad (C.98)$$

With the sign convention used here, the balance of payments is

$$CA_t - FA_t = 0. \quad (C.99)$$

Solving (C.98)–(C.99) for $\mathcal{T}_{A,t}$ proves (C.38).

Walras-law check. Actual U.S. claims clear through

$$N_t^H + N_t^F = 1, \quad B_t^H + B_t^F = \bar{B}. \quad (C.100)$$

The currency contract separately clears at zero net supply:

$$D_t^e + D_t^{e,c} = 0, \quad D_t^e \Pi_{t+1}^e + D_t^{e,c} \Pi_{t+1}^e = 0. \quad (C.101)$$

Both derivative counterparties are Foreign, so (C.101) vanishes when their positions and budgets are consolidated. It contributes neither a Home resource nor a balance-of-payments flow. Summing the Home budgets, using the actual-claim ownership conditions in (C.100), and imposing (C.98)–(C.99) therefore leaves the goods resource constraint with no residual financial term. This is the Walras law for the modeled Home markets and external accounts; the reduced-form Foreign customer schedule is not represented as a complete Foreign economy.

The exact equity-liability decomposition is

$$P_t N_t^F - P_{t-1} N_{t-1}^F = N_{t-1}^F (P_t - P_{t-1}) + P_t (N_t^F - N_{t-1}^F). \quad (C.102)$$

Because Treasurys mature after one period, their entire end-stock change is the rollover transaction $\mathcal{T}_{T,t}$, not a revaluation of an inherited long-lived claim. The corresponding ex-cash valuation component is therefore

$$VAL_t = Q_{A,t-1}^H (P_{A,t} - P_{A,t-1}) - N_{t-1}^F (P_t - P_{t-1}). \quad (C.103)$$

Combining the marked beginning positions, transactions, and (C.99) gives

$$NFA_t - NFA_{t-1} = CA_t + VAL_t,$$

which is (64).

Proof of Corollary 3. Within the acquisition component, (66) sets Home acquisition of the foreign asset equal to Foreign acquisition of U.S. liabilities. Hence the transaction changes gross positions but gives $d^{\text{acq}}\text{NFA} = 0$. Regardless of the contemporaneous change in N_t^F , the inherited-position term in (C.102) is $N_{t-1}^F dP_t$. Because U.S. equity is an external liability, its contribution to U.S. valuation is $-N_{t-1}^F dP_t$, proving (67). The current-price term $P_t dN_t^F$ is instead a financial transaction. This classification remains exact even when foreign demand helps determine P_t . Along a policy-regime comparison, $dP_t/d\psi = \bar{P}_\psi \pi_\psi$. Substitution into (67) gives (68). Its sign is negative when the right-hand side of (C.87) is positive and $N_{t-1}^F > 0$. $\square$

Foreign ownership and future valuation exposure. In the regime-level allocation, Foreign dollar equity demand is $E^F = \omega^F A^{F,0}$, and equity-market clearing implies $N^F = E^F / \bar{P}$. Log differentiation therefore gives

$$\frac{d \log N^F}{d\psi} = \frac{d \log A^{F,0}}{d\psi} + \frac{d \log \omega^F}{d\psi} - \frac{d \log \bar{P}}{d\psi}. \quad (\text{C.104})$$

The demand decomposition (53) and the portfolio response (C.75) give

$$\frac{d \log N^F}{d\psi} = \kappa_A \left(1 - \frac{A^{F,0}}{\mathcal{A}^F} \right) \frac{dQ^F}{d\psi} - \frac{\mathcal{D}_\omega}{\omega^F G_\omega} - \pi_\psi. \quad (\text{C.105})$$

Thus Foreign ownership rises exactly when the first two terms on the right-hand side exceed equity-price growth.

Once Foreign ownership N_ψ^F is established, the quantity is inherited at date $t + 1$. Applying the equity-liability identity (C.102) one date forward, its ex-cash valuation contribution is

$$\text{VAL}_{P,t+1} = -N_\psi^F (P_{t+1} - P_t).$$

Holding the inherited quantity fixed when differentiating with respect to P_{t+1} gives

$$-\frac{\partial \text{VAL}_{P,t+1}}{\partial P_{t+1}} = N_\psi^F.$$

Greater foreign ownership therefore raises exposure to subsequent appreciation one-for-one. The acquisition component of the change in N_ψ^F remains NFA-neutral by (66).

Treasury rollover interest. The quoted rate and issue price satisfy

$$r_t = \rho + \log(1 - \lambda_{T,t}^S), \quad q_{B,t} = e^{-r_t}.$$

Debt issued at t with face value B_t^F pays interest $I_{T,t+1}^F = (1 - q_{B,t})B_t^F$ at maturity. Holding the rolled-over face value fixed,

$$dr_t = -\frac{d\lambda_{T,t}^S}{1 - \lambda_{T,t}^S}, \quad dI_{T,t+1}^F = q_{B,t}B_t^F dr_t.$$

This proves (69). The policy-induced rate change affects newly issued debt; inherited debt pays the return fixed at its issue date.

Service and convenience-yield effects occupy separate accounting locations. Treasury interest paid to Foreign is part of NPI, and service-wedge erosion raises that payment upon rollover. The price paid for newly sold equity is a financial-account transaction, subsequent ex-cash appreciation is valuation, and collateral services enter utility.

C.8 Welfare Envelopes

For Home period utility

$$U(C, S) = \log C + \xi v(S/C),$$

the derivative along the ψ -indexed allocation is

$$\frac{dU}{d\psi} = U_C \left(\frac{dC}{d\psi} + \lambda_T^S \frac{dS}{d\psi} \right),$$

where $U_S/U_C = \lambda_T^S$. Discounting and taking expectations proves (73).

For the direct service term, let $z^H = S^H/C^H$. Home inverse service demand satisfies $\lambda_T^S = \lambda^S(z^H)$ with $\lambda_z^S < 0$. Hence

$$\frac{d\lambda_T^S}{d\psi} = \lambda_z^S \frac{dz^H}{d\psi}.$$

When current Home consumption is held fixed, $C_\psi^H = 0$,

$$S_\psi^H = C^H z_\psi^H.$$

Therefore $d\lambda_T^S/d\psi < 0$ implies $z_\psi^H > 0$ and $S_\psi^H > 0$. At the maintained interior $U_C^H \lambda_T^S > 0$, so the contemporaneous direct service contribution is positive. Along a general allocation path, its exact sign is the discounted weighted sum of service changes in (73).

Consumption incorporates cash income, Treasury interest, and the external account through (C.21). External valuation changes consumption resources through the budget and is not added again to utility.

For Foreign dollar-account customers, define participation surplus by

$$\mathcal{W}^F(q) = \int_{Q_0^F}^q A^{F,0}(u) du,$$

where Q_0^F is an arbitrary normalization point. The fundamental theorem of calculus gives

$$\frac{d\mathcal{W}^F}{dq} = A^{F,0}(q).$$

Setting $q = Q_\psi^F$ gives (74). Since $A^{F,0} > 0$, participation surplus rises exactly when $dQ_\psi^F/d\psi > 0$. The envelope theorem for (C.2) eliminates a separate first-order $d\omega^F$ term at an interior delegated portfolio. This object excludes the rents of Foreign currency specialists and therefore is not a measure of total Foreign welfare.

Proof of Proposition 8. Equation (73) gives the total Home condition. The inverse-demand argument above proves the fixed-consumption service statement. Equation (74) gives the Foreign dollar-account customer participation condition. The Home consumption and service terms can have opposite signs, so the service result does not sign total Home asset-holder utility. $\square$